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Enhancing Automatic Lyrics Transcription: Challenges, Insights, and Advances

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for the degree of Doctor of Philosophy

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Abstract

Automatic Lyrics Transcription (ALT) is the task of transcribing singing content into text. While Automatic Speech Recognition (ASR) has seen remarkable success in real-world applications, ALT remains a significant challenge due to the unique characteristics of the singing voice, which differ substantially from speech. ALT must address variations in pitch and rhythm, pronunciation differences, multilingual complexity, and background music interference. This thesis systematically investigates these challenges and proposes novel approaches to improve ALT across these dimensions, providing a comprehensive understanding of the factors that affect ALT and introducing techniques to mitigate these challenges.

First, we explore the role of pitch in lyrics alignment. By leveraging the shared onsets between pitch and lyrics, we improve alignment accuracy for English songs, motivating the extension of pitch-aware strategies to full ALT systems. Next, we deal with pronunciation and rhythm variations, which significantly impact transcription performance. Due to the difficulty of directly studying these factors, we introduce Singing-to-Speech Conversion (S2S), a novel cross-domain voice conversion task that transforms singing into speech while preserving phonetic content and enabling controllable duration manipulation. By evaluating ALT on converted singing, we gain valuable insights into the effects of pronunciation and rhythm variation.

Expanding beyond monolingual settings, we address the multilingual challenges of ALT, particularly the imbalance in language resources. We propose phonetic reinterpretation and language conditioning techniques to enhance transcription across multiple languages, introducing several methods where multilingual joint training benefits low-resource languages. Finally, we revisit the impact of background music on ALT, noting that previous conclusions often depend on the quality of source separation tools. We fine-tune a foundation ASR model for polyphonic mixtures and introduce a consistency loss to align vocal and mixture representations, leading to improvements in transcription robustness.

To support future ALT research, we also contribute to dataset annotation, curation, and benchmarking, including annotations for lyrics alignment (MUSDB18), expansion of multilingual datasets (MulJam), and a systematic evaluation of self-supervised features (integrated into the MARBLE benchmark).

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Contents

1	Introduction	18
1.1	Motivation and Aim	19
1.2	Thesis Structure	20
1.3	Contributions and Findings	21
1.4	Associated Publications	22
1.5	Visits and Placements	24
2	Background	25
	<i>Part I: Technical Foundations</i>	26
2.1	Introduction to Singing Voice	26
2.1.1	Speech and Singing Voice	26
2.1.2	Related Tasks and Research Area	28
2.2	Audio Representations for Deep Learning	29
2.2.1	Waveform	29
2.2.2	Time-Frequency Representations	30
2.2.3	Learned Neural Representations	30
2.2.4	Other Representations	31
2.3	Machine Learning Models and Architectures	31
2.3.1	Wave-U-Net	32
2.3.2	Transformer	33
2.3.3	Hybrid CTC/Attention	35
2.3.4	Wav2Vec2	36
2.3.5	Flow-based Generative Models	37
2.3.6	Diffusion Models	38
2.4	Machine Learning Paradigms	39
2.4.1	Supervised Learning	39
2.4.2	Semi-Supervised Learning	39
2.4.3	Self-Supervised and Unsupervised Learning	40
2.4.4	Multi-task Learning	41
2.4.5	Adapting Pre-trained Models	41

2.5	Data Augmentation for Singing Voice Research	42
2.5.1	Basic Augmentation Techniques	43
2.5.2	Advanced Augmentation Techniques	43
2.5.3	Beyond Existing Techniques	44
	<i>Part II: Related Tasks</i>	44
2.6	Speech Recognition	45
2.6.1	Task Definition	45
2.6.2	Evaluation Metrics	45
2.6.3	Related Work	46
2.6.4	Recent Advancements	49
2.7	Lyrics Alignment	50
2.7.1	Task Definition	50
2.7.2	Evaluation Metrics	51
2.7.3	Related Work	52
2.7.4	Limitations	55
2.8	Automatic Lyrics Transcription	55
2.8.1	Task Definition	56
2.8.2	Evaluation Metrics	57
2.8.3	Related Work	57
2.8.4	Key Challenges	61
2.9	Singing Voice Conversion	63
2.9.1	Task Definition	64
2.9.2	Evaluation Metrics	64
2.9.3	Related Work	65
2.9.4	Recent Advancements	67
	<i>Conclusion</i>	67
3	Datasets and Benchmarks	69
3.1	Introduction	69
3.2	Datasets: Existing Datasets	71
3.2.1	Large-Scale Datasets	71
3.2.2	Small-Scale Datasets	73
3.3	Datasets: MulJam v1.0 and v2.0	74
3.3.1	MulJam v1.0 Dataset	74
3.3.2	MulJam v2.0 Dataset	75
3.3.3	Further Updates	76
3.4	Datasets: Word-Level Lyrics Extension for MUSDB18 test set . .	76
3.5	Benchmarking: Architecture and Training	77
3.5.1	Method	77
3.5.2	Experiments	81

3.5.3	Results and Discussion	82
3.6	Benchmarking: MARBLE	85
3.6.1	MARBLE Benchmark	85
3.6.2	ALT in MARBLE	86
3.6.3	Language Model and Tokenizer	87
3.6.4	Training Details	87
3.6.5	Results and Discussion	87
3.7	Inconsistencies in Datasets and Evaluation	89
3.8	Conclusion	90
4	The Role of Pitch: Incorporating Musical Attributes	92
4.1	Introduction	92
4.2	Joint Phoneme Recognition and Pitch Detection	94
4.2.1	Acoustic Model	95
4.2.2	Loss Function	96
4.3	Alignment	97
4.3.1	Viterbi Forced Alignment	97
4.3.2	Incorporating Boundary Information	97
4.4	Experiments	98
4.4.1	Dataset	98
4.4.2	Training	99
4.4.3	Evaluation	100
4.5	Results and Discussion	100
4.5.1	English State of the Art Comparison	101
4.5.2	Boundary Detection and Pitch Detection	101
4.5.3	MTL Evaluations	103
4.6	Multilingual Lyrics Alignment Results and Discussion	103
4.6.1	Results on English Subset	103
4.6.2	MTL Evaluations	104
4.6.3	Multilingual State of the Art Comparison	104
4.7	Results on MUSDB18 Test	105
4.8	Conclusion	106
5	Pronunciation and Rhythm: Studying Singing Voice through Singing-to-Speech Conversion	109
5.1	Introduction	109
5.2	Singing to Speech Conversion	111
5.3	Proposed Method	112
5.3.1	Training	113
5.3.2	Inference	116

5.3.3	Architecture	116
5.4	S2S for data augmentation	117
5.5	Experiments	118
5.5.1	Datasets	118
5.5.2	Baselines	119
5.5.3	Training and Inference	119
5.5.4	Evaluation	120
5.5.5	S2S for data augmentation	121
5.6	Results and Discussion	124
5.6.1	Objective results	124
5.6.2	Subjective results	126
5.6.3	S2S for data augmentation results	128
5.7	Conclusion	128
6	Multilingual ALT: Addressing the Resource Imbalance	130
6.1	Introduction	130
6.2	Related Work	132
6.2.1	Language Conditioning for Multilingual ASR	132
6.2.2	Phonetic Reinterpretation	132
6.3	Motivation	133
6.3.1	Joint Multilingual Training and Language Conditioning	133
6.3.2	Phonetic Augmentation for Multilingual Data	133
6.4	Implementation and Experimental Setup	135
6.4.1	Datasets	135
6.4.2	Baseline and State-of-the-Art Models	136
6.4.3	Training and Inference	139
6.4.4	Language Model	140
6.4.5	Evaluation	141
6.5	Multilingual Joint Training	142
6.5.1	Method	142
6.5.2	Results and Discussion	143
6.6	Language Conditioning	144
6.6.1	Method	144
6.6.2	Results and Discussion	146
6.7	Phonetic Reinterpretation	147
6.7.1	Method	148
6.7.2	Experiments	149
6.7.3	Results and Discussion	149
6.7.4	With Unseen Languages	151
6.8	Limitations and Key Insights	152

6.9	Conclusion	152
7	Enhancing ALT for Music Mixtures: Adapting Foundation Models	154
7.1	Introduction	154
7.2	Revisiting the impact of background music	156
7.3	Probing Wav2Vec2 with a Lightweight Back-End	157
7.3.1	Method	158
7.3.2	Experiments	159
7.3.3	Results and Discussion	159
7.4	Fine-Tuning Whisper with a Consistency loss	160
7.4.1	Method	161
7.4.2	Experiments	162
7.4.3	Results and discussion	167
7.5	Conclusion	169
8	Conclusions and Further Work	170
8.1	Summary	170
8.1.1	Challenges	172
8.1.2	Insights	174
8.1.3	Advances	175
8.2	What Makes Singing Harder than Speech	176
8.3	Limitations	177
8.4	Future Work	178
8.4.1	Learning Better Data Representations	178
8.4.2	Leveraging Data More Effectively	178
8.4.3	Understanding and Modeling Singing Voice	179
8.4.4	Reconsidering the Role of Background Music	180
A	Screening questions for phonetic similarity ratings	181
B	Objective evaluation on NHSS	183
C	Ablation study on Intelligibility	185

List of Figures

2.1	Intersection of text, speech, and singing domains.	28
2.2	Architecture of the Wave-U-Net model with L layers for K sources, adapted from Figure 1 in Stoller et al. [2018]. The prediction for the K -th source is computed as the difference between the input mixture and the sum of the other $K - 1$ predicted sources. . . .	33
2.3	The Transformer architecture (Vaswani et al. [2017], Figure 1). Used with permission from Google.	34
2.4	Hybrid CTC/attention-based end-to-end architecture, simplified from Figure 1 in Watanabe et al. [2017].	35
2.5	Example of word-level alignment and phoneme-level alignment. .	50
2.6	Example of automatic lyrics transcription.	55
2.7	Example of voice conversion. The same linguistic content is rendered in different visual styles to represent changes in vocal attributes.	63
3.1	Diagram of ALT model architectures.	78
4.1	System overview.	95
4.2	The network architecture for the baseline, boundary detection, and the MTL models. The only differences among them are the dimensions of the BiLSTMs and the last fully-connected layer. .	95
4.3	Multi-task loss computation.	96
4.4	Lyrics alignment, pitch detection, and boundary detection results on a clip of <code>Average_-_Embers.mp3</code> from Jamendo. The GT (ground truth) notes are slightly shifted for visualization purposes. Two notes have the same pitch if there is an overlap. The MTL model successfully associates the onset of the word “close” with the corresponding note change.	102
4.5	AAE and PCO across source separation methods and models. . .	106
5.1	Example of phonetic ambiguity in singing.	111

5.2	Diagram of the proposed S2S model.	114
5.3	Architecture of the ALT model. It consists of a convolutional block <i>CNNBlock</i> , a Transformer encoder <i>TfmEnc</i> , a Transformer decoder <i>TfmDec</i> , and two fully connected layers FC_{ctc} , FC_{s2s}	122
5.4	Comparison of different methods in subjective evaluation. All values are Mean Opinion Score (MOS). All pair-wise significance level is $p < 0.001$ except for those indicated above the plot. A dotted line means no significance.	126
5.5	S2S for data augmentation results on different training sets. On the left: WERs on the DAMP test set, with numbers in parentheses indicating the duration of the subsets. The numbers in the parentheses indicate the duration of the subsets. On the right: CERs on the OpenSinger test set.	128
6.1	Example of phonetic reinterpretation for ALT.	134
6.2	Language conditioning through language embeddings.	145
6.3	Language conditioning through FiLM layers.	145
6.4	Process for generating phonetic proxies for singing samples in language A to language B using a language-configurable multilingual ALT model.	148
6.5	Process for training with phonetic augmentation.	148
7.1	Diagram of Wav2Vec2 probing for ALT.	158
7.2	Diagram of dual-domain fine-tuning with consistency loss.	161
A.1	Example shown to participants to explain the concept of phonetic similarity.	182
A.2	Interface of a screening question.	182

List of Tables

2.1	Comparison of lyrics alignment performance in the literature. . .	53
2.2	The current state-of-the-art methods for lyrics alignment (ResConv: residual convolutional blocks; ContrL: contrastive learning). All methods use Viterbi forced alignment, although applied to different types of score matrices.	53
2.3	Comparison of architectures and learning strategies in E2E ALT. <i>Hybrid</i> refers to a combination of CTC and attention-based sequence-to-sequence learning [Watanabe et al., 2017].	59
2.4	Comparison of music-related domain knowledge and their integration strategies in ALT.	59
3.1	Number of songs in MulJam1.0 and MulJam2.0 datasets. The relatively small number of test songs in some languages results from limited human resources for manual annotation.	75
3.2	Overview of the three ALT model architectures explored in this section, and the chapters where these models are applied.	79
3.3	Results of CRNN model with different CTC branch types.	82
3.4	Results of Transformer model with different CTC branch types.	83
3.5	Results of Wav2Vec2-based ALT model with different fine-tuning strategies, loss combinations, and decoding methods. Fine-tuned results (second row) are from Ou et al. [2022], while frozen-encoder results are from our experiments.	83
3.6	Multilingual lyrics transcription results on MulJam v2.0 and Multi-Lang Jamendo v1.0.	88
3.7	Overview of datasets model training and evaluation granularity in this thesis. Unless specified, the default task is ALT. The SVS column indicates the Singing Voice Separation tool used for the task. NA indicates that the task does not involve SVS.	91

4.1	Comparison with the state-of-the-art systems for lyrics alignment. The SS column indicates if the input has been source-separated (Yes or No). The FW column indicates if the method takes an end-to-end (E2E), standard (STD) ASR, or contrastive learning (Contrl) framework. The dashed line indicates when our ICASSP paper [Huang et al., 2022] was published.	100
4.2	Boundary detection results.	101
4.3	Pitch detection results.	101
4.4	English lyrics alignment results. Substantial improvement in AAE on Jamendo for the MTL model can be observed for word-level metrics. All line-level metrics increase after introducing boundary information.	102
4.5	Multilingual lyrics alignment results on the Multi-Lang Jamendo dataset (Overall) and the English subset (English). The English subset is equivalent to the Jamendo (English) set in Table 4.4.	103
4.6	Multilingual lyrics alignment results on the Multi-Lang Jamendo dataset, with comparison to the DSE models in Durand et al. [2023]. The Overall column is the overall result, while the other columns are the results on each language subset.	104
4.7	Lyrics alignment results on the MUSDB18 test set. We compare the model from Gupta et al. [2020], our Baseline models, and the MTL models. The header specifies the inference input source, indicating whether clean vocal stems or separated vocals obtained using the stated source separation method were used.	105
5.1	Objective evaluation on the DSing test set. Sing indicates the original singing voice.	125
5.2	Demographic information and singing experience of participants, including gender, age, and amount of singing experience.	127
6.1	The number of lines in the training, validation, and test sets for each language.	135
6.2	Comparison of model complexity, required GPU VRAM, and relative inference time. The Whisper model used here is large-v3 . SpecTfm denotes the spectrogram-based Transformer. Inference time is the relative time to it.	137
6.3	WER (%) of monolingual, four-, five-, and six-language models. Significant differences are found between monolingual and 4Lang models, and between 5Lang and 6Lang models, but not between 4Lang and 5Lang models.	143

6.4	WER (%) of two Wav2Vec2-based multilingual models and Whisper. The initial Wav2Vec2 feature extractors are large-xlsr-53 for Wav2Vec2 Cross-lingual and large-960h-1v60-self for Wav2Vec2 English.	144
6.5	WER (%) of language-informed models. The numbers indicate the activated conditioning positions depicted in Figure 6.2 and Figure 6.3. Significant differences are found between baseline~Emb12 and baseline~FiLM123.	146
6.6	WER (%) of language-configurable models. On/off in the configuration row means whether language conditioning is activated for inference. No significant difference is found between Emb12 and FiLM123 under the same conditioning settings.	147
6.7	WER (%) of the FiLM123 models trained with phonetic augmentation. Under the Tran. Aug. column, N means no phonetic proxy used, which is the same model as the column FiLM123 in Table 6.5. Y means that the model is trained with phonetic augmentation generated by FiLM123 , followed by the number of languages involved. The Lang column indicates if language conditioning and LMs are used during inference. After introducing phonetic augmentation, significant differences are observed in conditioned overall (<i>Cond. All</i>) WERs only between N and Y 5lang , and between N and Y 6lang	150
6.8	WER (%) of the FiLM123-switch models through phonetic reinterpretation iterations. Under the Tran. Aug. column, N means no phonetic proxy used, which is the same model as the column FiLM123-switch in Table 6.6. Y means that the model is trained with phonetic augmentation generated by FiLM123-switch , followed by the number of languages involved. The Lang column indicates if language conditioning and LMs are used during inference. None means blind inference. After introducing phonetic augmentation, significant differences are observed for blind inference (None) in overall (<i>All</i>) WERs between N and the augmented versions Y *lang	150
7.1	A comparison of Whisper large-v2 performance on mixtures, HTDemucs-separated vocals, and vocals separated by Music.AI SVS model.	157
7.2	The performance of Wav2Vec2 on different acoustic domains during probing.	160

7.3	Dataset Statistics of DALI v2.0, MulJam v2.0, Multi-lang Jamendo, and the new Italian (IT) and Portuguese (PT) test sets. .	162
7.4	Word Error Rates (WER) evaluated on both mixture (Mix) and separated vocals (Voc) conditions. The bold text highlights the best performance within each section, whereas underlined bold text indicates the overall best performance. The L_{CNS} column specifies the consistency loss function used (L1 or L2), and the w column denotes the corresponding weighting parameter.	166
B.1	Objective evaluation on the NHSS set. Sing indicates the original singing voice.	184
C.1	WER/CERs↓ of ALT models trained in their own domains. Sing indicates that the model is trained and tested on singing samples. M1-nodur means that the model is trained and tested on M1-nodur -converted samples. This table displays the change in linguistic information before and after conversion.	186

List of abbreviations

AAE	Average Absolute Error
ASR	Automatic Speech Recognition
AUC	Area Under Curve
ALT	Automatic Lyrics Transcription
CER	Character Error Rate
CNN	Convolutional Neural Network
CRNN	Convolutional Recurrent Neural Network
CTC	Connectionist Temporal Classification
DCT	Discrete Cosine Transform
DE	German
DNN	Deep Neural Network
DTW	Dynamic Time Warping
E2E	End-to-End
EN	English
ES	Spanish
F0	Fundamental Frequency
FFT	Fast Fourier Transform
FR	French
GMM	Gaussian Mixture Model
HMM	Hidden Markov Model
IT	Italian
LM	Language Model
LoRA	Low-Rank Adaptation
LSTM	Long Short-Term Memory
MER	Mixed Error Rate
MFCC	Mel-Frequency Cepstral Coefficients
MIR	Music Information Retrieval
MSE	Mean Squared Error
MOS	Mean Opinion Score
NN	Neural Network
PCO	Percentage of Correct Onsets
PEFT	Parameter-Efficient Fine-Tuning
PT	Portuguese
RNN	Recurrent Neural Network
RU	Russian
S2S	Singing to Speech
SDR	Signal-to-Distortion Ratio
SNR	Signal-to-Noise Ratio
SOTA	State of the Art
SSL	Self-Supervised Learning
STFT	Short-Time Fourier Transform
SVC	Singing Voice Conversion
SVS	Singing Voice Separation
TTS	Text-to-Speech
WER	Word Error Rate
WFST	Weighted Finite-State Transducers

Chapter 1

Introduction

Music information retrieval (MIR) is a research area that encompasses both technical challenges and artistic considerations. Within this field, the singing voice holds a unique position: Unlike many instruments, singing is an activity accessible to nearly everyone, regardless of formal training. Listening to singing often triggers a sense of resonance in the listener, as they imagine how the sounds are produced using their own voice [Taylor, 1908].

Recent progress in deep learning has helped machines understand music more effectively, leading to rapid growth in MIR. Modern techniques have enabled applications in entertainment, education, and creative production. In singing voice research, tasks such as singing synthesis and conversion allow the generation of new performances and creative reuse of existing recordings. Transcription and performance analysis contribute to music education, while lyrics alignment and transcription support karaoke systems and content-based music retrieval.

This thesis investigates automatic lyrics transcription (ALT) using deep learning methods. While ALT has received increasing attention in recent years, it remains a challenging task with several unresolved issues. Building on recent advances in deep learning and music information retrieval (MIR), this work identifies, analyzes, and addresses key challenges in ALT, contributing to the broader field of singing voice research.

This thesis is written with researchers and practitioners in music information retrieval and speech processing in mind, particularly those interested in bridging the gap between the two domains. It may also be of interest to machine learning researchers exploring cross-domain generalization, developers of music education or creative tools, and linguists or phonetics researchers studying singing voice characteristics. By combining insights from both technical and perceptual perspectives, this work aims to advance the study of singing voice modeling and to inform future research on data, model design, and evaluation

for automatic lyrics transcription.

1.1 Motivation and Aim

Automatic lyrics transcription (ALT) and alignment are important tasks in music information retrieval (MIR). By extracting linguistic content from music recordings, these technologies enable a range of applications such as karaoke generation, lyrics-based retrieval, and singing voice analysis. Once considered “not yet feasible” [Fujihara and Goto, 2012], ALT has seen significant progress in recent years, evolving into a task with mature models and even commercial deployment.

Research in ALT and lyrics alignment has its roots in automatic speech recognition (ASR), which is the task of transcribing spoken content from speech audio. Since singing and speech are both produced by the human vocal system, it is natural to begin from the standpoint of ASR and ask: Why do methods developed for speech often fail when applied to singing? What makes singing fundamentally more challenging?

These questions have traditionally been answered by listing the key differences between singing and speech: a wider pitch range, irregular and expressive rhythm, variations in pronunciation, and the presence of background music. However, these factors do not contribute equally to the difficulty of ALT, and some are better addressed by existing techniques than others. To move beyond that, this thesis aims to analyze the relative impact of these factors on model performance. This kind of detailed investigation has only recently become possible with the availability of sufficient training data, a turning point that occurred around the beginning of this research.

At the same time, recent advances have led to high ALT performance for English singing under controlled conditions, particularly when using source-separated vocals. This opens the door to new challenges, most notably multilingual ALT. As in multilingual ASR, we aim to investigate whether established strategies transfer to singing voice, and what novel methods might be needed to account for its unique characteristics. This also motivates the development and refinement of new datasets.

Lastly, this thesis revisits a longstanding and unresolved issue in ALT research: the role of background music. For many years, systematic study was hindered by the lack of tools to isolate the singing voice. With the emergence of open-source source separation models such as Spleeter [Hennequin et al., 2020] and Open-Unmix [Stöter et al., 2019], it became common practice to separate vocals as a preprocessing step for ALT. Yet some studies suggest that the presence of background music may even improve transcription [Gupta et al., 2020,

Gao et al., 2022a, 2023a], while others show degradation, particularly in large pre-trained speech models [Cířka et al., 2024]. These contradictory findings call for a revisit in light of modern tools and datasets, which this thesis covers.

Together, these efforts contribute new insights into the nature of singing voice and the challenges it poses for ALT. The aim is not only to improve ALT performance, but also to deepen our understanding of what makes singing distinct, and how this affects computational modeling for singing voice.

In summary, this thesis aims to achieve the following objectives:

- Analyze both existing and emerging challenges of singing voice for ALT through systematic computational studies.
- Contribute to a deeper understanding of the singing voice by investigating how its characteristics affect model performance in the context of ALT.
- Advance end-to-end ALT systems with recent deep learning methods informed by empirical insights gained from the above analysis.

1.2 Thesis Structure

Chapter 1 (this chapter) provides an introduction to the thesis, including the motivation and aims, structure, and the contributions of the PhD work.

Chapter 2 provides the necessary background for the thesis. It discusses the characteristics of the singing voice and its relationship to related tasks. The chapter also reviews signal processing techniques and machine learning methods relevant to the thesis and surveys related work on ALT and associated tasks.

Chapter 3 presents contributions to dataset creation and benchmarking. It includes the development of the curation of a multilingual dataset for ALT, a manually annotated dataset for lyrics alignment, and benchmark studies on the use of self-supervised learning features, network architectures, and training strategies for ALT.

Chapter 4 proposes incorporating pitch detection as an auxiliary task via multi-task learning to improve acoustic modeling for lyrics alignment. The approach is extended to the multilingual setting. The chapter also evaluates model performance on clean vocal stems and source-separated vocals, analyzing the sensitivity to different source separation tools.

Chapter 5 introduces a novel task, Singing-to-Speech Conversion, to investigate pronunciation and rhythm in singing voice. The goal is to convert singing into speech while preserving phonetic content. This chapter explores insights gained from converted audio and demonstrates the method’s effectiveness for data augmentation for ALT.

Chapter 6 focuses on the multilingual challenges of ALT. It presents several approaches for adapting ALT models to multiple languages and studies methods that allow language configurations. A phonetic reinterpretation method is also proposed to enhance cross-lingual robustness.

Chapter 7 revisits the role of background music in ALT by probing and fine-tuning foundation ASR models. The chapter introduces a consistency loss to guide fine-tuning and provides a contemporary perspective on the ongoing debate around background music’s effect on transcription accuracy.

Chapter 8 summarizes the challenges addressed, insights gained, and contributions made throughout the thesis. It also discusses the limitations of the work and outlines potential directions for future research.

1.3 Contributions and Findings

Below is a list of contributions to knowledge made as part of this thesis:

- Chapter 3: Development and release of the updated *MulJam* dataset for multilingual ALT research.
- Chapter 3: Development of a word-level alignment extension for the MUSDB18 test set.
- Chapter 3: Systematic benchmarking of ALT performance across architectures, training strategies, and self-supervised learning features (as implemented in the MARBLE framework).
- Chapter 4: A multi-task learning approach for lyrics alignment, incorporating pitch detection and structural information to enhance acoustic modeling and alignment algorithm.
- Chapter 4: Comprehensive evaluation of lyrics alignment performance across different source separation tools on the MUSDB18 test set, highlighting model sensitivity to vocal extraction quality.
- Chapter 5: Proposal of a generative flow-based Singing-to-Speech (S2S) conversion system that preserves phonetic content while producing speech-like output.
- Chapter 5: Use of S2S-converted data as a novel form of data augmentation to improve ALT model generalization.
- Chapter 6: In-depth study of multilingual ALT, including adaptation strategies for extending English ALT models to multiple languages.

- Chapter 6: Introduction of phonetic reinterpretation as a data augmentation strategy to enhance cross-lingual performance in ALT.
- Chapter 7: Analysis of Wav2Vec2 representations through probing experiments to assess their cross-domain generalizability from speech to singing.
- Chapter 7: Fine-tuning of Whisper with a consistency loss between vocal and mixture features to improve robustness to background music.

Building on the contributions outlined above, this thesis provides several insights into what makes singing voice more challenging than speech for ALT. Across multiple experiments, pronunciation variability emerges as a core difficulty, interacting with factors such as pitch, rhythm, multilingual content, and background music. Our results highlight the importance of phonetic modeling, multilingual adaptation strategies, and careful handling of musical context, while also demonstrating the benefits of large-scale datasets, data augmentation, and pre-trained models. These findings not only advance practical ALT performance but also deepen our understanding of the computational modeling of singing voice, laying the groundwork for future research directions.

1.4 Associated Publications

Below is a list of publications for which I was the main contributor during my PhD:

- (i) Jiawen Huang, Emmanouil Benetos, and Sebastian Ewert. Improving Lyrics Alignment through Joint Pitch Detection. In *IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP 2022, Virtual and Singapore, 23-27 May 2022*, pages 451–455. IEEE, 2022.
- (ii) Jiawen Huang and Emmanouil Benetos. Towards Building an End-to-End Multilingual Automatic Lyrics Transcription Model. In *32nd European Signal Processing Conference, EUSIPCO 2024, Lyon, France, August 26-30, 2024*, pages 146–150. IEEE, 2024.
- (iii) Jiawen Huang and Emmanouil Benetos. Singing to Speech Conversion with Generative Flow. *EURASIP Journal on Audio, Speech, and Music Processing*, 2025(1):12, 2025.
- (iv) Jiawen Huang, Felipe Sousa, Emir Demirel, Emmanouil Benetos, Igor Gadelha. Enhancing Lyrics Transcription on Music Mixtures with Consistency Loss. In *26th Annual Conference of the International Speech Communication Association, Interspeech 2025, Rotterdam, The Netherlands, August 17-21, 2025*.

- (v) (Under Review) Jiawen Huang and Emmanouil Benetos. Enhancing Multilingual Lyrics Transcription: Leveraging Language Information and Phonetic Augmentation. Under review of Frontiers in Signal Processing - Audio and Acoustic Signal Processing.

The following poster presentations were given at workshops during my PhD:

- (vi) Jiawen Huang and Emmanouil Benetos. Lyrics Alignment for Polyphonic Music. DMRN+15: Digital Music Research Network Workshop 2020.
- (vii) Jiawen Huang and Emmanouil Benetos. Singing to Speech Conversion with Generative Flow. Sheffield Speech Synthesis Workshop 2023.
- (viii) Jiawen Huang and Emmanouil Benetos. Multilingual Integration in Lyrics Transcription: Data, Language Conditioning, and Transliteration Augmentation. UK and Ireland Speech Workshop 2024.
- (ix) Jiawen Huang and Emmanouil Benetos. Evaluating Lyrics Alignment under Source Separated Conditions. Late-Breaking Demo Session, International Society for Music Information Retrieval Conference 2025.

Below is a list of the works I contributed to through collaboration:

- (x) Ruibin Yuan, Yinghao Ma, Yizhi Li, Ge Zhang, Xingran Chen, Hanzhi Yin, Le Zhuo, Yiqi Liu, Jiawen Huang, Zeyue Tian, Binyue Deng, Ningzhi Wang, Chenghua Lin, Emmanouil Benetos, Anton Ragni, Norbert Gyenge, Roger B. Dannenberg, Wenhua Chen, Gus Xia, Wei Xue, Si Liu, Shi Wang, Ruibo Liu, Yike Guo, and Jie Fu. MARBLE: Music Audio Representation Benchmark for Universal Evaluation. In Advances in Neural Information Processing Systems 36: Annual Conference on Neural Information Processing Systems 2023, NeurIPS 2023, New Orleans, LA, USA, December 10-16, 2023.
- (xi) (Under Review) Yinghao Ma, Anders Øland, Anton Ragni, Bleiz Macsen Del Sette, Charalampos Saitis, Chris Donahue, Chenghua Lin, Christos Plachouras, Emmanouil Benetos, Elio Quinton, Elona Shatri, Fabio Morreale, Ge Zhang, György Fazekas, Gus Xia, Huan Zhang, Ilaria Manco, Jiawen Huang, Julien Guinot, Liwei Lin, Luca Marinelli, Max W. Y. Lam, Megha Sharma, Qiuqiang Kong, Roger B. Dannenberg, Ruibin Yuan, Shangda Wu, Shih-Lun Wu, Shuqi Dai, Shun Lei, Shiyin Kang, Simon Dixon, Wenhua Chen, Wenhao Huang, Xingjian Du, Xingwei Qu, Xu Tan, Yizhi Li, Zeyue Tian, Zhiyong Wu, Zhizheng Wu, Ziyang Ma, and Ziyu Wang. Foundation Models for Music: A Survey. Under review of ACM Computing Surveys.

For (i)-(iii) and (v)-(ix), I was responsible for conceptualizing the study, designing and implementing the methodology, conducting all experiments, analyzing and interpreting the results, and drafting the manuscript under the supervision of Dr. Emmanouil Benetos and Dr. Sebastian Ewert. For (iv), I also conceptualized the study, designed and implemented the methodology, conducted all experiments, analyzed and interpreted the results, and drafted the manuscript. The second author, Felip Sousa, contributed to the implementation and evaluation, and drafted the corresponding sections of the manuscript. This work was conducted with guidance and feedback from Dr. Emir Demirel, Dr. Emmanouil Benetos, and Dr. Igor Gadelha.

For (x) and (xi), they are relevant to my PhD research and were carried out as part of collaborative projects. In (x), I contributed the ALT task to a shared benchmark. In (xi), I contributed a literature review on singing voice research.

1.5 Visits and Placements

In February 2022, I took part in a group visit at the Institute for Research and Coordination in Acoustics/Music (IRCAM - Paris, France), organized as part of the UKRI Centre for Doctoral Training in AI and Music (AIM CDT). During this visit, I contributed to collaborative work that led to a paper [Bindi et al., 2023].

From October 3, 2022, I was selected for the Alan Turing Institute’s Enrichment Scheme. This scheme supported a 9-month period of independent research development.

During my PhD, I also completed two industry internships: one at Sony Group Corporation and another at Moises. In both placements, I worked on projects related to singing voice manipulation and automatic lyrics transcription. The work carried out during the Moises internship is described in Chapter 7.

Chapter 2

Background

This chapter provides the technical and conceptual foundation for the work presented in this thesis. We begin by examining the unique characteristics of singing voice, motivating the need for specialized approaches (Section 2.1). We then review the audio representations commonly used in deep learning, highlighting their suitability for singing voice tasks (Section 2.2).

Next, we explore the machine learning models, architectures, and learning paradigms relevant to our research, discussing their strengths, limitations, and roles in our system design (Section 2.3 and Section 2.4). We also introduce commonly used data augmentation techniques that address data scarcity issues in singing voice research (Section 2.5).

Finally, we look into the related research in automatic speech recognition (ASR) (Section 2.6), lyrics alignment (Section 2.7), automatic lyrics transcription (ALT) (Section 2.8), and singing voice conversion (SVC) (Section 2.9), which provide important context and methodologies that inform our work.

While this thesis addresses multiple tasks, this chapter does not aim to provide a comprehensive survey of all related work. Instead, it focuses on aspects most relevant to the central theme of ALT. It selectively highlights concepts and related work that are directly connected to the challenges addressed in subsequent chapters, particularly those related to modeling and analyzing the singing voice. Where appropriate, brief overviews of related developments and recent advancements are provided. Together, these sections establish the background necessary to understand the core concepts and techniques addressed in later chapters. They also help position our contributions within the broader landscape of singing voice research.

Part I: Technical Foundations

2.1 Introduction to Singing Voice

Singing voice is a form of human vocalization. Human voice is produced through a complex interaction of the respiratory system, the vocal folds, and the vocal tract. The lungs provide air pressure that drives the vocal folds to vibrate, creating a sound source [Lindblom and Sundberg, 2014]. This is then shaped by vocal tract resonances, where the throat, mouth, and nasal cavity act as a filter to modify the sound.

Although singing and speech share the same basic voicing mechanisms, they differ in both physiological and acoustic aspects. These distinctions motivate the need to treat singing voice as a distinct domain in computational research.

2.1.1 Speech and Singing Voice

The difference between speech and singing has been discussed extensively in the literature on vocal physiology and voice perception [Sundberg, 2013, Lindblom and Sundberg, 2014]. Researchers have explored these differences across production mechanisms, perceptual attributes, and acoustic features. Understanding these distinctions is essential for tasks such as automatic lyrics transcription (ALT), singing synthesis, and singing voice beautification.

Physiological Differences

Lindblom and Sundberg [2014] provides a comprehensive overview of differences in respiration, phonation, and articulation between speech and singing. Singing requires precise coordination across the respiratory system, glottal source, and articulatory gestures. Singers must actively regulate subglottal pressure and control vocal fold adduction, particularly when aiming for pitch accuracy, dynamic range, and timbral consistency.

From a performance perspective, singing demands additional layers of control compared to speech. This includes stabilization of pitch, precise timing, clarity of articulation under musical constraints, and expressivity through vibrato, dynamics, and intonation. Advanced singing techniques also require mastery of different vocal registers (e.g., chest, head, falsetto) and stylistic elements.

Acoustic and Perceptual Differences

In the field of singing voice analysis, researchers often characterize the differences between speech and singing from an acoustic and perceptual perspective [Fujihara and Goto, 2012, Mesaros, 2013, Kruspe, 2024, O’Connor, 2024]. These are commonly summarized as:

- **Pitch:** Singing typically spans a wider pitch range than speech, with sustained and stable tones. Speech pitch varies due to intonation and emotion.
- **Rhythm:** Singing follows a more regular rhythmic structure aligned with the music. Phoneme durations are often lengthened or shortened to fit the musical notes.
- **Pronunciation:** Singers adapt vowel and consonant articulation based on tempo, pitch, and other musical context. Vowels may be lengthened for musical phrasing, and articulation may be more exaggerated for clarity.
- **Dynamics:** Singing involves a wider dynamic range than speech, often with deliberate loudness variation for expressiveness.
- **Techniques:** Singing employs specialized vocal modes and techniques, such as vibrato, falsetto, and belting, which are uncommon in speech.
- **Musical Accompaniment:** Singing is often accompanied by music, which harmonically and temporally correlates with singing voice.

These features, except musical accompaniment, overlap with the domain of speech prosody. However, prosody in speech interacts with linguistic structure and emotion, whereas in singing, it aligns with musical context. The distinctions support the hypothesis that singing is not merely “speech with pitch and rhythm”, but a distinct vocal behavior.

Singing Expertise

However, it does not mean speech and singing are far apart. In fact, the definition of singing is broad and ambiguous. The above differences, are observed more in professional singers than non-professional singers [Brown et al., 1988, 2000]. Yet singing is not limited to professional performers or the entertainment industry. It also occurs in everyday creative and social contexts. In fact, we are exposed to a lot of singing voices by amateur singers and those with limited training on social media. The amount of training could affect singing voice production and perception [Hunter et al., 2006, Brown et al., 1988, 2000]. Vocal training enhances control over phonation and rhythm, leading to phenomena like vibrato and the singer’s formant, which are key markers of trained singing voices [Brown et al., 2000]. Furthermore, training has cognitive benefits: it improves sensitivity to timing and pitch, and enhances the ability to imitate unfamiliar sounds such as foreign languages [Christiner and Reiterer, 2013].

Implications for Singing Voice Research

In computational research, the distinction between trained and untrained singers is often ignored. Models for singing voice analysis and synthesis are typically trained on datasets containing various levels of vocal expertise.

Recent work on singing voice beautification explicitly targets amateur singers [Liu et al., 2022b], aiming to enhance voice quality by approximating features found in professional singing. However, most tasks do not account for singer expertise as a modeling factor. Still, it is reasonable to expect that performance on singing voice analysis tasks may vary depending on the singer’s level of expertise.

To summarize, singing differs from speech across physiological, acoustic, and expressive dimensions. These differences become more pronounced with formal training. The literature implies a range of perspectives: some treat singing as distinct from speech, while we view it as existing along a continuum shaped by vocal training and usage context.

In this thesis, we adopt a broad definition of singing that includes voices from various levels of expertise. We do not discriminate based on expertise in our models, but acknowledge its potential impact on task difficulty and system performance.

2.1.2 Related Tasks and Research Area

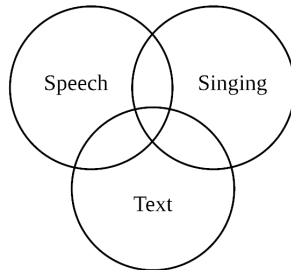


Figure 2.1: Intersection of text, speech, and singing domains.

We now turn to computational tasks that aim to model, analyze, or generate voice in relation to text, speech, and singing. Figure 2.1 illustrates how various tasks lie at the intersection of these domains. Tasks such as ASR and TTS occupy the speech–text intersection, while ALT, lyrics alignment, and singing synthesis belong to the singing–text domain. Singing–speech intersection includes voice conversion and cross-domain modeling.

While this thesis focuses primarily on ALT (Section 2.8), we also discuss lyrics alignment (Section 2.7) and singing voice conversion (Section 2.9). As these tasks are strongly influenced by ASR techniques, we introduce the basics of ASR in Section 2.6. Although we do not directly work on TTS or singing synthesis, their architectures and training strategies have influenced the models explored in this thesis.

2.2 Audio Representations for Deep Learning

This section provides a brief overview of the audio representations used in our models. We categorize them into three types: waveform, spectral representations, and learned neural representations. Each representation has distinct characteristics and must be carefully paired with an appropriate model architecture and training strategy to be effective.

In later chapters, we primarily use Mel-spectrograms (a time-frequency representation) and learned neural representations as input features for our models. Mel-spectrograms offer interpretability and capture rich acoustic information, while learned representations are particularly effective when labeled data is scarce. Although waveform and other representations are less common in our experiments, we discuss them here for completeness.

2.2.1 Waveform

The waveform is the raw audio signal in the time domain. It serves as the most fundamental representation of audio, capturing all the information contained in the sound wave. Waveforms are typically represented as sequences of amplitude values sampled at a fixed rate. For singing voice, where high-frequency content plays a significant role, sampling rates of 16 kHz or higher are commonly used.

Depending on the architecture, resampling may be necessary to satisfy constraints related to the receptive field or computational efficiency. Waveform normalization is often applied to ensure consistent amplitude levels across recordings.

Waveform representations are well-suited to tasks that require preservation of fine-grained signal detail, such as music source separation [Défossez et al., 2019, Lluís et al., 2019]. They are also used in self-supervised representation learning [Schneider et al., 2019, Oord et al., 2018], where retaining the full signal content is important for learning rich features. In these models, it is common to begin with a 1D convolutional layer, followed by pooling or downsampling operations.

2.2.2 Time-Frequency Representations

Time-frequency representations [Müller, 2015a] are typically computed using the short-time Fourier transform (STFT). They offer a compact view of the frequency content over time and allow the signal to be treated similarly to an image. The STFT produces a complex spectrogram that retains both magnitude and phase information. However, in most applications, only the magnitude or power spectrogram is used. Discarding phase information results in a more compact representation but may limit its utility for certain tasks where phase is important.

The computation of a spectrogram involves setting parameters such as window size, hop size, and window function. Among various types of spectrograms, the Mel-spectrogram is common because of its alignment with human auditory perception. It is computed by applying a Mel filter bank to the STFT magnitude spectrum, often followed by logarithmic scaling. Mean-variance normalization is sometimes applied to stabilize model training.

Spectral representations are widely used in tasks where frequency structure is important, including pitch tracking, voice conversion, lyrics transcription, and speech recognition. Their image-like structure makes them suitable for 2D convolutional neural networks (CNNs). Popular architectures in computer vision, such as ResNet [He et al., 2016] are frequently applied to spectrogram inputs. A prominent example is PANNs [Kong et al., 2020], a CNN model for general audio classification, which achieves strong performance across a variety of audio classification tasks.

2.2.3 Learned Neural Representations

Learned neural representations refer to feature embeddings produced by deep neural networks, often pre-trained using large-scale unlabeled audio data. Recent self-supervised learning (SSL) models such as Wav2Vec2 [Baevski et al., 2020], HuBERT [Hsu et al., 2021a], and MERT [Li et al., 2023b] learn representations using objectives such as contrastive learning or masked prediction. These embeddings are rich in information and can be transferred to a variety of downstream tasks. An overview of SSL is provided in Section 2.4.3.

Before the emergence of SSL, learned representations in music information retrieval (MIR) were typically obtained through supervised training on large labeled datasets, with features extracted from the final network layers [Choi et al., 2017]. Early forms of contrastive learning, such as Siamese networks [Bromley et al., 1993] and models trained with triplet loss [Schroff et al., 2015] were also explored in MIR applications [Kim et al., 2021c, Park et al., 2018]. These methods learn embeddings by comparing pairs or triplets of samples, aiming to

bring semantically similar samples closer and dissimilar ones farther apart in the latent space. While still used in some cases, these representations tend to generalize less effectively across tasks compared to self-supervised alternatives.

Learned representations are typically used without further processing, although they may be reshaped or pooled to match the input requirements of downstream models.

Learned neural representations are advantageous when labeled data is limited. They give strong performance on a wide range of tasks, including classification, transcription, and voice conversion. Depending on the application, the representations may be fixed, partially fine-tuned, or fully fine-tuned during training.

2.2.4 Other Representations

Several additional feature types are used in music signal processing [Müller, 2015b], though they are less suitable for singing voice analysis. For example, chroma features represent the energy distribution across 12 pitch classes and are useful for chord and key detection. However, singing voices often deviate from discrete pitch classes, limiting the effectiveness of chroma features in such contexts. Mel-frequency cepstral coefficients (MFCCs) apply a discrete cosine transform (DCT) to Mel-spectrograms to compress the information further. They have been used in early singing voice analysis research, including singer identification [Mesaros and Astola, 2005, Patil et al., 2012], phoneme recognition [Hansen, 2012, Mesaros and Virtanen, 2010], keyword spotting [Kruspe, 2014, 2015, 2016], and lyrics transcription [Tsai et al., 2018, Dabike and Barker, 2019]. While MFCCs are widely used in speech recognition, they discard much of the fine-grained spectral detail that is often crucial for modeling harmonic structure and timbre in music. As a result, they are less suited for singing-related tasks than richer features such as spectrograms and waveforms, especially now that modern computational resources and efficient models are available to process these higher-dimensional features.

2.3 Machine Learning Models and Architectures

This section introduces the machine learning architectures commonly used in automatic lyrics transcription (ALT) and provides essential background for the models developed in subsequent chapters. These architectural components form the foundation for understanding the design choices throughout this thesis.

To clarify how these architectures are used throughout the thesis, we briefly indicate their applications in later chapters. Among these architectures, Wave-U-Net [Stoller et al., 2018] is introduced as the basis for the first end-to-end lyrics alignment and transcription model. The Transformer [Vaswani et al., 2017], together with the hybrid CTC/attention architecture, is included due to its wide adoption in ASR and ALT systems. We employ Transformers as the core ALT models in Chapters 5 and 6. Wav2Vec2 [Baeovski et al., 2020], which also uses a Transformer backbone, was the first widely successful self-supervised learning framework for ASR (see Section 2.4.3), and its learned features have been applied in state-of-the-art ALT methods. We also experiment with the features in Chapter 7. Finally, flow-based generative model [Kingma and Dhariwal, 2018] is introduced for their use in Chapter 5 for a voice conversion task, and diffusion models [Ho et al., 2020, Yang et al., 2024] are included as they represent the latest advances in singing voice conversion.

2.3.1 Wave-U-Net

Wave-U-Net [Stoller et al., 2018] is an adaptation of the U-Net architecture [Ronneberger et al., 2015] designed for processing raw audio waveforms. It is illustrated in Figure 2.2. Like the original U-Net, it follows a symmetric encoder-decoder structure with skip connections that link corresponding layers in the encoder and decoder. The encoder progressively downsamples the input signal to extract high-level features, while the decoder upsamples these features to reconstruct the output at the original resolution. Wave-U-Net performs all operations in the time domain. This design choice avoids the need for a short-time Fourier transform (STFT) and enables direct preservation of phase information, which is often lost or distorted in time-frequency domain approaches.

Besides, it carefully designs a method to avoid aliasing artifacts due to up-sampling due to zero-padding. It replaces the transposed strided convolutions for upsampling with learned linear interpolation, implemented using 1D convolution.

Wave-U-Net was originally proposed for music source separation [Stoller et al., 2018], but its architecture has since been applied to other MIR tasks, including lyrics alignment and transcription [Stoller et al., 2019a], speech enhancement [Macartney and Weyde, 2018], and beat tracking [Cheng and Goto, 2023].

Wave-U-Net is effective for tasks such as lyrics alignment and transcription. Its time-domain processing avoids errors from imperfect phase reconstruction, and its skip connections preserve fine-grained temporal detail. However, it is computationally heavy when modeling raw waveforms. Its limited receptive

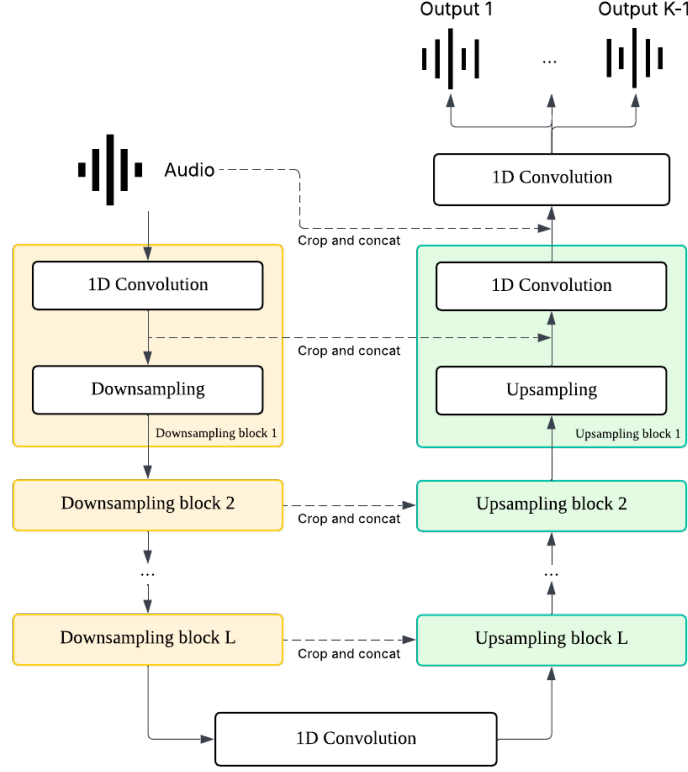


Figure 2.2: Architecture of the Wave-U-Net model with L layers for K sources, adapted from Figure 1 in Stoller et al. [2018]. The prediction for the K -th source is computed as the difference between the input mixture and the sum of the other $K - 1$ predicted sources.

field also reduces its ability to capture long-range dependencies.

2.3.2 Transformer

The Transformer architecture, introduced by Vaswani et al. [2017], is a neural network design based on attention mechanisms. It was proposed as an alternative to recurrent neural networks (RNNs), which process inputs sequentially and therefore struggle with efficient parallelization. This limitation becomes particularly significant for long sequences, such as those found in audio processing. The Transformer uses self-attention mechanisms, enabling parallel computation and better modeling of long-term dependencies.

An attention function maps a query and a set of key-value pairs to an output, where the output is a weighted sum of the values. The weights are determined by a similarity function between the query and the keys. This operation is

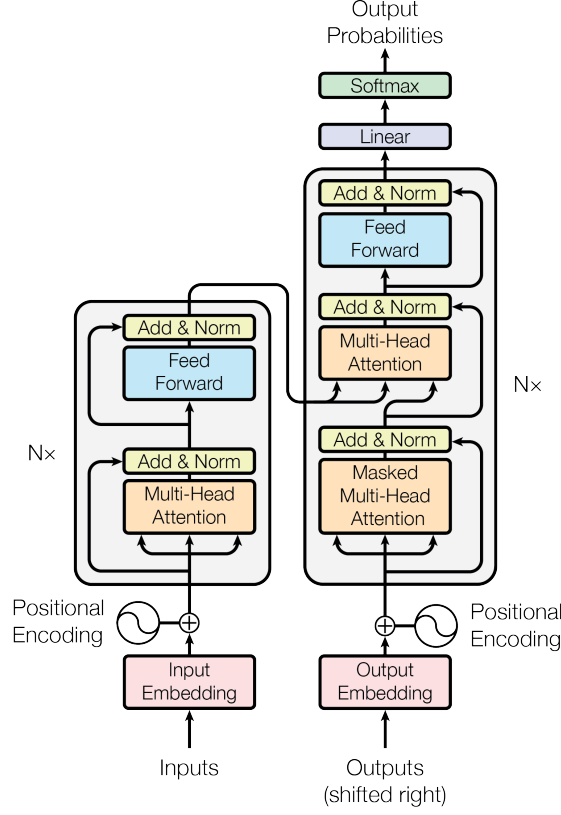


Figure 2.3: The Transformer architecture (Vaswani et al. [2017], Figure 1). Used with permission from Google.

defined as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \quad (2.1)$$

where Q is the query, K is the keys, V is the values, and d_k is the dimension of the keys. The softmax function ensures that the weights sum to 1. When the query, keys, and values are derived from the same input sequence, the operation is known as self-attention. If the query comes from one sequence and the key-value pairs from another, it is called cross-attention.

The Transformer model is composed of an encoder-decoder architecture, illustrated in Figure 2.3. Both the encoder and decoder stacks consist of multiple layers of self-attention and feedforward networks. In the decoder, cross-attention layers allow the model to attend to the encoder outputs. This design enables effective sequence-to-sequence modeling.

The Transformer architecture has been widely adopted in MIR tasks, including piano transcription [Hawthorne et al., 2021], music tagging [Lu et al., 2021], and symbolic music generation [Huang et al., 2019]. It has also been applied

to lyrics transcription [Gao, 2022], where its ability to model long-term dependencies is especially beneficial. Recent state-of-the-art models such as Whisper [Radford et al., 2023] are also built on the Transformer architecture. Whisper has demonstrated strong performance in both automatic speech recognition (ASR) and ALT. For further discussion of Whisper in the context of ALT, see Section 2.8.3.

Transformers are effective in ALT because their self-attention mechanism models long-range dependencies better than RNNs. This is important for capturing the complex structure of singing voice, which often spans extended context. However, Transformers are computationally demanding. They also cannot process arbitrarily long inputs, so audio must be chunked, which can disrupt temporal continuity.

2.3.3 Hybrid CTC/Attention

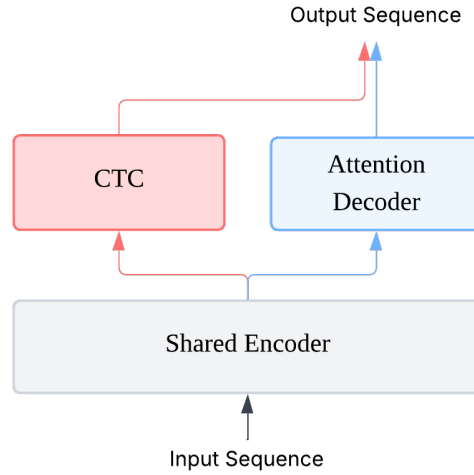


Figure 2.4: Hybrid CTC/attention-based end-to-end architecture, simplified from Figure 1 in Watanabe et al. [2017].

Connectionist Temporal Classification (CTC) [Graves et al., 2006] is a loss function designed for sequence modeling tasks where the alignment between input and output sequences is unknown. It computes the total probability of the correct output sequence by summing over all valid alignment paths that could produce that sequence, allowing the model to learn alignment implicitly. The model learns to prefer correct alignments without frame-level annotation. CTC loss has become a standard loss function in ASR and other sequence-to-sequence modeling tasks.

Hybrid CTC/Attention architecture [Watanabe et al., 2017] is a common practice in ASR and an extension of end-to-end ASR systems that combines the strengths of CTC and attention-based training. It is shown in Figure 2.4. End-to-end systems typically use either CTC loss or attention-based sequence-to-sequence (seq2seq) loss. The CTC loss models the alignment between input and output sequences, while the attention-based loss models the sequence generation process. Attention-based models offer flexible alignments, but this flexibility can sometimes result in unstable training or non-monotonic alignments. In contrast, CTC enforces a monotonic alignment constraint between input and output.

By jointly training with both CTC and attention-based losses, the hybrid model benefits from the alignment constraints of CTC and the flexibility of the attention mechanism, resulting in improved performance and more stable training. This approach is not paired with any specific architecture but is commonly applied in transformers and RNNs.

Hybrid CTC/attention training is effective because it combines the monotonic alignment of CTC with the flexible sequence modeling of attention. This often leads to more stable and accurate transcriptions. However, the balance between the two losses requires careful tuning.

2.3.4 Wav2Vec2

Wav2Vec2 [Baevski et al., 2020] is a self-supervised learning framework for speech representation learning. Raw waveforms are fed into a 1D convolutional block to obtain latent speech representations. During pre-training, the model uses product quantization [Jégou et al., 2011] on the latent features to create discrete targets for contrastive learning. The latent representations are passed into a Transformer encoder, which is trained to predict the quantized representations. The model is optimized using a contrastive loss function that discriminates between the true quantized representation and negative samples. The negative samples are randomly drawn from the same utterance.

The training strategy is briefly discussed in Section 2.4.3. This framework has produced multiple variants [Conneau et al., 2021, Babu et al., 2022, Hsu et al., 2021b]. The resulting models have had a significant impact on speech and singing tasks.

A music version of Wav2Vec2 has been trained and evaluated for instrument classification and pitch classification [Ragano et al., 2023]. A speech-pre-trained Wav2Vec2 has been adapted for lyrics transcription [Ou et al., 2022]. Yamamoto [2023] investigates the effectiveness of a pre-trained Wav2Vec2 checkpoint on three singing voice tasks. Recently, a new SSL feature called `voc2vec` was trained under the Wav2Vec2 framework on non-verbal audio data [Koudounas

et al., 2025].

Wav2Vec2 is effective for ALT because its self-supervised pretraining captures phonetic information in a label-efficient way. This reduces the need for large annotated singing corpora. However, since it was trained mainly on speech, its features may not capture pitch variation and prolonged vowels well. Fine-tuning on singing data is usually necessary.

2.3.5 Flow-based Generative Models

Flow-based generative models [Kingma and Dhariwal, 2018] are a class of deep generative models that enable exact likelihood computation and efficient sampling by transforming simple probability distributions through invertible mappings. It transforms a simple tractable probability distribution, such as a standard normal distribution, to the data distribution, through a series of invertible transformations. The resulting transformation can be applied in both directions. As a consequence, it is possible to generate new samples from the data distribution and compute the exact likelihood of existing data.

The generative process can be defined as:

$$z \sim \pi(z) \quad (2.2)$$

$$x = f(z) \quad (2.3)$$

where z is a latent variable, x is a data sample, and π is a tractable density, such as a multivariate Gaussian: $\pi(z) = \mathcal{N}(z; \mu, \sigma^2)$. The function f consists of L invertible functions:

$$f = f_1 \circ f_2 \circ \dots \circ f_L \quad (2.4)$$

The exact log-likelihood of data x can be calculated as follows:

$$\begin{aligned} \log p(x) &= \log \pi(z) + \log \left| \det \frac{\partial f^{-1}(x)}{\partial x} \right| \\ &= \log \pi(z) + \sum_{i=1}^L \log \left| \det \mathcal{J}(f_i^{-1}(x)) \right| \end{aligned} \quad (2.5)$$

where $\det \mathcal{J}(f_i^{-1}(x))$ is the Jacobian determinant of the function f_i^{-1} . Flow-based models choose functions f_i whose Jacobian is a triangular matrix, such that the log-determinant calculation is straightforward and efficient.

The advantage of flow-based models is that they directly model the log-likelihood of data. The latent variables are inferred without approximation, superior to Variational Autoencoders (VAEs) [Kingma and Welling, 2019] and Generative Adversarial Networks (GANs) [Goodfellow et al., 2014]. Flow-based

models are also invertible, allowing for efficient sampling and exact likelihood computation.

The advantage of flow-based models is their direct modeling of the log-likelihood of data, with latent variables inferred without approximation. Compared to VAEs (Variational Autoencoders) [Kingma and Welling, 2019] and GANs (Generative Adversarial Networks) [Goodfellow et al., 2014], flow-based models offer exact likelihood computation and invertibility, enabling efficient sampling and likelihood evaluation.

In audio generation tasks, flow-based generative models have attracted considerable interest due to their exact likelihood computation, invertibility, and efficiency [Hoogetboom et al., 2019]. These models have been successfully applied to TTS [Kim et al., 2020, Valle et al., 2021], singing synthesis [Zhang et al., 2022c, Choi et al., 2023], and voice conversion [Choi et al., 2023, Proszewska et al., 2022].

Flow-based models are effective for singing voice synthesis and conversion. They can model complex distributions, allow exact likelihood computation, and generate high-fidelity audio. However, they are often less expressive than diffusion models. They may also struggle to produce natural prosody without extensive conditioning.

2.3.6 Diffusion Models

Diffusion models [Ho et al., 2020, Yang et al., 2024] are a class of generative models that produce data by learning to reverse a gradual noise addition process, enabling high-quality and flexible generation. They operate by gradually adding noise to data in a forward diffusion process until the data becomes nearly Gaussian noise. The model then learns the reverse process, which iteratively denoises this noisy sample to recover data from the noise-like distribution. This reverse process is modeled as a Markov chain, where each denoising step depends on the previous one. Conditioning information can be incorporated to control the generation.

Diffusion models have been successfully applied in MIR tasks such as music generation [Huang et al., 2023b], singing voice synthesis [Liu et al., 2022a], and singing voice conversion [Liu et al., 2021b].

Diffusion models are effective for singing voice synthesis and conversion. Their iterative denoising process produces high-quality, natural-sounding audio. They also support flexible conditioning signals. However, they are computationally expensive. Inference is slow because of the many sampling steps, and training requires large amounts of data and compute.

2.4 Machine Learning Paradigms

In this section, we introduce machine learning paradigms relevant to the tasks addressed in this thesis, including supervised learning, semi-supervised learning, self-supervised learning, unsupervised learning, and multi-task learning. We also discuss strategies for adapting pre-trained models to new tasks through probing and fine-tuning. These paradigms form the methodological foundation for later chapters.

To clarify how the machine learning paradigms introduced in this section are applied throughout the thesis, we briefly indicate their usage in later chapters. Most ALT systems in this thesis rely on supervised learning, with semi-supervised approaches explored in Chapter 6. Building on the Wav2Vec2 self-supervised learning (SSL) framework introduced earlier, we present SSL as a general paradigm and summarize widely used methods for ASR and ALT. Multi-task learning is applied in Chapter 4 for joint lyrics alignment and pitch detection, and in Chapter 5 for simultaneous voice conversion and phoneme recognition. We also review strategies for adapting pre-trained models, which is explored in Chapters 3 and 7.

2.4.1 Supervised Learning

In supervised learning [Bishop and Nasrabadi, 2006], the model is trained on a labeled dataset, and optimized to minimize an objective function that measures the difference between the predicted output and the label, such as cross-entropy loss for classification or mean squared error for regression.

When a task is well-defined and sufficient labeled data is available, supervised learning is the preferred approach. In music information retrieval, where most cases suffer from data scarcity, supervised learning is often a first step to establish a baseline. Other approaches, including semi-supervised or unsupervised learning methods, can be used to improve the performance or leverage additional data.

2.4.2 Semi-Supervised Learning

In contrast to supervised learning, semi-supervised learning [Van Engelen and Hoos, 2020] leverages both labeled and unlabeled data. Typically, a model is trained with labeled data and then used to assign pseudo-labels to the unlabeled data. The model is then trained again on the combined dataset. This process can be iterated several times, which is often referred to as self-training [Scudder, 1965].

Teacher-student paradigms [Hinton et al., 2015, Tarvainen and Valpola,

2017] generalize this by decoupling the label generator and learner. Using separate models allows a more stable learning target for the student and can prevent error accumulation through iterations. Self-training can be seen as a special case where the teacher and the student are the same model.

Semi-supervised learning is useful when labeled data is limited but unlabeled data is accessible. However, the performance strongly depends on the quality of the pseudo-labels. Sometimes the pseudo-labels can be filtered based on confidence to ensure the quality. Semi-supervised approaches have not yet been widely explored in ALT. A notable exception is the study by Gao et al. [2023b], which employed self-training and reported clear effectiveness for this task.

2.4.3 Self-Supervised and Unsupervised Learning

Unsupervised learning [Hastie et al., 2001] broadly refers to methods that learn patterns from unlabeled data, such as clustering, dimensionality reduction, or representation learning (e.g., k-means, principal component analysis, autoencoders).

Self-supervised learning (SSL) [Liu et al., 2023] is a subcategory of unsupervised learning, where the model learns from automatically generated supervisory signals derived from the data itself. This is typically done by solving a pretext task, such as predicting missing parts of the input.

SSL takes advantage of the large amount of unlabeled data available in the wild. It learns from the data itself, allowing the model to capture the underlying structure that could not be captured by supervised learning. These learned features can be used for downstream tasks by probing or fine-tuning, or as fixed feature extractors. Approaches to adapt learned representations are discussed in Section 2.4.5.

Below we discuss three representative SSL methods in the context of this thesis:

Wav2Vec2

Wav2Vec2 [Baevski et al., 2020] (also discussed in Section 2.3.4) is trained using a contrastive loss. During training, raw audio input is partly masked, and the model is trained to identify the correct latent representation of the masked frames among distractors sampled from other time steps. The learned features are context-aware.

Wav2Vec2 is effective especially for ASR tasks. There are variants for multilingual pre-training [Conneau et al., 2021], robust for domain shift [Hsu et al., 2021b], and fine-tuned versions on different languages ¹.

¹<https://huggingface.co/collections/facebook/xlsr-651e8a5bb947065cccb62c6c>

HuBERT

Unlike Wav2Vec2’s contrastive loss, HuBERT [Hsu et al., 2021a] predicts discrete cluster assignments of masked audio segments, where the clusters are obtained through pre-trained k-means clustering of acoustic features. The initial cluster labels are generated from a pre-trained model, and then HuBERT is trained to predict these labels. The cluster labels are then updated through iterations. The learned representations improve as the clustering and training process is repeated, the pseudo-labels are refined and the model learns rich audio representations.

MERT

MERT [Li et al., 2023b] extends HuBERT in two ways. First, EnCodec [Défossez et al., 2023], a high fidelity neural audio compression model, is used to replace the k-means clustering step for more flexible mapping. The learned codebook replaces the clusters in HuBERT to assign discrete labels. Second, the model has two branches for mask prediction, one for acoustic features and one for Constant-Q transform (CQT) features. The added CQT branch emphasizes pitch information, making the learned feature more suitable for music tasks.

2.4.4 Multi-task Learning

Multi-task learning [Caruana, 1997] is a paradigm where multiple tasks are trained simultaneously for the same model. This facilitates knowledge sharing and can improve generalization when tasks are related.

A common implementation involves sharing the first few layers across tasks, while using separate branches for task-specific outputs. The model is optimized using a weighted sum of task-specific loss functions. Careful tuning of these weights is necessary to balance contributions from each task.

2.4.5 Adapting Pre-trained Models

Foundation models are the models trained for general purposes, as opposed to specific tasks. They capture rich representations of data, and can be adapted to various tasks. Some of them are SSL features, while others are pre-trained on large datasets with supervised learning. Foundation models have achieved great success in MIR [Ma et al., 2024]. To adapt these models to specific tasks, they either serve as feature extractors, or get fine-tuned. We briefly introduce probing and fine-tuning methods, while other methods such as knowledge distillation [Hinton et al., 2015] are not discussed, as they primarily aim at improving model efficiency, which is not our focus.

Probing

Probing is a technique to evaluate the quality of pre-trained representations on downstream tasks, by training a simple, usually linear classifier on frozen features extracted by the model (called the front end). The classifier (the back end) is trained on small labeled datasets to verify if the representations encode relevant information. This process does not modify the pre-trained model parameters.

Standard Fine-Tuning

In contrast, fine-tuning updates both the pre-trained model (front end) and task-specific head (back end) jointly. Fine-tuning transfers the pre-trained knowledge to a target task and domain. To prevent catastrophic forgetting [Kirkpatrick et al., 2017], the front end is often updated with a smaller learning rate than the back end.

Parameter Efficient Fine-Tuning

Full fine-tuning can be computationally expensive and prone to overfitting when labeled data is limited. Parameter-efficient fine-tuning (PEFT) techniques address this by updating only a small set of parameters while keeping most of the model frozen. PEFT methods, such as Low-Rank Adaptation (LoRA) [Hu et al., 2022], add low-rank matrices to be updated. This is effective because most knowledge in large models is stored along a few principal directions, so low-rank updates allow efficient specialization without retraining all parameters. A limitation is that LoRA may fail to capture task-specific information if the adaptation requires changes beyond the low-rank subspace.

Other commonly used PEFT methods include prompt/prefix tuning [Liu et al., 2022c, Li and Liang, 2021], and adapters [Houlsby et al., 2019].

PEFT reduces computational cost and memory requirements, making it practical to adapt large foundation models. It has shown promising results in multilingual ASR and emotion recognition [Prasad et al., 2024, Gao et al., 2024].

2.5 Data Augmentation for Singing Voice Research

Data availability plays a central role in lyrics alignment and transcription research. While a detailed survey of existing datasets is provided in Chapter 3.2, this section focus on data augmentation strategies that address the scarcity and imbalance of singing voice corpora. Some methods are adopted from general audio processing research [Wen et al., 2021], while others are specifically tailored

for singing voice applications. The primary objective of data augmentation is to enhance model robustness and generalization by introducing controlled variations into the training data, which is a critical strategy in low-resource settings.

2.5.1 Basic Augmentation Techniques

Basic augmentation methods apply signal processing techniques to audio waveforms or spectrograms. These techniques are widely used in both speech and singing voice tasks and include:

- **Time stretching:** Modifying the speed of the audio without affecting pitch.
- **Pitch shifting:** Changing the pitch of the audio while maintaining duration.
- **Noise injection:** Adding background noise to the audio to simulate real-world conditions.
- **Mixing:** Combining one or multiple audio signals and applying gain, EQ, compressor, pan, and other effects to the mixed signal.
- **Time and frequency masking:** Randomly masking parts of the audio in the time and frequency domains to create variations and prevent overfitting.

These techniques are easily implemented using tools such as librosa [McFee et al., 2015], SpecAugment [Park et al., 2019], sox [Norskog and Bagwell, 2015], torchaudio [Hwang et al., 2023], and pymixconsole [Steinmetz et al., 2021]. Their main goal is to increase data variation and improve model robustness. While they can incidentally modify features like pitch range and vowel duration, they are not designed specifically for singing voice and do not systematically capture the unique characteristics of singing. This motivates the use of more advanced augmentation methods.

2.5.2 Advanced Augmentation Techniques

More advanced methods exploit the specific characteristics of voice signals. In these approaches, the audio signal may be decomposed, altered, and then reconstructed to enable fine-grained manipulation of pitch, timing, or timbre. These methods are especially beneficial for singing voices, where pitch variation and duration are key elements. Notable techniques include:

- **WORLD vocoder augmentation** [Morise et al., 2016]: The WORLD vocoder decomposes voice signals into separate components for spectral envelope, F0, and aperiodicity. This decomposition is specifically designed for voice and allows pitch and duration to be modified independently while preserving the natural timbre. In contrast, basic time stretching and pitch shifting introduce more artifacts or distort the voice characteristics.
- **PDAugment** [Zhang et al., 2022a]: Aligns pitch and timing of speech recordings to music scores, effectively converting speech to singing-like signals to reduce domain mismatch and improve performance.
- **Mixup** [Zhang et al., 2018]: A sample interpolation technique where two audio signals (and their corresponding labels) are mixed in the latent space. This approach, originally proposed for image data, has been successfully applied to speech [Meng et al., 2021] and adapted for singing voice tasks [Guo et al., 2022].

2.5.3 Beyond Existing Techniques

While most data augmentation efforts for singing voice focus on modifying the audio signal, text-level augmentation has not been explored. For instance, transliteration has been applied in ASR to convert text between scripts for multilingual training [Thomas et al., 2020, Khare et al., 2021] and a similar approach can be adapted for lyrics. In this thesis, we explore this direction for multilingual lyrics transcription, as discussed in Chapter 6.

In addition, recent work in speech-to-singing conversion [Wu and Yang, 2020, Parekh et al., 2020] can be used to augment singing voice datasets (as demonstrated by PDAugment [Zhang et al., 2022a]). However, conversion in this direction tends to undervalue the phonetic variation inherent in singing. To address this, we propose a complementary method: singing-to-speech conversion. This approach transforms singing signals into speech-like representations while preserving phonetic information, making it suitable for data augmentation in low-resource scenarios. Our method is described in detail in Chapter 5.

Part II: Related Tasks

2.6 Speech Recognition

Automatic speech recognition (ASR) [Yu and Deng, 2016] has been a major research topic for decades. From the very early spoken digits recognition systems at Bell labs, to the voice assistant on smart phones, ASR has evolved significantly. It has various applications and has been successfully deployed in many real-world scenarios, such as virtual assistants, voice-controlled devices, and real-time transcription.

ASR is a foundational field upon which automatic lyrics transcription (ALT) research is built. The techniques and models developed for ASR have been adapted and extended to the singing domain, as discussed in Section 2.8.

2.6.1 Task Definition

Automatic speech recognition (ASR) is a well-established field that has been extensively studied. As a result, the definition of the task is less mentioned in the recent literature, as it is generally well understood and can be defined from different perspectives. For example, in *Fundamentals of Speech Recognition* [Rabiner and Juang, 1993], a foundational work covering the basic knowledge over decades, characterizes ASR as an effort “to design an intelligent machine that can recognize the spoken word and comprehend its meaning” [Rabiner and Juang, 1993]. Yu and Deng [2016] describe ASR as a task “aimed to enable natural human-machine interaction”. According to *A Dictionary of Computer Science* [Butterfield and Ngondi, 2016], speech recognition is “the process of recognizing elements of speech by analysis of acoustic signal”. Although the core task remains unchanged, recent research has focused on advancing model architectures, improving scalability, and enhancing robustness, rather than redefining the task itself.

In this thesis, we adopt the following working definition of ASR, framed in the context of singing voice and lyrics transcription: *given an audio signal of speech or singing, the task is to produce a sequence of linguistic units (phonemes, characters, or words) that accurately represents the spoken content.*

2.6.2 Evaluation Metrics

Word Error Rate (WER) and Character Error Rate (CER) are the most commonly used metrics for evaluating ASR systems. **WER** is derived from the Levenshtein distance [Levenshtein et al., 1966]. It first aligns the recognized word sequence with the reference and computes the number of substitutions S , deletions D , and insertions I . The WER for an utterance (a line or a song) is

then computed as:

$$\text{WER} = \frac{S + D + I}{N}$$

where N is the number of words in the reference. The final WER is averaged over all utterances. CER could be computed in the same way, replacing all the word-level counts with character-level counts.

In a multilingual context, the choice of evaluation metric could differ based on the language. For example, in languages with a large character set, CER might be more informative than WER. In languages with a small character set, WER might be more informative. In practice, character-based languages such as Chinese often use CER as the primary metric. For alphabetic languages such as English or Spanish, WER is more common.

Mixed Error Rate (MER) is another metric that combines WER and CER. It is defined similarly, but the counts are at a mixture of word- and character-level. This is to deal with the situation where multiple languages are involved in the test set, referring to the code-switching scenario in ASR. For example, when dealing with English-Chinese code-switching, the MER is computed by summing the word-level counts for English and the character-level counts for Chinese.

2.6.3 Related Work

ASR systems can be broadly categorized into two paradigms based on their modeling approach: traditional systems and end-to-end (E2E) systems. In addition, we also cover multilingual and low-resource ASR. These areas have been studied in depth and provide important insights and techniques that inform our work on ALT. Finally, we point out the role of prosody in ASR, as it connects to the expressive nature of singing.

Traditional ASR

Traditional systems are typically based on the Gaussian Mixture Model–Hidden Markov Model (GMM-HMM) or Deep Neural Network–Hidden Markov Model (DNN-HMM) framework. These systems consist of several separate components: an acoustic model, a pronunciation model, a language model, and a decoding algorithm. Each component is trained or designed independently and integrated at inference time. For a comprehensive overview, readers can refer to Yu and Deng [2015].

Modeling. The goal is to estimate the most probable word sequence W

given an input audio signal X , typically modeled as:

$$\hat{W} = \arg \max_W P(W|X) = \arg \max_W P(X|W)P(W),$$

where $P(X|W)$ is governed by the acoustic model and $P(W)$ by the language model. In practice, $P(X|W)$ is further decomposed into:

$$P(X|W) = \sum_{\Phi} P(X|\Phi) \cdot P(\Phi|W) \quad (2.6)$$

where: $P(X|\Phi)$: Acoustic model — how well the audio matches the phone sequence Φ . $P(\Phi|W)$: Pronunciation model — how likely a phone sequence Φ corresponds to word sequence W .

Substituting back into the original decoding objective:

$$\hat{W} = \arg \max_W \sum_{\Phi} P(X|\Phi) \cdot P(\Phi|W) \cdot P(W) \quad (2.7)$$

This probabilistic formulation has the advantage of explicitly decomposing the modeling process into three interpretable components:

- Acoustic modeling: $P(X|\Phi)$
- Pronunciation modeling: $P(\Phi|W)$
- Language modeling: $P(W)$

Typically, $P(\Phi|W)$ is implemented using a pronunciation lexicon. In decoding, this is realized through a cascade of weighted finite-state transducers (WFSTs) [Mohri et al., 2002].

While traditional systems offered modularity and interpretability, they required significant linguistic resources, making them less adaptable to multilingual or low-resource settings.

End-to-End ASR

End-to-end (E2E) systems are based on deep neural architectures that directly map audio to text, without explicitly modeling intermediate phonetic representations or requiring a pronunciation lexicon. These systems can be broadly categorized into CTC models, sequence-to-sequence (seq2seq) models and hybrid CTC/attention models [Watanabe et al., 2017] (introduced in Section 2.3.3).

Modeling. E2E models directly estimate $P(W|X)$, often using a combination of CTC loss and sequence-to-sequence (cross-entropy) loss. The system typically consists of: an *encoder* that processes the input audio into high-level feature representations, a *decoder* that autoregressively generates output tokens

(characters or subwords), and a *decoding algorithm* (e.g., beam search) that may integrate optional language models.

As a result, E2E systems learn a joint latent space where both linguistic and acoustic information are fused, improving performance when sufficient data is available. The decoder inherently learns language modeling by predicting the next token in a sequence given past tokens, conditioned on the encoded audio. This joint optimization enables the model to learn both acoustic and linguistic features simultaneously.

E2E systems typically use beam search to generate hypotheses. Integration of external language models or CTC alignment scores is straightforward via shallow fusion. The lack of dependence on a pronunciation lexicon allows easier adaptation to multilingual and low-resource settings.

Multilingual ASR

As the first steps towards multilingual ASR, early works have extensively examined and compared mono-, cross²-, and multilingual models using a single model [Heigold et al., 2013, Toshniwal et al., 2018]. These investigations indicate that multilingual models generally outperform their mono- and cross-lingual counterparts, especially in scenarios with limited resources. Moreover, some research notes performance enhancements by integrating language information through conditioning [Toshniwal et al., 2018] or predicting language-specific tokens [Zhou et al., 2018]. Recent advancements leverage unlabelled data via self-supervised learning methods, such as Wav2Vec2 [Baevski et al., 2020, Conneau et al., 2021].

A particularly challenging aspect of multilingual ASR is code-switching, where speakers alternate between languages within a single utterance [Mustafa et al., 2022]. It is a challenging task because the vocabulary can vary significantly between languages, and the model must determine when to switch between languages. Approaches to handle code-switching include incorporating language identification [Li et al., 2019, Lyu and Lyu, 2008, Huang et al., 2021b], as well as designing unified systems by merging or mapping language-specific vocabularies or phoneme sets [Yilmaz et al., 2016, Emond et al., 2018, Hara and Nishizaki, 2017].

While this issue has been studied primarily in speech recognition, it is also likely to be relevant for ALT, especially in multilingual songs, as lyrics may similarly mix languages and require models to handle word- and character-level sequences across different scripts.

²Cross-lingual learning involves training on data from other languages or adapting models from different languages to achieve improved performance.

Low-Resource ASR

Low-resource ASR focuses on building effective ASR models for languages with limited annotated data. Transfer learning has been a dominant strategy in this domain, where knowledge from high-resource languages is leveraged to improve performance on low-resource languages [Kunze et al., 2017, Conneau et al., 2021]. A common approach is to pre-train the whole or part of the model on high-resource language data and fine-tune it on the target low-resource language.

In addition to transfer learning, other methods for low-resource ASR include data augmentation [Kim et al., 2021b], text-wise transliteration [Khare et al., 2021], using phone-based target units [Yusuyin et al., 2023], and meta-learning [Hsu et al., 2020].

Prosody and ASR

Prosody plays an important role in speech perception. It refers to the rhythm, stress, and intonation of speech, which can convey meaning beyond the words themselves. Prosodic features such as pitch, duration, and energy can provide cues for word segmentation, disambiguation, and emphasis in speech, and has been shown effective in early ASR systems [Waibel, 1988, Vicsi and Szaszák, 2010].

Modern ASR systems do not explicitly model prosody, the features are likely captured implicitly with learned representations, given the large amount of training data. Moreover, prosody is not evaluated by standard ASR metrics such as WER. However, given the definition of ASR being to not only recognize the words, but also to support natural and meaningful human-machine interaction, prosodic information remains important.

As discussed in Section 2.1.1, prosody in singing voice is largely shaped by musical context rather than linguistic meaning. This misalignment may contribute to the increased difficulty of ALT.

2.6.4 Recent Advancements

Recent research in ASR has expanded to address more diverse user populations, challenging environments, and real-time demands.

Pathological speech recognition aims to personalize ASR for individuals with speech disorders, such as dysarthria and speech impairments. General purpose ASR systems often struggle with these cases due to the limited data available for training. Recent studies have adapted techniques from low-resource ASR to create more effective and personalized systems, such as transfer learning [Xiong et al., 2020], data augmentation [Geng et al., 2020], and self-supervised learning

[Violeta et al., 2022]. These studies offer insights relevant to our work on singing voice, which can be viewed as a heterogeneous subdomain of human voice.

Recent work in real-time ASR has focused on reducing latency while maintaining robustness and enabling integration with large language models (LLMs). Online variants of hybrid CTC/Attention architectures have been proposed to convert offline models for real-time use [Miao et al., 2020]. A real-time LLM is proposed to produce the flow of time and integrated with recurrent neural network transducers (RNN-T) for streaming ASR [Seide et al., 2024].

2.7 Lyrics Alignment



Figure 2.5: Example of word-level alignment and phoneme-level alignment.

Given the lyrics to a song, audio-to-lyrics alignment aims at identifying for each lyric the corresponding position in a recording of the song, at a line, word or phoneme level [Fujihara and Goto, 2012]. Applications include the generation of karaoke-style lyrics for music players and subtitles for music videos. Moreover, lyrics alignment can serve as a basic block for singing voice analysis and can benefit other tasks such as cover song identification and music structure analysis. As our first step towards singing voice analysis, we work on lyrics alignment in Chapter 4.

2.7.1 Task Definition

Lyrics alignment is the task of obtaining the temporal relationship between lyrics and audio Fujihara and Goto [2012]. More specifically, it is to find the exact timing in music of a given line, word, or phoneme of the corresponding lyrics, depending on the granularity level. Figure 2.5 shows an example of word-level alignment and phoneme-level alignment.

Alignment could be performed at various granularity levels. Taking the word-level alignment as an example: For a given song S , $\{w_1, w_2, \dots, w_N\}$ is the word sequence that composes the lyrics L . The goal is to obtain the onset timestamp t_i for each word w_i , $i \in \{1, 2, \dots, N\}$ where N is the total number of words (Figure 2.5). Sometimes the offset is also taken into account. In the case where the word sequence from L is not available, one needs to deal with structural analysis of lyrics L before it can be formulated as above.

2.7.2 Evaluation Metrics

Three evaluation metrics are commonly used for lyrics alignment: Average Absolute Error (AAE), percentage of correct segments (Perc), and percentage of correct onsets (PCO). All the three metrics are adopted by the Music Information Retrieval Evaluation eXchange (MIREX) lyrics-to-audio alignment challenge³. Let $\hat{t}_i$ be the predicted time for word w_i .

AAE is first proposed in Mesaros and Virtanen [2008]. It is the absolute difference between the actual word onsets and estimated ones, averaged over individual errors:

$$AAE = \frac{1}{N_d} \sum_{i=1}^{N_d} |\hat{t}_i - t_i|$$

where N_d is the total number of words in the dataset. Sometimes the deviation of the absolute error is also reported. The authors mentioned the drawback of the absolute error: the perception of the same AAE can be different for songs at different tempos.

Perc is the percentage of total length of the correctly labeled segments over the duration of the song ([Fujihara and Goto, 2012], Figure 9), where the segments are split according the onset times:

$$Perc_{s_j} = \frac{\text{duration of correctly labeled segments}}{\text{duration of the song}} \times 100$$

where s_j is the j^{th} song. It is then averaged over all songs in the dataset. Comparing to AAE, Perc is less biased by the tempo, but less intuitive because the duration of a correctly labeled segment depends on the prediction of the current word and the next word.

PCO calculates the percentage of correct onsets with a tolerance window [Mauch et al., 2012]. The estimated onsets that fall within the tolerance window

³https://www.music-ir.org/mirex/wiki/2017:Automatic_Lyrics-to-Audio_Alignment

are considered as correct:

$$PCO_{s_j} = \frac{1}{N_s} \sum_{i=1}^{N_s} 1_{if \ abs(\hat{t}_i - t_i) < \tau} \times 100$$

where N_s is the number of words in the j^{th} song, and τ is the tolerance. The final PCO is averaged over all songs. In Mauch et al. [2012], the authors use a tolerance window of 1 second, which is very large. In the MIREX evaluations, τ is set to 0.3 seconds, which is still much larger than the tolerance level for transcription tasks (≤ 0.1 seconds).

All the three metrics consider only the onset of a word. While this is sufficient for generating prompt-level alignment, the offset is also important if one uses the alignment system as a basic module for singing analysis. Reporting the AAE and PCO for offsets could be helpful.

Alignment could also be evaluated in phoneme [Schulze-Forster et al., 2021] or line level [Wang et al., 2004, Huang et al., 2022], although not very common. It is worth pointing out that the alignment unit can be different from the evaluation unit. For example, alignment is commonly performed at character or phoneme level, but the evaluation is performed at word level.

2.7.3 Related Work

Early Work

A comprehensive survey on lyrics alignment was published in 2012 [Fujihara and Goto, 2012]. The authors concluded that automatic lyrics transcription was not yet feasible due to limitations in acoustic modeling and data availability. Prior to 2012, most efforts focused on feature engineering and integrating auxiliary information, such as musical structure or phoneme duration models, to improve alignment performance [Wang et al., 2004, Fujihara et al., 2006, Mesaros and Virtanen, 2008].

In 2017, the MIREX introduced the Automatic Lyrics-to-Audio Alignment challenge. The MIREX Automatic Lyrics-to-Audio Alignment benchmark provided a standardized dataset and evaluation protocol, enabling systematic comparison across different methods. This attracted interest from both academic and industrial researchers and brought innovation in alignment techniques. Early submissions relied on HMM-based methods [Kruspe and Fraunhofer, 2017], while more recent approaches employ end-to-end neural architectures such as Wave-U-Net and the use of self-supervised learning features [Stoller et al., 2019b, Fan et al., 2024b], leading to improved accuracy, robustness to polyphonic music, and multilingual alignment capabilities.

One-Way Encoding Alignment

We can categorize lyrics alignment methods into two main approaches: one-way encoding and latent space alignment. The distinction between these two approaches is based on where alignment is performed. In one-way encoding, the model is trained to directly map audio to text, while in latent space alignment, the model learns to align audio and text in a shared latent space. The latter approach is relatively novel and has shown promising results in recent studies. Comparisons of performance across recent works and their architectural design choices can be found in Table 2.1 and 2.2.

Dataset	Jamendo		Mauch		Hansen	
System	AAE	PCO	AAE	PCO	AAE	PCO
SDE [Stoller et al., 2019a]	0.82	85	0.35	77	-	-
GYL [Gupta et al., 2020]	0.22	94	0.19	91	0.10	97
VHMRA [Vaglio et al., 2020b]	0.37	92	0.22	91	0.18	95
SDRB [Schulze-Forster et al., 2021]	0.88	81	-	-	0.79	85
DAD [Demirel et al., 2021b]	0.31	93	-	-	-	-
HBE [Huang et al., 2022]	0.23	94	0.20	91	-	-
DSE [Durand et al., 2023]	0.15	92	-	-	-	-
KGLW [Kick et al., 2025]	0.20	94	-	-	-	-

Table 2.1: Comparison of lyrics alignment performance in the literature.

System	Learning Strategy	Granularity
SDE [Stoller et al., 2019a]	Wave-U-Net + CTC	Char
GYL [Gupta et al., 2020]	TDNN-F	Phone
VHMRA [Vaglio et al., 2020b]	BiLSTM + CTC	Char
SDRB [Schulze-Forster et al., 2021]	BiLSTM, DTW-attention	Phoneme
DAD [Demirel et al., 2021b]	CTDNN	Phone
HBE [Huang et al., 2022]	BiLSTM + CTC	Phone
DSE [Durand et al., 2023]	ResConv, ContrL	Char
KGLW [Kick et al., 2025]	ResConv, ContrL	Phone

Table 2.2: The current state-of-the-art methods for lyrics alignment (ResConv: residual convolutional blocks; ContrL: contrastive learning). All methods use Viterbi forced alignment, although applied to different types of score matrices.

A one-way encoding alignment system comprises two key components: an acoustic model and an alignment algorithm. The acoustic model maps input audio to posterior probabilities for text units, usually phoneme or character posterior, while the alignment algorithm (e.g., Viterbi decoding, dynamic time warping, or CTC-based marginalization) infers the most probable alignment between the audio frames and reference sequences.

Along with the improvement in ASR in recent years, network architectures designed for ASR, such as time-delay neural networks (TDNN) [Peddinti et al.,

2015] and bi-directional long short-term memory (BiLSTM) [Graves et al., 2013], are adopted for lyrics alignment and transcription [Gupta et al., 2019, 2020, Demirel et al., 2020, Dabike and Barker, 2019, Vaglio et al., 2020b].

Among them, the E2E systems are typically trained using CTC loss [Graves et al., 2006]. It allows weak supervision with line level annotations, which is more accessible than more precise annotations. However, CTC loss is optimizing the summation of all probable alignment paths, which introduces ambiguity in the training and may lead to peaky distributions [Liu et al., 2018]. It is based on a strong assumption of monotonicity, which may be violated with ill-transcribed lyrics. Alternatives are proposed to avoid this issue.

Latent Space Alignment

Schulze-Forster et al. [2021] propose a joint approach to phoneme-level lyrics alignment and text-informed singing voice separation, where the model is only trained with the singing voice separation objective, but able to perform both tasks. It proposes DTW-attention mechanism, which produces a learnable score matrix, from which alignment path can be extracted. The architecture used for encoding text and audio are standard BiLSTM and Conv-BiLSTM models, but no alignment objective is directly optimized. At inference, hard alignment is performed in phoneme-level. Although the alignment performance is not state-of-the-art, this work demonstrates the potential of using non-alignment losses to learn time-aligned representations, and alignment in the latent space.

Durand et al. [2023] propose a contrastive learning approach for lyrics alignment that jointly embeds audio and text into a shared space. The model is trained using a contrastive loss that pushes embeddings of matching audio-text pairs closer and mismatched pairs apart. No alignment supervision or transcription loss (e.g., CTC) is used. Instead, alignment is decoded from a soft similarity matrix between audio and text embeddings. This enables accurate and multilingual lyrics alignment with only weak line-level supervision. This approach achieves competitive results and demonstrates strong generalization across multiple languages. Kick et al. [2025] further extended the method by introducing a timestamp-informed box loss and achieved competitive results with less training data.

Broader Directions

Beyond model architecture, recent work has expanded the language, genre, and real-time application for lyrics alignment systems. Wang et al. [2023] adapted Whisper for Mandarin lyrics transcription and alignment. Their follow-up work [Wang et al., 2024] introduced a multilingual dataset MIR-MLPop for Man-

darin, Cantonese, and Taiwanese Hokkien. Park et al. [2024] and Brazier and Widmer [2021] advanced real-time alignment for live performances and opera. Other works addressed alignment with incomplete lyrics [Han et al., 2023] and developed memory-efficient methods for long recordings [Demirel et al., 2021b]. These contributions reflect a growing interest in building lyrics alignment systems that are more flexible, scalable, and broadly applicable across musical genres, languages, and deployment settings.

2.7.4 Limitations

Most of the approaches discussed above rely on the monotonicity assumption: the order of lyrics matches the temporal progression of the audio. While this generally holds true, it can break down in cases involving overlapping vocal lines, such as duets or layered backing vocals. In such scenarios, multiple words may be sung simultaneously, violating the strict left-to-right alignment assumption. The dataset introduced by Schulze-Forster et al. [2021] includes labels for different voice properties (see Section 3.2), which could facilitate research into these phenomena, but no method has yet been proposed to directly address the challenge.

Another practical limitation arises from inconsistencies in lyrics formatting. In real-world scenarios, lyrics are often poorly structured or contain metadata and annotations irrelevant to the sung content. For example, phrases such as “repeat,” “repeat 2x,” or credits and performance notes are sometimes embedded in the lyrics text. Moreover, songs that end with repeated phrases and fade-outs pose additional difficulty, as the alignment target becomes ambiguous in the absence of clearly articulated lyrics. These cases remain a challenge for deploying lyrics alignment systems in user-facing applications.

2.8 Automatic Lyrics Transcription



Figure 2.6: Example of automatic lyrics transcription.

Access to lyrics not only enhances the intelligibility of sung content but also deepens emotional and cognitive engagement with music. Psychological

studies have shown that lyrics play a crucial role in shaping listeners’ affective responses. Ruth and Schramm [2021] found that prosocial lyrics, especially when combined with acoustic arrangements, significantly increased empathetic emotions and prosocial thoughts among listeners. Neuroimaging research by Brattico et al. [2011] showed that lyrics intensify emotional processing in the brain. These findings further motivate the development of ALT systems.

Automatic Lyrics Transcription (ALT) refers to the task of extracting sung lyrics from audio recordings. It facilitates applications such as synchronized lyrics display and karaoke-style interaction, while also supporting research and development in music content analysis. ALT contributes to a variety of downstream music information retrieval (MIR) tasks. These include lyrics alignment (discussed in Section 2.7), singing pronunciation analysis [Demirel et al., 2021a], and cover song identification [Vaglio et al., 2021]. Furthermore, transcribed lyrics can serve as language learning material [Israel, 2013], particularly when aligned with audio, offering learners an engaging and immersive environment to improve pronunciation and vocabulary.

2.8.1 Task Definition

Formally, Automatic Lyrics Transcription (ALT) is the task of converting an audio recording containing singing voices into a sequence of words or characters that represent the lyrical content. This definition parallels that of automatic speech recognition (ASR), but is adapted to account for the unique characteristics of singing voice and lyrics. Figure 2.6 illustrates an example of ALT.

However, such a definition leaves implicit what is considered part of the “lyrics”. In practice, ALT systems deal with several ambiguities that distinguish it from traditional ASR: for instance, whether to include non-lexical vocalizations (e.g., “ooh”, “ah”), how to represent exaggerated or expressive forms (e.g., “whaaaat”), and how to format overlapping vocal parts. These elements often convey paralingual information such as emotion, emphasis, or stylistic expression, which ALT may capture in addition to the purely linguistic content targeted by ASR. No universal guidelines exist for handling them. In particular, overlapping vocals from simultaneous singers raise questions about segmentation, attribution, and transcription formatting. While singing diarization [Suda et al., 2022, 2024] may offer one possible solution by identifying “who sings when”, this remains an open research challenge. As a result, linguistic information from non-overlapping vocals remains the primary focus of ALT.

In this thesis, we adopt the standard input-output definition of ALT but acknowledge its limitations. We retain non-lexical elements and overlapping vocals if they are present in the dataset annotations, without enforcing additional

formatting conventions.

2.8.2 Evaluation Metrics

ALT systems may produce outputs at various linguistic levels, depending on the task formulation and training data. Common output units include words and characters. Accordingly, the most widely used metrics for evaluating ALT systems are Word Error Rate (WER) and Character Error Rate (CER) inherited from ASR. For details on the calculation, please refer to Section 2.6.2.

Lyrics formatting can significantly affect evaluation results. Before computing error rates, certain normalization steps are commonly applied to both reference and predicted lyrics, such as lowercasing all text and removing punctuation. However, there is no consensus in prior work on what normalization rules to apply. Some studies report error rates on raw text, while others normalize the text first. Similarly, error rates may be reported at different levels (per line, per segment, or per song), further complicating comparisons. Jam-ALT [Cífka et al., 2024] addresses this issue by examining standard formatting practices in the music industry and establishing a benchmark that adheres to those guidelines. While this benchmark was released after most of the work in this thesis was conducted and is therefore not adopted in our main experiments, we follow the Jam-ALT conventions when formatting our newly released Portuguese and Italian evaluation sets, as described in Chapter 7.4.2.

2.8.3 Related Work

This section reviews the development of ALT, examining its evolution through distinct research phases. For background on core ASR components and concepts, we refer readers to Section 2.6.

ALT has been an active research topic in recent years [Demirel et al., 2021b,c, Stoller et al., 2019a, Gao et al., 2022b, 2023a, Ou et al., 2022]. Early studies primarily focused on English ALT. More recently, research has shifted toward improving performance through pre-trained models [Ou et al., 2022, Zhuo et al., 2023], self-training [Gao et al., 2023b], and multilingual adaptation [Wang et al., 2023, Han et al., 2023]. These developments, along with remaining challenges in the field, are discussed in more detail in Section 2.8.4.

Before ALT Is Considered Feasible

Before ALT was considered viable, research focused on simpler tasks such as keyword spotting (KWS) and lyrics alignment. KWS, adapted from speech processing, aimed to detect specific words in music audio, which is a less de-

manding task to full transcription [Kruspe, 2014, Vaglio et al., 2020a, Kruspe, 2016]. Around the same time, lyrics alignment emerged as a related but more tractable problem (See Section 2.7 for more details). Unlike transcription, alignment tolerates recognition errors if the timing remains accurate.

Shift to End-to-End Models

Early ALT systems were primarily built using traditional ASR pipelines based on GMM-HMM or DNN-HMM architectures. These systems typically relied on handcrafted acoustic features (e.g., MFCCs), forced alignments, pronunciation lexicons, and phoneme-level decoding, resulting in complex and rigid pipelines that were often sensitive to musical variability. An overview of early efforts in this space is provided in Mesaros [2013], which discusses the foundational work that shaped the field before (end-to-end) E2E ALT became mainstream. Mesaros and Virtanen [2009] adapted a pre-trained GMM-HMM-based speech recognizer to the singing domain. Later, DNN-HMM models became more popular for ALT and lyrics alignment tasks [Kruspe, 2016, Gupta et al., 2018b]. Demirel et al. conducted a series of works using DNN-HMM systems [Demirel et al., 2020, 2021c], marking the end of the GMM/DNN-HMM era and the beginning of a transition toward E2E learning in ALT.

E2E models have gained popularity in ALT due to their ability to learn directly from raw audio and text data, eliminating the need for feature engineering and intermediate representations. This approach simplifies the model architecture and allows easy scaling. The first E2E ALT models were designed for English ALT. Stoller et al. [2019a] introduced a model based on Wave-U-Net with CTC loss [Graves et al., 2006]. Comparing to traditional ASR models that use HMMs, E2E systems do not need a pronunciation lexicon for inference, as they directly map audio to text. This is particularly beneficial for ALT, as the pronunciation of lyrics can vary significantly from speech (discussed in Section 2.1.1). However, a key challenge is that such models typically require large amounts of labeled data to learn such mappings.

Consequently, E2E models were not widely adopted in ALT until the introduction of large-scale datasets (DALI and DSing). Traditional methods relied on domain knowledge, which performs better with limited data. As the amount of data increased, E2E models became more effective. It is worth noting that, in Gupta et al. [2020] the authors found that traditional methods still perform better when training on music mixtures. However, this comparison was limited to a single traditional system and one E2E model, so the generalizability of the conclusion remains uncertain. Following this trend, Gao et al. conducted a series of works [Gao et al., 2022b, 2023a, 2022a], with a Transformer architecture

and a hybrid CTC/attention model. Table 2.3 summarizes representative E2E ALT architectures, including both earlier and more recent approaches.

All the work presented in this thesis follows the E2E paradigm, leveraging recent advances in E2E architectures to address the unique challenges of ALT.

Architecture	Learning Strategy	Work
Wav-U-Net	CTC	Stoller et al. [2019a]
CRNN	hybrid	Gupta et al. [2020]
Transformer	hybrid	Gao et al. [2022b]
Wav2Vec2 ALT	fine-tuning; hybrid	Ou et al. [2022]
Whisper	fine-tuning; seq2seq (multi-task)	Radford et al. [2023]

Table 2.3: Comparison of architectures and learning strategies in E2E ALT. *Hybrid* refers to a combination of CTC and attention-based sequence-to-sequence learning [Watanabe et al., 2017].

Incorporating Domain Knowledge

The use of domain knowledge in ALT became prominent between 2020 and 2023, coinciding with the shift toward E2E models. Domain knowledge serves not only to improve performance but also to enable more interpretable analyses by incorporating musical attributes. Common forms of domain knowledge include genre and chord labels [Gao et al., 2022a,b, Gupta et al., 2020], which are closely related to the background music, as well as the use of lyrics corpora for training language models [Demirel et al., 2020].

During this period, domain knowledge was frequently integrated into ALT systems via mechanisms such as conditioning and multi-task learning. These strategies enriched the learning process by providing relevant contextual signals, leading to more robust models and deeper insights into the task. Representative works are summarized in Table 2.4.

Domain Knowledge	Integration	Work
Lyrics	Lyrics corpus for LM training	Demirel et al. [2020]
Genre	Conditioning	Gupta et al. [2020], Gao et al. [2022a]
Chord	Multi-task learning	Gao et al. [2022b]
Background Music	Multi-task learning	Gao et al. [2023a]
Music Transcription	Multi-task learning	Gu et al. [2024]

Table 2.4: Comparison of music-related domain knowledge and their integration strategies in ALT.

Separate Paths for ALT and Alignment

The model training pipelines for lyrics alignment and automatic lyrics transcription (ALT) were largely similar prior to the emergence of E2E methods. Traditional approaches relied on GMM-HMM frameworks, where GMMs captured the acoustic features and HMMs modeled the temporal structure of the lyrics. This method could be used both for alignment—when the text was provided—and for transcription—when coupled with a language model. As a result, improvements in one task often lead to gains in the other. This synergy continued in some later works that replaced the GMM component with a Deep Neural Network (DNN).

However, this connection breaks down with the shift to E2E models. In these architectures, acoustic, lexical, and linguistic representations are learned jointly, eliminating the intermediate separation found in traditional pipelines. Even in CRNN-based systems without attention mechanisms, temporal correspondence is not inherently preserved [Teytaut et al., 2022]. For models using a seq2seq decoder, temporal alignment becomes even more ambiguous: the attention mechanism is the only proxy for time, but attention weights are not guaranteed to be monotonic and are often dispersed across multiple layers and heads.

These architectural changes have led to a divergence in modeling strategies for ALT and lyrics alignment. In current practice, the two tasks are typically addressed by separate models, with few exceptions utilizing Whisper [Wang et al., 2023, 2024].

Foundation Models for ALT

Recent advances in large-scale pre-trained models, also referred to as foundation models, have significantly influenced the field of ASR. These models are trained on vast amounts of data and exhibit strong generalization across domains and languages. Their robustness makes them attractive to ALT, where training data is often scarce and audio conditions vary. Among these, Wav2Vec2 [Baevski et al., 2020] and Whisper [Radford et al., 2023] are the most commonly used foundation models in ALT research.

Ou et al. [2022] were the first to fine-tune wav2vec 2.0 features on source-separated singing, achieving significant improvements. Following this, Gao et al. [2023b] applied self-training with noisy student augmentation.

Whisper is an open-source, multilingual ASR model trained on a large and diverse audio dataset. It can perform multilingual ASR, speech translation, and language identification. Its robustness to various noise conditions has led to promising results in ALT [Zhuo et al., 2023, Cifka et al., 2024]. Performance

varies depending on the quality of singing voice separation (SVS) used as a pre-processing step, and this relationship is further explored in Chapter 7.

Whisper has been adapted to low-resource languages, such as Mandarin Chinese [Wang et al., 2023] and Korean [Han et al., 2023]. These adaptations demonstrate Whisper’s flexibility and potential for multilingual ALT tasks.

2.8.4 Key Challenges

This section summarizes key challenges in automatic lyrics transcription (ALT), which also serve as the primary focus areas of this thesis. Each challenge is discussed in the context of prior research and how it is addressed in the subsequent chapters.

Dealing with Phonetic Variations

In Section 2.1.1, we mentioned the phonetic difference between singing and speech. This is often underexplored due to limited phonemic awareness in the ALT literature. Research in psycholinguistic and cognitive neuroscience has shown that human listeners can often access lexical or semantic information directly, without explicit phoneme-level analysis [Luce and Pisoni, 1998, Liberman et al., 1974, McClelland and Elman, 1986]. This suggests that the human brain might not be sensitive to the phonetic variations in singing voice.

While phonetic variation is frequently acknowledged, it is difficult to model effectively. One early attempt is the NUS Sung and Spoken Lyrics Corpus [Duan et al., 2013], which provides phoneme-level annotations for both modalities. However, the dataset is too small for training ALT models, limiting its impact. Another approach by Demirel et al. [2021a] introduced a singing-adapted pronunciation model based on phoneme confusion analysis. These works highlight the phonetic variations in singing voice and their potential impact on ALT.

In Chapter 5, we present the phonetic differences between singing and speech in a more directly way through singing-to-speech conversion. We show that this task not only makes the differences more tractable but also serves as a useful data augmentation strategy for ALT.

Addressing Data Scarcity: Pre-Trained Models and Self-Training

Since labeled training data for ALT remains much smaller than those for ASR, researchers have explored ways to address the data scarcity issue. One common approach is leveraging pre-trained speech or language models. Ou et al. [2022] fine-tuned Wav2Vec2, achieving strong results with relatively limited data. The MARBLE benchmark [Yuan et al., 2023] also examined this approach, although its pre-trained model performance remained limited without fine-tuning. Gao

et al. [2023b] applied self-training [Scudder, 1965], improving word error rates (WERs) by training on a small labeled subset. These works marked a shift from training ALT models from scratch to leveraging large-scale ASR models for singing voice transcription for better performance.

This thesis explores data scarcity in several ways. We benchmark pre-trained features in Chapters 3 and 7, and in Chapter 6 we introduce a phonetic reinterpretation strategy that augments the lyrics, which inherits the idea of self-training.

Expanding Beyond English: Multilingual ALT Challenges

Although several datasets have been introduced for multilingual ALT, such as DALI v2 [Meseguer-Brocal et al., 2020] and MulJam v2 [Yuan et al., 2023], they are still insufficient for training robust multilingual ALT models. The majority of available training songs are in English—over 80% in DALI v2 and more than 60% in MulJam v2—making it challenging to develop systems that generalize across languages. Some studies have attempted to address this issue. Vaglio et al. [2020b] trained a lyrics alignment model using a universal phoneme set on multilingual data from DALI, but transcription of non-English lyrics remains relatively unexplored.

Multilingual ALT presents several additional challenges compared to English ALT: Firstly, models must implicitly recognize the underlying language of the singing to ensure the predicted lyrics match the appropriate character set, unless specified. Secondly, there is a language imbalance in datasets, with languages like English typically dominating, while other languages remain low-resource in ALT development. Thirdly, some characters appear in multiple languages’ alphabets but follow different pronunciation rules, further complicating the task.

In Chapter 6, we present practical guidelines for training data selection and language conditioning strategies. Our results demonstrate the benefits of joint multilingual training combined with appropriate conditioning techniques. Based on these insights, we introduce a phonetic augmentation method specifically designed for multilingual settings.

Dealing with Background Music

Background music is a major challenge for ALT comparing to ASR. Some prior work treats background music as noise [Stoller et al., 2019a, Gu et al., 2024]. In this setup, models are trained either on clean vocals or with strategies to ignore the accompaniment. An alternative view treats background music not as interference but as contextual input. Since musical accompaniment and vocals are usually coordinated, background music may provide genre, rhythm, or

mood cues that guide the transcription model. Three background conditions are typically considered: clean vocals, singing voice separated (SVS) vocals, and mixtures.

Singing voice separation (SVS) tools allow researchers to isolate vocals for training or evaluation. However, inconsistent findings have emerged: some studies suggest that models trained on clean/SVS vocals perform better on mixtures [Demirel et al., 2021c, Wang et al., 2023], while others show the opposite [Gupta et al., 2020, Cifka et al., 2024]. These discrepancies likely stem from varying SVS tool quality and dataset characteristics.

In Chapter 7, we examine how background music affects ALT performance, focusing on the role of singing voice separation (SVS) tools. We evaluate the impact of fine-tuning strategies across single and dual domains, training models on either mixtures or source-separated vocals. To bridge the gap between these domains, we propose a consistency loss that encourages alignment between vocal and mixture representations.

2.9 Singing Voice Conversion



Figure 2.7: Example of voice conversion. The same linguistic content is rendered in different visual styles to represent changes in vocal attributes.

Singing voice conversion (SVC) has been studied both for creative applications, including voice cloning, and as a means to investigate how vocal attributes such as pitch, timbre, rhythm, and style can be disentangled and controlled. This technical perspective is particularly relevant for ALT, where understanding and manipulating singing voice characteristics can offer significant benefits. This section reviews key developments in singing voice conversion, beginning with in-domain approaches for speech and singing. It then turns to the more challenging problem of cross-domain conversion, where the source and target modalities differ. The challenges of cross-domain conversion come from the distinct characteristics of each domain, as discussed in Section 2.1.1. These foundations support our work on singing-to-speech conversion, introduced in Chapter 5, which aims to better capture phonetic variation in singing and en-

hance ALT systems.

2.9.1 Task Definition

Singing voice conversion (SVC) is a specialized form of voice conversion (VC), which involves modifying specific attributes of a source voice to match those of a target voice while preserving other aspects of the signal. These attributes can include speaker identity, emotion, pitch, and others. In the context of singing, the term SVC typically refers to the conversion of singer identity, without changing the linguistic content [Sisman et al., 2021]. Figure 2.7 illustrates an example of voice conversion.

Although the term *voice conversion* can apply to a wide range of vocal attributes, most work in SVC has focused on identity conversion. Nonetheless, other dimensions of the singing voice can also be manipulated. These include pitch [Wager et al., 2020, Luo et al., 2018, Morrison et al., 2021], singing techniques [O’Connor et al., 2021], and prosody [Qian et al., 2020, Shechtman and Sorin, 2019], offering a broader view of SVC for controlling expressive features in singing.

2.9.2 Evaluation Metrics

The performance of voice conversion (VC) systems, including singing voice conversion (SVC), can be assessed using both subjective and objective evaluation metrics.

Subjective evaluation typically involves listening tests, where human participants rate the naturalness and similarity of the converted voice to a target voice. The results are aggregated using the Mean Opinion Score (MOS), a widely adopted metric for perceptual quality assessment.

Objective evaluation depends on whether a reference signal is available. When a reference is present, Mel-cepstral distortion (MCD) [Kubichek, 1993] is commonly used to measure the spectral distance between the converted and reference signals. In addition, pitch similarity can be assessed using metrics such as the Pearson Correlation Coefficient and Root Mean Squared Error (RMSE) between pitch contours.

When no reference is available, objective evaluation often relies on proxy tasks. These may include classification accuracy (e.g., speaker identification) or reconstruction error in autoencoding frameworks, depending on the system design. If the linguistic content is preserved during conversion, automatic speech recognition (ASR) metrics such as word error rate (WER) can be used to assess intelligibility. Furthermore, neural networks trained to predict human perceptual ratings, such as MOSNet [Lo et al., 2019], have been proposed as automated

alternatives to subjective evaluations.

2.9.3 Related Work

Speech Conversion

Various methods have been explored in the field of speech conversion, including using a pre-trained Automatic Speech Recognition (ASR) model to extract speaker-invariant linguistic features (text [Zhang et al., 2020, Park et al., 2020] or phonemes [Proszewska et al., 2022, Zhou et al., 2019]), followed by speech synthesis for the target speaker. Others employ autoencoders to disentangle speaker and linguistic information during reconstruction [Qian et al., 2019, Chou and Lee, 2019]. GAN-based (Generative Adversarial Network) models [Li et al., 2021a, Kaneko and Kameoka, 2018] have been applied to alter the speaker identity while retaining the content with the help of discriminators.

Beyond speaker identity, recent work has extended speech conversion to more expressive aspects of prosody. These include emotion conversion [Zhou et al., 2022], accent conversion [Jia et al., 2024], and more specialized applications such as transforming whispered speech [Rekimoto, 2023] and pathological voices to normal speech [Chu et al., 2023].

Singing Voice Conversion

Singing voice conversion (SVC) research generally follows that of speech conversion: GAN-based models (ROSVC [Takahashi et al., 2022], FastSVC [Liu et al., 2021a]) and disentanglement-based methods (VQVAE and its variants [Luo et al., 2020, Takahashi et al., 2021]) have also been explored.

AutoVC [Qian et al., 2019] is an autoencoder designed for speech conversion conditioned on speaker embeddings, where tuning the bottleneck dimension allows the encoder to focus on speaker-independent features. This model was adapted for SVC in Nercessian [2020], where AutoVC applies only on the harmonic spectral envelope produced by WORLD [Morise et al., 2016]. Subsequent work further enhanced the AutoVC model with a latent regressor loss for improved identity embedding [O'Connor and Dixon, 2023]. Other approaches based on disentanglement use Variational Autoencoders (VAE), phonetic posteriorgrams (PPG), and adversarial training. For example, Luo et al. [2020] achieve many-to-many SVC by training a Gaussian mixture variational autoencoder (GMVAE) on non-parallel data. Another method [Li et al., 2021b] condenses phonetic information into PPGs while encoding other content information with a separate encoder. In Lu et al. [2020], the authors adapted VAW-GAN [Hsu et al., 2017] to the SVC task. The adversarial losses in Li et al. [2021b]

and Lu et al. [2020] enhance the robustness.

Cross-Domain Voice Conversion

While some of the previously discussed voice conversion pipelines incorporate alignment or time-warping techniques (either explicitly or implicitly) the transformations involved are generally moderate, as they operate within the same domain. However, in cross-domain scenarios where speech is converted to singing or vice versa, alignment becomes significantly more difficult. For instance, Vijayan et al. [2017] report that alignment errors between speech and singing using Dynamic Time Warping (DTW) are substantially higher than those observed in speech-to-speech alignment, primarily due to the large differences in rhythm and timing. Substantial local tempo differences are also challenging for time warping methods.

One specific cross-domain task is speech-to-singing conversion, where the prosody of natural speech is modified to resemble singing [Vijayan et al., 2019]. This type of conversion can be used to improve the vocal performance of amateur singers by enhancing expressiveness.

To enable conversion between speech and singing with deep learning presents several challenges. These include the scarcity of paired training data (recordings of speech and singing that share the same linguistic content) and the difficulty of obtaining accurate temporal alignment between modalities. Early methods [Vijayan et al., 2017, Cen et al., 2012, Sharma and Li, 2019] relied on signal processing techniques, using DTW or its variants to compute alignment between speech, singing, and text. More recent work by Li et al. [2023a] employs cross-attention combined with alignment regulation loss to encourage diagonal and monotonic alignment paths.

Recent studies [Agarwal et al., 2022, Parekh et al., 2020] have shown a growing interest in deep learning approaches without explicit synchronization input. These approaches aim to learn a mapping from speech to singing, or vice versa, without requiring explicit alignment between the two modalities, making cross-domain voice conversion more flexible and scalable.

Duration Prediction

Duration prediction [Ren et al., 2019, Yu et al., 2020, Kim et al., 2020] has been introduced to address robustness issues commonly observed in neural text-to-speech (TTS) models, such as word skipping, repetition, and attention collapse [Tan et al., 2021]. A duration predictor can be used in two primary ways: (1) to estimate the overall length of the output sequence, and (2) to predict the number of output frames corresponding to each input token.

The former approach helps constrain the model from producing overly long outputs, while the latter provides a deterministic alignment between input and output tokens by replacing the attention mechanism at inference time. This substitution mitigates the risk of attention collapse and improves alignment robustness, particularly in cases where the input and output sequences differ significantly in length. By removing reliance on soft attention modules, duration prediction contributes to more stable and controllable synthesis.

2.9.4 Recent Advancements

Self-supervised features have been applied to SVC for extracting rich and meaningful features. Jayashankar et al. [2023] explores the use of Wav2Vec 2.0 [Baevski et al., 2020] and HuBERT [Hsu et al., 2021a] together with an f_0 harmonic generation module. Additionally, Wang et al. [2022a] demonstrates that HuBERT features outperform Mel-spectrogram and PPG-like features when using a contrastive predictive coding module. Huang et al. [2023a] replaces the traditional acoustic features input with WavLM [Chen et al., 2022] to extract content features and for reconstruction. Furthermore, Ning et al. [2023] utilises Whisper’s [Radford et al., 2023] encoder to extract bottleneck features as content representation.

The introduction of diffusion decoders has also led to substantial improvements in conversion quality. DiffSVC [Liu et al., 2021b] is the first method to apply a diffusion-based decoder to SVC, although it is limited to any-to-one conversion. CoMoSVC [Lu et al., 2024] extends this framework to many-to-many settings using a consistency model, enabling fast and high quality inference. The `so-vits-svc` [Team, 2024] is an impactful open-source project providing state-of-the-art singing voice conversion tools based on VITS [Kim et al., 2021a]. It supports training with various SSL-based features, including HuBERT, Whisper PPGs, and WavLM. Additionally, it incorporates shallow diffusion techniques [Liu et al., 2022a] as a post-processing module to further enhance audio quality.

Conclusion

This chapter presented the foundational background for automatic lyrics transcription (ALT). We began by introducing the characteristics of singing voice and its differences from speech. We then reviewed key concepts in deep learning for audio processing, including commonly used audio representations, neural network architectures, and learning paradigms.

Next, we discussed practical considerations such data augmentation strategies to address data scarcity. The chapter then surveyed related computational tasks and research areas, including automatic speech recognition (ASR), lyrics alignment, automatic lyrics transcription (ALT), and singing voice conversion (SVC). ASR serves as the basis for many singing voice modeling approaches. Lyrics alignment is closely related to transcription and provides a testbed for modeling linguistic information. ALT is the core focus of this thesis, while SVC offers a perspective for analyzing and manipulating specific attributes of singing voice.

The following chapter begins the presentation of the original research work conducted during this PhD.

Chapter 3

Datasets and Benchmarks

3.1 Introduction

In this chapter, we examine two pillars in ALT: datasets and model design. Datasets are essential for training and evaluating ALT systems, and their quality, diversity, and accessibility play a critical role in model performance. Model design refers to the architectural choices and training strategies that define how the system process and learn from data. By addressing both pillars, this chapter lays a foundation for the subsequent chapters.

ALT depends heavily on the quality and quantity of available data. We discuss how public datasets have enabled research at scale, and present our contributions to dataset curation and evaluation.

Before 2018, ALT research was limited by data scarcity. Studies often relied on small, private datasets with inconsistent annotation formats and evaluation protocols, making it difficult to achieve performance suitable for real-world application and compare results across works [Kawai et al., 2017, Tsai et al., 2018]. This changed with the release of DSing [Smule, 2015] and DALI v1.0 [Meseguer-Brocal et al., 2018], which provided the first large-scale public corpora for ALT and quickly became foundational resources in the field [Dabike and Barker, 2019, Demirel et al., 2020, Gupta et al., 2020]. As a result, progress has been accelerated, and the research community studied the task from diverse angles, while having generally comparable results. In this chapter, we begin by reviewing datasets commonly used for lyrics alignment and ALT. We categorize the datasets based on their scale and purpose, describe their characteristics and limitations, and note how they have been used in prior work.

Despite their impact, these datasets come with limitations. DSing is focused exclusively on English songs and only includes clean vocal tracks recorded via a karaoke app, resulting in variable quality and limited acoustic diversity. While

robustness to variable quality is important for real-world systems, the diversity here is confined to recordings from smartphone microphones and does not reflect other acoustic environments. Moreover, the uncontrolled recording conditions often introduce artifacts and inconsistencies that may mislead training. DALI, while multilingual, relies on audio sourced from YouTube, which presents access and licensing issues. The timestamp annotation is not accurate, as it is created by matching lyrics with the audio according to singing voice detection (so are the pitch annotations). And there has not been an established split, so researchers tend to create their own splits, making results less comparable.

To address these challenges, together with the Multimodal Art Projection (m-a-p) team ¹, we developed MulJam [Zhuo et al., 2023], a multilingual ALT dataset designed to expand language coverage, provide standardized splits, and offer legally accessible audio. MulJam v2.0 is built specifically for ALT training, offering line-level annotations and improved design for model development and evaluation. This dataset adds value to the research community by enabling systematic multilingual benchmarking and by providing shareable audio and lyrics, which has been a major bottleneck for ALT datasets. My contribution focused on curating, cleaning, and updating MulJam v2.0, including the release of standardized splits and additional evaluation sets. We believe this dataset adds value to the research community, extending ALT to multilingual songs and diverse genres. This chapter discusses my work related to MulJam curation and update.

We also note that both DALI and MulJam are synthetic datasets, meaning that the annotations are not verified by humans. While this enables large-scale data collection, it introduces quality issues that remain largely unaddressed in current literature. Developing guidelines or methods for improving annotation quality remains an open and important research problem.

In addition to data, architectural and training choices significantly influence model performance. This chapter also surveys common ALT modeling approaches and presents a series of benchmarking experiments that examine these design choices. These architectures and training strategies are not only representative of current practice, but also form the foundation for the models used in later chapters. These findings help inform the architecture of the MARBLE ALT backend, which is part of the broader MARBLE framework [Yuan et al., 2023]. MARBLE provides an unified benchmark for evaluating SSL features across a range of music information retrieval (MIR) tasks, including ALT. Our results show that SSL models trained on general-purpose music often underperform on ALT, pointing to a need for more targeted modeling strategies.

¹An open-source AI community on machine learning for multimodal art. Website: <https://m-a-p.ai/>

While this chapter primarily focuses on data and model design for automatic lyrics transcription (ALT), we also include a complementary contribution to the task of lyrics alignment. Specifically, we annotated the MUSDB18 [Raffi et al., 2017] test set with word-level timestamps to address the lack of high-quality, manually verified alignment data. In this work we aim to support lyrics alignment evaluation on a larger and more diverse set of tracks. This dataset is used in our lyrics alignment experiments presented in Chapter 4.

The remainder of this chapter is organized as follows: Section 3.2 provides a survey of existing datasets for lyrics alignment and ALT. Section 3.3 introduces the MulJam dataset and its updates, including the annotation process and dataset statistics. Section 3.4 presents the annotation work done for the MUSDB18 test set. Section 3.5 discusses architectural and training choices for ALT. Section 3.6 introduces the ALT part in the MARBLE evaluation benchmark. Section 3.7 points out and explains the inconsistencies in the datasets and evaluation of ALT models throughout the thesis. Finally, Section 3.8 concludes the chapter. The ALT for MARBLE benchmark was developed in collaboration with the m-a-p team and was published in Yuan et al. [2023].

3.2 Datasets: Existing Datasets

The datasets used for automatic lyrics transcription (ALT) and lyrics alignment can be broadly divided into two categories: *large-scale datasets* and *small-scale datasets*. Large-scale datasets are typically collected from the web using fully- or semi-automated pipelines, and may include noisy or weakly aligned labels. They are mainly used for training and validation. Small-scale datasets are usually manually annotated with higher accuracy and are more appropriate for evaluation. This section covers publicly available datasets used in recent research.

3.2.1 Large-Scale Datasets

DSing (DAMP Sing! 300x30x2) [Smule, 2015] is constructed from vocal recordings collected via the Smule karaoke mobile application, where users sing along to instrumental backing tracks while lyrics are presented on screen. It is the third and most refined release in a series derived from Smule’s user-generated content, and has been widely adopted in ASR and ALT research. The dataset provides line-level (or utterance-level) lyrics annotations, obtained through a combination of metadata and alignment procedures. These annotations are suitable for training alignment and transcription models. All recordings are in English, covering 300 popular song arrangements performed by one male

and one female singer from each of 30 countries, totaling 5,690 singing tracks. This results in balanced gender and song arrangement representation, but the dataset lacks linguistic diversity beyond English. The most widely used data splits (DSing1, DSing3, and DSing30) were introduced by Dabike and Barker [2019]. These subsets organize the recordings by user region and are commonly used for training and evaluation in lyrics alignment tasks. However, DSing consists exclusively of a cappella recordings by amateur singers in uncontrolled environments, leading to variation in recording quality and vocal style. These factors, along with the absence of accompaniment and limited language coverage, constrain its generalizability.

DALI [Meseguer-Brocal et al., 2018, 2020] is a large-scale dataset collected from YouTube, with metadata and lyrics sourced from the WASABI music knowledge base [Meseguer-Brocal et al., 2017]. The dataset provides audio as downloadable YouTube links and aligns lyrics to the music using automated singing voice activity detection. DALI offers hierarchical annotations at multiple levels, including notes, words, lines, and paragraphs. These annotations are automatically generated and reflect the structure of the lyrics in relation to the musical content, although the precision of alignment varies due to the automatic nature of the process. The dataset has 30 languages in v1.0 and extends to 32 in v2.0. Nevertheless, English songs are dominant, resulting in an imbalanced distribution. Several splits have been introduced in prior work [Vaglio et al., 2020b, Demirel et al., 2021c], but none have become standard due to a combination of version inconsistencies, the instability of YouTube links, and the lack of officially maintained splits. DALI is primarily used for training and validation purposes with customized splits. Despite its breadth and annotation richness, DALI poses challenges for reproducibility and evaluation. The dependency on external video links, which may become unavailable, and the imperfect accuracy of automated alignments limit its reliability for fine-grained analysis or evaluation.

MulJam [Zhuo et al., 2023, Yuan et al., 2023] is a multilingual dataset derived from the MTG-Jamendo music platform [Bogdanov et al., 2019], with annotations generated using Whisper [Radford et al., 2023] and refined using ChatGPT², which compared multiple candidate transcriptions and selected the most accurate version. In this thesis, we extend MulJam as described in Section 3.3. The dataset comprises 5,797 songs across six languages, offering a more balanced English/non-English distribution compared to earlier resources. The initial annotations were generated at the song level and refined to the line level using automatic text alignment methods. An official training and evaluation split is provided with the dataset. However, given that MulJam was intro-

²<https://openai.com/blog/chatgpt>

duced recently and the multilingual ALT task remains relatively underexplored, it has not yet seen widespread adoption in the literature. The error introduced by text alignment for refining annotations leads to inaccurate line-level labels.

3.2.2 Small-Scale Datasets

Jamendo [Stoller et al., 2019a] and **Multi-Lang Jamendo** [Durand et al., 2023] are two benchmark datasets for lyrics alignment and ALT. The latter is a multilingual extension of the former. Jamendo includes 20 English songs, while Multi-Lang Jamendo expands coverage to 20 songs each in English, Spanish, and German, and 19 songs in French ³. Both provide line-level annotations, word-level timestamps and are publicly available. They are among the most widely used datasets for evaluation in both lyrics alignment and transcription tasks.

Mauch [Mauch et al., 2012] and **Hansen** [Hansen, 2012] are two English datasets frequently used for evaluation. Mauch contains 20 songs and Hansen has 10 songs, both with line-level and word-level annotations. Although not publicly available, they have been widely adopted in earlier work for lyrics alignment and ALT.

MIR-MLPOP [Wang et al., 2024] provides time-aligned lyrics in Mandarin, Cantonese, and Taiwanese Hokkien, with 30 songs per language. The dataset was introduced to support ALT in low-resource languages and serves as a valuable benchmark for multilingual ALT evaluation.

MIR-1k extension [Wang et al., 2023] is an annotated extension of a subset of the original MIR-1k [Hsu and Jang, 2010]. It is labeled in character level for alignment evaluation in [Wang et al., 2023]. The subset contains 17 songs in Mandarin.

MUSDB18 lyrics extension [Schulze-Forster et al., 2021] augments the MUSDB18 source separation dataset [Rafii et al., 2017] with line-level segmentation lyrics annotations for 141 English songs. Vocal conditions such as solo singing, unison, and non-unison multiple voices are labeled. While this extension only provide segment boundary timestamps and corresponding lyrics, we further annotate the test set (45 songs) with word-level lyrics in Section 3.4, enabling its use in lyrics alignment.

NHSS [Sharma et al., 2021] and **NUS-48E** [Duan et al., 2013] are parallel corpora of speech and singing recordings, used primarily for comparative analysis and voice conversion studies. NUS-48E contains recordings of 48 English songs sung and spoken by 12 singers, with phoneme-level annotations with durations.

³The initial release v1.0 has 80 songs in total. One song was removed due to legal issues in v1.1.

NHSS consists of 100 songs performed in both speech and singing by 10 singers. These datasets are used in our evaluation of singing-to-speech conversion in Chapter 5.

Opencpop [Wang et al., 2022b], **OpenSinger** [Huang et al., 2021a], and **M4Singer** [Zhang et al., 2022b] are professional Mandarin singing voice synthesis datasets. Opencpop comprises 100 songs from a single professional singer, OpenSinger includes 1,146 songs by 76 singers, and M4Singer consists of 700 songs performed by 20 singers. While originally intended for singing synthesis, these datasets have also been adapted for ALT evaluation, as demonstrated in prior work [Wang et al., 2023] and in our experiments in Chapter 5.

JamendoLyrics++ [Kick et al., 2025] is a recent extension of the Jamendo dataset that includes word-level timestamp annotations. It comprises 80 English songs selected from the Jamendo platform, spanning over 20 musical styles. While the availability of fine-grained annotations for additional songs is a valuable contribution, it introduces the potential risk of song overlap with MulJam (commonly used for training), which is also derived from the same collection. Such overlap previously occurred between MulJam v1.0 and initial English Jamendo dataset. Researchers should therefore carefully verify and remove any overlapping songs to avoid data leakage and ensure proper evaluation.

3.3 Datasets: MulJam v1.0 and v2.0

3.3.1 MulJam v1.0 Dataset

The first version of MulJam was released in Zhuo et al. [2023], which is introduced in Section 3.2. It is a synthetic dataset whose annotations are generated by Whisper⁴ and then refined by ChatGPT. However, the authors did not originally consider the need of line-level annotation to make the dataset useful for ALT training, and they only prepared song-level annotation. My work as part of MulJam v1.0 was to curate the dataset and prepare line-level annotations from the song level; the latter are essential in order to prepare the dataset for training. Timestamps can be obtained from the output of Whisper by aligning the lyrics both before and after ChatGPT processing. For the alignment of strings, the Levenshtein distance [Levenshtein et al., 1966] is employed. To exclude aligned lines of lower confidence, the distance is normalized, setting a threshold at 0.2. The quality of annotation is further enhanced through two subsequent filtering stages. In the first stage, lines that exhibit unusually high character rates, exceeding 37.5 Hz, are eliminated. The second stage encompasses another Whisper iteration; segments yielding a transcription of “Thank

⁴Whisper is a robust speech recognition model, described in Section 2.8.3

you.” are excluded. These segments, which typically represent instrumental sections, are believed to originate from Whisper’s training on data similar to video transcripts.

Following the construction process outlined above, we ultimately obtained a multilingual lyric transcription dataset, MulJam v1.0, consisting of 6,031 songs with 182,429 lines and a total duration of 381.9 hours in the training set. There are six languages in the dataset: English, French, Spanish, German, Italian and Russian. The test set contains 40 songs, for which are manually annotated at song level. The number of lines in the training and validation set can be found in Table 3.1.

This version of the dataset is used in training for our EUSIPCO paper [Huang and Benetos, 2024], which served as a preliminary study for the experiments presented in Chapter 6. To download MulJam v1.0 and prepare audio for ALT training, scripts are accessible online ⁵.

3.3.2 MulJam v2.0 Dataset

MulJam v2.0 is derived from MulJam v1.0, featuring larger and more refined human annotations on the test set. We selected 34 songs from the training set and obtained human lyrics annotations to expand the test set. Native speakers from our research lab were recruited. Each song is transcribed by one person. For each language, 20 songs are randomly selected from the original training set to form the validation set. A few songs are excluded due to poor alignment for obtaining the line-level annotations. We also exclude the songs in the training and validation sets that were present in Jamendo (3 songs in training and 1 song in validation), ensuring that the songs in the evaluation datasets remain unseen during training. The numbers of songs by language can be found in Table 3.1.

Table 3.1: Number of songs in MulJam1.0 and MulJam2.0 datasets. The relatively small number of test songs in some languages results from limited human resources for manual annotation.

Dataset Split	MulJam1.0			MulJam2.0		
	Train	Valid	Test	Train	Valid	Test
English (en)	3533	20	20	3557	20	28
French (fr)	979	20	7	977	19	19
Spanish (es)	585	19	5	584	19	13
German (de)	107	20	1	107	20	3
Italian (it)	278	20	3	278	20	7
Russian (ru)	106	16	4	106	16	4
Total	5588	115	40	5609	114	74

The human annotation is performed at the song level. We applied similar

procedures to obtain line-level annotations, as was done for the training set in MulJam v1.0. We use the timestamps provided by Whisper [Radford et al., 2023], and align the lines predicted by Whisper with the human annotation. As in Zhuo et al. [2023], lines with unusually high character rates (exceeding 37.5 Hz) are removed. However, for the test set we choose not to filter by the similarity between the aligned text pairs, to prevent introducing excessive bias in favor of Whisper predictions.

MulJam v2.0 can be downloaded and processed using the same repository ⁵. This version of dataset is used in Chapter 6, which is extended and updated from the EUSIPCO paper [Huang and Benetos, 2024].

3.3.3 Further Updates

Since the release of MulJam v2.0, its annotations were further refined with tools from company Music.AI, as described in Section 7.4.2. This version enhances the accuracy of lyrics annotations and timestamps, because the timestamps are obtained from alignment results, rather than indirect text alignment as for previous versions. This version of dataset is used in the work presented in Chapter 7. It can be accessed online ⁶.

3.4 Datasets: Word-Level Lyrics Extension for MUSDB18 test set

The commonly used lyrics alignment datasets, including Jamendo, Mauch, and Jensen, are relatively small, containing only 20, 20, and 10 songs, respectively ⁷. To support more robust evaluation and enable broader research on automatic lyrics transcription (ALT), we annotated the MUSDB18 test set, which consists of 45 professionally produced English songs spanning multiple genres. This work serves as an alignment extension of the *MUSDB18 lyrics extension* [Schulze-Forster et al., 2021], introduced in Section 3.2.

MUSDB18 [Rafii et al., 2017] is a widely used benchmark for source separation and already includes additional annotations such as pitch and vocal activity. The pitch annotations are provided as continuous f_0 , where each short time frame is assigned a frequency value in Hz. These represent monophonic melodic lines, which may not always correspond to a strictly monophonic audio source. Adding word-level timestamp annotations allows us to assess lyrics alignment on a more diverse set of tracks. This expansion could also facilitate

⁵<https://github.com/jhuang448/ALT-Preprocess>

⁶<https://github.com/weAreMusicAI/alt-datasets-interspeech2025>

⁷See Section 3.2 for details.

future research on the impact of background music and separation artifacts on alignment performance.

We hired a professional music producer to manually annotate word-level timestamps under my supervision. The annotator was provided with source-separated vocals (obtained using Spleeter [Hennequin et al., 2020]) and a preliminary alignment generated by the model from Huang et al. [2022] (described in Chapter 4). The task involved adjusting the start times of each word and editing the lyrics text when necessary to ensure alignment accuracy. The entire annotation process took approximately 80 hours. The resulting annotations are released on Zenodo ⁸ to support future research.

This dataset is used for evaluation in Chapter 4.

3.5 Benchmarking: Architecture and Training

In this section, we benchmark architecture and training strategies, focusing on end-to-end (E2E) models. As discussed in Section 2.8.3, most E2E ALT systems are adapted from ASR architectures and are typically trained with either CTC loss, sequence-to-sequence loss (seq2seq, essentially a cross-entropy loss), or a combination of both. Common model choices include Wave-U-Net, CRNNs, Transformers, and RNNs with attention components.

These modeling strategies are adopted throughout the chapters. Chapter 4 employs a CRNN trained with CTC loss for phoneme recognition. Chapter 6 adopts a Transformer-based architecture trained with a hybrid CTC and seq2seq loss for multilingual ALT. In Chapter 5, we train a phoneme predictor (CRNN) with CTC loss in the singing-to-speech model, and the ALT model to test data augmentation is a Transformer trained with hybrid loss. In Chapter 7, we use Whisper (also a Transformer architecture), trained with seq2seq loss only. In each case, architecture and training choices are guided by task objectives, resource constraints, and context.

3.5.1 Method

⁸<https://zenodo.org/records/15547046>

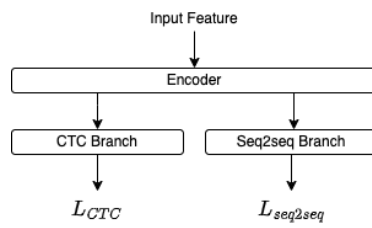


Figure 3.1: Diagram of ALT model architectures.

Model	Input Feature	Core Layers	Loss Type	Used In
CRNN	Mel-spectrogram	CNN + BiLSTM	CTC	Ch. 4, Ch. 5
Transformer	Mel-spectrogram	CNN + Transformer	CTC + Seq2Seq (hybrid)	Ch. 6, Ch. 5
Wav2Vec2 ALT	Wav2Vec2 features	Linear + GRU + Attn	CTC + Seq2Seq	Ch. 7

Table 3.2: Overview of the three ALT model architectures explored in this section, and the chapters where these models are applied.

We compare representative E2E architectures that are widely adopted in ALT modeling and used throughout this thesis. A detailed breakdown of the hyperparameters, including layer configurations and the loss weighting, will be provided in the following subsection. They can be represented by the general diagram shown in Figure 3.1. The input features are either Mel-spectrograms or Wav2Vec2 features, which are then processed by an encoder. It is followed by two branches (they can be optional depending on the purpose) for decoding. We outline the general architecture and design choices for each model in Table 3.2.

- **CRNN:** This architecture combines convolutional layers with bidirectional LSTM layers. It is well-suited for phoneme recognition tasks and was commonly used in early E2E ASR systems. We adopt this model in Chapter 4 for phoneme alignment and in Chapter 5 as part of the singing-to-speech system. Both are trained using the CTC loss. It is adaptable to the ALT task by adding a seq2seq branch, and thus fits into the general diagram shown in Figure 3.1.
- **Transformer:** This model follows the hybrid CTC/attention framework that has become standard in recent ASR systems. It includes a convolutional frontend, Transformer encoder and decoder blocks, and is trained with a weighted combination of CTC and seq2seq (cross-entropy) loss. We use this architecture in Chapter 6 for multilingual ALT and in Chapter 5 for evaluating data augmentation strategies.
- **Wav2Vec2 ALT:** This architecture uses a pre-trained Wav2Vec2 model (introduced in Section 2.3.4) as a frozen frontend, followed by a GRU-based decoder with location-aware attention. It produces outputs via both CTC and seq2seq branches. It is the same as the model proposed in Ou et al. [2022] except that the Wav2Vec2 frontend is frozen. We also use this setup as a baseline for benchmarking and as a case study for adapting ASR foundation models to ALT in Chapter 7.
- **Whisper:** Whisper [Radford et al., 2023] is also a Transformer model. However, it is pre-trained on a large amount of data, so it can work without a CTC branch. Whisper is not included in the architecture benchmarking comparison due to its model and pre-training scale, which makes it unsuitable for controlled studies on architectural choices and training strategies. Background information on Whisper in the context of ALT is provided in Section background:alt:related_work. We study this model in Chapter 7 for fine-tuning on music mixture data, where we have access to the required computational resources and efficient fine-tuning strategies.

3.5.2 Experiments

We begin by comparing the three architectures. This comparison allows us to evaluate how different modeling choices affect performance under consistent training conditions. We also compare character-level and phoneme-level CTC branches. Phoneme-level prediction is motivated by its success in prior work, including our lyrics alignment experiments (Chapter 4) and related literature [Vaglio et al., 2020b]. In addition, phoneme-level modeling is employed in the singing-to-speech chapter (Chapter 5) to align latent features across modalities. This setup enables us to assess the effectiveness of phoneme-level supervision for lyrics transcription.

Next, we conduct a focused study on the Wav2Vec2-based model. We assess the impact of fine-tuning the pre-trained Wav2Vec2 encoder on ALT performance. We also examine different loss strategies by comparing models trained with the CTC branch alone, the seq2seq branch alone, or both combined.

Below are the details of the three architectures we study, corresponding to the components illustrated in Figure 3.1.

CRNN

Input Feature: 80-dimensional Mel-spectrograms.

Encoder: The encoder consists of a convolutional block and an attention block. The convolutional block includes 5 Conv1D layers with group normalization, followed by 3 additional Conv1D layers to reduce the feature dimension. The attention block includes 3 Conv1D layers with LayerNorm, followed by a 6-layer attention module. Each attention block contains multi-head attention, Conv1D layers, and LayerNorm. A final Conv1D layer is applied after the attention stack.

CTC Branch: A linear layer, followed by 3 BiLSTM layers, and another linear layer that projects to the phoneme vocabulary. A log-softmax is applied before computing the CTC loss.

Seq2Seq Branch: A 6-layer Transformer decoder with causal attention. Decoder output is passed through a linear projection layer to obtain token probabilities.

Transformer

Input Feature: 80-dimensional Mel-spectrograms.

Encoder: A convolutional frontend with 3 Conv1D layers to process the input. A 12-layer Transformer encoder, where each layer includes multi-head attention and a position-wise feed-forward network.

CTC Branch: A linear layer projects the encoder output to the character vocabulary.

Seq2Seq Branch: A 6-layer Transformer decoder with causal attention and feed-forward layers. Decoder output is passed through a linear projection to obtain token probabilities.

Wav2Vec2 ALT

Input Feature: Raw audio waveform.

Encoder: A pre-trained Wav2Vec2 encoder ⁹ is used as the front-end. It remains frozen in benchmarking experiments unless specified (in Table 3.5). The output features are passed through a linear layer with non-linear activation.

CTC Branch: A linear layer projects the encoded features to the target vocabulary for CTC training.

Seq2Seq Branch: A single-layer GRU decoder with hidden dimension 1024 and location-aware attention with attention dimension 256. The GRU operates autoregressively, and the output is passed through a linear projection layer to compute token probabilities.

By default, all models are trained with a weighted combination of CTC and seq2seq losses. For detailed hyperparameters, optimizer, learning rate scheduling, and loss weight, please refer to corresponding chapters (Table 3.2). In some experiments, only one of the two losses is used. In some experiments, the char-level CTC branch is replaced with a phoneme-level CTC branch.

All models are trained and tested using the DSing30 [Dabike and Barker, 2019] split. During inference, beam search is used for all seq2seq decoding unless otherwise stated. For the one experiment that does not use the seq2seq branch, greedy decoding is used on the CTC branch.

3.5.3 Results and Discussion

Model	CTC Loss	Seq2seq Loss	Decoding	CER (%)	WER (%)
CRNN	char	char	seq2seq BS	31	58
	phone	char	seq2seq BS	27	55

Table 3.3: Results of CRNN model with different CTC branch types.

⁹Checkpoint: `large-960h-1v60-self`

Model	CTC Loss	Seq2seq Loss	Decoding	CER (%)	WER (%)
Transformer	char	char	seq2seq BS	18	42
	phone	char	seq2seq BS	18	43

Table 3.4: Results of Transformer model with different CTC branch types.

Model	Wav2Vec2	CTC Loss	Seq2seq Loss	Decoding	CER (%)	WER (%)
Wav2Vec2 ALT	freeze	char	char	seq2seq BS	11	19
	fine-tune	char	char	seq2seq BS	7	13
	freeze	phone	char	seq2seq BS	46	53
	freeze	no	char	seq2seq BS	44	50
	freeze	char	no	CTC Greedy	20	50

Table 3.5: Results of Wav2Vec2-based ALT model with different fine-tuning strategies, loss combinations, and decoding methods. Fine-tuned results (second row) are from Ou et al. [2022], while frozen-encoder results are from our experiments.

Architecture Comparison

Tables 3.3, 3.4, and 3.5 present the results of CRNN, Transformer, and Wav2Vec2 ALT models respectively, each with different loss strategies and CTC branch types. When comparing architectures trained with both char-level CTC and seq2seq losses, Wav2Vec2 ALT achieves the best performance, followed by Transformer and then CRNN. This ranking aligns with the historical development of ALT models: CRNN represents earlier approaches, while Wav2Vec2 reflects more recent advances (note that Whisper is excluded from this comparison). It is noteworthy, however, that the Transformer model outperforms Wav2Vec2 variants trained with single-branch losses or phone CTC, highlighting that training strategies and loss configurations can have a greater impact on performance than the architecture itself. These choices must be carefully tailored to the task.

CTC Branch Type Comparison

The Wav2Vec2 ALT model is particularly sensitive to the type of CTC branch (Table 3.5). In contrast, the CRNN and Transformer models, both trained from scratch, do not exhibit a consistent preference for either char-level or phoneme-level CTC. This difference may also reflect the influence of the Wav2Vec2 checkpoint’s original CTC-based fine-tuning on speech data (see Section 2.4.3 for more details). While either target type can yield slightly better results depending on the setup, aligning the CTC target with the pre-training objective appears advantageous when leveraging pre-trained features like Wav2Vec2.

Wav2Vec2 ALT with and without Fine-Tuning

Table 3.5 also includes a comparison of the Wav2Vec2-based ALT model with and without fine-tuning. Fine-tuning significantly improves both CER and WER, demonstrating the benefit of adapting the pre-trained model to in-domain data. This supports the discussion in Chapter 7, emphasizing the importance of fine-tuning foundation models for downstream tasks like ALT.

Wav2Vec2 ALT with Different Loss Strategies

Table 3.5 further reports the performance of the Wav2Vec2 ALT model under different loss configurations. When trained with only one of the two losses (CTC or seq2seq), WER increases dramatically, reaching up to 50%. This indicates the importance of the hybrid CTC/attention training strategy for ALT.

Interestingly, while beam search decoding is typically more effective than greedy decoding, the results here suggest that using either branch alone leads to similarly degraded performance. Additionally, training with only CTC loss achieves a lower CER than training with seq2seq loss alone. This could indicate that the seq2seq decoder struggles to resolve ambiguities in the encoded features, while greedy CTC decoding remains more robust at recovering individual characters.

Insights and Model Choices Across Chapters

Based on the results and discussions presented above, we draw several insights regarding effective architectural and training choices for automatic lyrics transcription (ALT). First, the Wav2Vec2 ALT model achieves the best performance among the evaluated architectures, though it requires fine-tuning to reach optimal accuracy. Second, the hybrid CTC/attention training strategy is crucial for achieving good performance. Third, the choice of CTC branch type can significantly impact performance, especially when using pre-trained features with a shallow backend.

The insights on Wav2Vec2 ALT informs our design choices for the MARBLE ALT task, which is discussed in the next section. While both systems leverage self-supervised learning (SSL) features, the SSL models used in MARBLE are trained on general-purpose music data, without any supervision from ASR or ALT tasks. As a result, the learned representations are less aligned with linguistic units such as characters, and thus a more complicated mapping is needed to obtain the final output. This leads to the use of a Transformer backend in MARBLE for MERT (an SSL feature for acoustic music understanding; see Section 2.4.3 for details), in Section 3.6.2.

It is important to note that Wav2Vec2 is not the only viable architecture for ALT. While performance is a key factor, it is not the only criterion for architectural choice. Long-term considerations such as computational cost, training and inference efficiency, development cycle, model interpretability, and ease of deployment are also critical. In this regard, the Wav2Vec2 model is the most resource-intensive, requiring significant data and compute for training and fine-tuning. The Transformer-based model presents a compelling alternative, offering competitive performance with faster iteration and reduced computational demands. This consideration leads to the choice of a Transformer architecture in Chapter 6. Although the CRNN model underperforms on full ALT tasks, it remains well-suited for alignment scenarios, where capturing fine-grained temporal structure is more critical than transcribing content accurately. The convolutional layers in CRNNs are effective at modeling local acoustic patterns, while the recurrent layers capture temporal dependencies with frame-level precision.

These findings also provide a rationale for the architectural choices made across chapters of this thesis. In Chapter 4, we employ the CRNN because its recurrent structure and efficient CTC training align well with the requirements of phoneme alignment, where temporal precision outweighs transcription accuracy. In Chapter 6, we adopt the Transformer with hybrid CTC/attention loss, as it balances strong performance with computational feasibility and is particularly suited to the multilingual setting. Chapter 5 combines both directions: a CRNN phoneme predictor with CTC loss is used within the singing-to-speech system for duration modeling, while a Transformer is employed to evaluate data augmentation strategies for ALT. Finally, Chapter 7 focuses on Whisper, reflecting the importance of adapting large-scale foundation models for ALT. Whisper is not directly included in the benchmarking here, but it already demonstrates a degree of robustness to singing, which allows us to concentrate on the more challenging problem of adapting it to music mixtures. Together, these insights and design choices reflect both the historical development of ALT modeling and pragmatic considerations of task requirements, computational cost, and interpretability.

3.6 Benchmarking: MARBLE

3.6.1 MARBLE Benchmark

MARBLE (Music Audio Representation Benchmark for universaL Evaluation) [Yuan et al., 2023] is a large-scale benchmark for evaluating self-supervised audio features on downstream music-related tasks. It comprises 18 MIR tasks across 12 publicly available datasets, grouped into four categories based on abstrac-

tion level: acoustic-level (e.g., source separation), performance-level (e.g., vocal technique detection), score-level (e.g., beat tracking, lyrics transcription), and high-level (e.g., music tagging). This benchmark provides a standardized framework for reproducible comparisons of self-supervised learning (SSL) methods in MIR.

Among the tasks, lyrics transcription is categorized as a score-level task within MARBLE. My contribution has been to develop and evaluate a multilingual lyrics transcription system within this benchmark. This section focuses on the specific implementation details and results in the MARBLE paper. For a broader background on ALT and sequence modeling, please refer to Section 2.8. The pre-trained SSL features involved in the ALT task are MAP-Music2Vec [Li et al., 2022b] and MERT variations [Li et al., 2022a, 2023b].

3.6.2 ALT in MARBLE

We perform the task using the MulJam v2.0 dataset, which comprises 6031 songs in 6 languages: English, French, Spanish, Italian, German, and Russian. This dataset offers a rich repository of around 153k lines with lyrics annotations. The training, validation and testing sets contain 147k, 3k and 2k lines respectively (see Section 3.3.2 for details). Additionally, we also report WERs on the Multi-Lang Jamendo dataset v1.0 [Durand et al., 2023]. This dataset consists of 80 songs in 4 languages: English, French, Spanish, and German. Further details on the MulJam and Multi-Lang Jamendo datasets are provided in Section 3.3.2 and Section 3.2.

For comparison, we refer to Whisper [Radford et al., 2023], a model trained on a large-scale, multilingual, and multitask supervised corpus collected from the internet. Whisper also contributed to the construction of the MulJam dataset. The potential biases introduced by Whisper are mentioned in Section 3.6.5.

Following the insights from Section 3.5, the backend adopts a hybrid CTC/Attention architecture design [Watanabe et al., 2017]. Given the task’s complexity and the necessity to capture long-term dependencies, we use a Transformer with 3 encoder layers and 3 decoder layers. The output from the encoder is further processed by a fully connected layer to map it to the target dimension for the character CTC loss computation. We also calculate a seq2seq loss between the output from the decoder and the true lyrics text. The final loss is a balanced combination of the CTC loss and the seq2seq loss. For validation and testing, we employ beam search on the Transformer decoder to iteratively select the best predictions. Additionally, a Transformer language model is trained from the same data split to incorporate language knowledge at test time. Performance evaluation is conducted using Character Error Rate (CER) and Word

Error Rate (WER).

3.6.3 Language Model and Tokenizer

The language model (LM) is trained using a speechbrain [Ravanelli et al., 2021] language model recipe ¹⁰. The model consists of 12 Transformer encoder layers, with an attention dimension of 768, 12 attention heads, and a position-wise feed-forward layer dimension of 3072. The LM is trained using cross-entropy loss for 20 epochs, and the model with the lowest loss is selected. More details on the LM training can also be found in Section 2.6.

The target character set is the union of the character sets from 6 languages, resulting in a total of 91 tokens: ϵ , <bos>, <eos>, <unk>, A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z, À, Á, Â, Ã, Ä, Å, Æ, Ç, È, É, Ê, Ë, Ì, Í, Î, Ï, Ñ, Ò, Ó, Ô, Õ, Ù, Ú, Û, Ü, Æ, ÿ, È, A, Б, В, Г, Д, Е, Ж, З, И, Й, К, Л, М, Н, О, П, Р, С, Т, У, Ф, Х, Ц, Ч, Ш, Щ, Ъ, Ы, Ь, Э, Ю, Я.

3.6.4 Training Details

The beam search used for validation and testing incorporates a combination of CTC probabilities, LM probabilities (applied only at test time), and seq2seq probabilities. We assign a weight of 0.4 to the CTC probabilities and 0.3 to the LM probabilities. During validation, we utilize a beam size of 10 and calculate WER every 5 epochs and monitor seq2seq output accuracy for early stopping and model selection. At test time, beam size is increased to 40 for more robust decoding.

3.6.5 Results and Discussion

¹⁰<https://github.com/speechbrain/speechbrain/blob/develop/recipes/LibriSpeech/LM/hparams/transformer.yaml>

Language Metric	English		French		Spanish		German		Italian		Russian		Whole	
	CER	WER	CER	WER	CER	WER	CER	WER	CER	WER	CER	WER	CER	WER
MulJam2.0 test														
MAP-Music2Vec [138]	54.7	79.2	58.3	90.9	43.2	83.7	63.4	99.5	53.0	91.9	101.6	125.6	56.4	87.8
MAP-MERT-v0-95M [Li et al., 2022a]	48.7	71.2	55.5	85.4	41.0	80.1	65.9	100.9	49.1	86.3	99.5	124.9	52.6	82.3
MAP-MERT-v0-95M-public [Li et al., 2022a]	49.0	71.2	55.3	85.4	39.0	76.6	63.5	99.9	50.3	90.3	104.7	129.3	52.5	82.7
MAP-MERT-v1-95M [Li et al., 2023b]	45.5	66.5	52.5	81.9	38.2	73.9	58.8	93.2	44.4	81.6	96.1	117.8	49.4	77.9
MAP-MERT-v1-330M [Li et al., 2023b]	45.5	65.9	50.7	79.6	35.9	71.9	58.3	93.1	42.4	80.3	100.5	125.5	48.5	77.0
SOTA [Radford et al., 2023]	<u>33.2</u>	<u>44.8</u>	<u>52.9</u>	<u>70.1</u>	<u>29.9</u>	<u>43.8</u>	<u>36.5</u>	<u>53.0</u>	<u>38.1</u>	<u>58.5</u>	<u>34.7</u>	<u>53.7</u>	<u>39.5</u>	<u>54.8</u>
Jamendo														
MAP-Music2Vec [138]	49.0	73.6	55.3	87.1	50.3	90.7	67.8	108.8	-	-	-	-	55.7	89.6
MAP-MERT-v0-95M [Li et al., 2022a]	48.5	71.8	54.0	85.1	49.3	87.6	67.6	108.1	-	-	-	-	54.8	87.6
MAP-MERT-v0-95M-public [Li et al., 2022a]	46.9	71.5	52.0	81.5	44.8	82.8	66.3	106.8	-	-	-	-	52.6	85.2
MAP-MERT-v1-95M [Li et al., 2023b]	43.6	67.2	49.4	79.6	43.2	80.6	62.1	103.3	-	-	-	-	49.6	82.2
MAP-MERT-v1-330M [Li et al., 2023b]	45.7	68.8	50.2	80.1	44.1	82.8	61.0	102.3	-	-	-	-	50.3	83.1
SOTA [Radford et al., 2023]	<u>24.9</u>	<u>39.3</u>	<u>29.2</u>	<u>49.9</u>	<u>21.2</u>	<u>41.7</u>	<u>25.8</u>	<u>46.6</u>	-	-	-	-	<u>25.4</u>	<u>44.4</u>

Table 3.6: Multilingual lyrics transcription results on MulJam v2.0 and Multi-Lang Jamendo v1.0.

The results of multilingual lyrics transcription using different pre-trained features can be found in Table 3.6. Overall, lyrics transcription remains a challenging task, especially under multilingual and music mixture conditions. Having a strong performance requires the model to isolate singing voice and apply appropriate language knowledge to decode meaningful text. Achieving this with SSL features is a significant challenge. The WERs around 80% or more in Table 3.6 indicate that there is much room for improvement in this task.

Among the six languages we considered, English, French, and Spanish, which have a larger number of songs than the other three, yield better results. This suggests that there may be an impact from the imbalanced training data. Russian, on the other hand, produces the worst result for two main reasons: First, Russian employs the Cyrillic writing system, which has its own set of characters. Second, the training data for Russian is insufficient for the model to establish a connection between the pronunciation rules of Cyrillic and Latin alphabets.

The MulJam v2.0 test set is human-annotated at the song level but relies on the alignment with Whisper results to derive line-level annotations (described in Section 3.3.2). Therefore, it is worth noting that bias is introduced, as the alignment is reliable only when the human annotation closely matches Whisper’s prediction.

This experiment marks the first evaluation in this thesis on data that is both multilingual and includes music mixtures, which are two difficult dimensions. The high error rates highlight the difficulty of applying SSL models to these settings. From here, we break down the problem and explore directions to improve on each aspect in the following chapters: In Chapter 6, we focus on the multilingual aspect and provides effective guidelines for developing a multilingual ALT model. In Chapter 7, we explore the potential of adapting foundation models (including SSL and Whisper) to ALT on music mixture data.

3.7 Inconsistencies in Datasets and Evaluation

We are aware of some inconsistencies regarding datasets and evaluation in this and the following chapters. We summarize these inconsistencies in Table 3.7.

Different versions of datasets and SVS tools are used for training and evaluation. This is due to the availability when the work was conducted. As mentioned in Section 3.2 and Section 3.3, some datasets, including MulJam and Multi-Lang Jamendo, have gone through multiple updates. These version changes directly impact the training and evaluation over time.

Different granularity of evaluation is used. Some evaluate on line-level, while others evaluate on song-level. This is due to the purpose of the evaluations. When we are more interested in models’s ability to understand the lyrics, es-

pecially when the results are not so good, we evaluate on line-level to avoid introducing additional complexity (Chapter 5 and 6). When we are more interested in the application, or we have a robust pipeline to merge line-level results into song-level, we evaluate on song-level (Chapter 7).

3.8 Conclusion

This chapter has established the foundation for the rest of the thesis by addressing two key pillars of ALT: data and model design.

On the data side, we introduced the MulJam dataset and its updates, which serve as the primary data source for ALT training in later chapters. We also discussed our annotation efforts for the MUSDB18 test set, aiming to assess lyrics alignment with different background music conditions. This dataset is used for evaluation in the next chapter.

On the model side, we conducted a series of benchmarking experiments to explore the impact of various architectural and training choices on ALT performance. These experiments provide insights into the design of effective models for the task, which directly inform our decisions in later chapters. For E2E ALT models, the Wav2Vec2-based model outperforms others. The hybrid CTC/attention architecture is more effective than single-branch. We also found that there is no clear winner between char-level and phoneme-level CTC branches, but it might be better to align with the pre-training task when pre-trained features are used.

ALT in MARBLE benefits from both the development of MulJam dataset and the insights gained from our model design experiments. Our work ensures that the ALT task is properly integrated into the MARBLE framework, with multilingual ALT as a core component, providing an accurate evaluation of model performance across different languages and music mixtures.

These findings serve as a critical foundation for the subsequent chapters. They inform the model architectures, training strategies, and evaluation protocols used in our lyrics alignment and ALT experiments. This chapter provides an overview of the available options and the rationale for selecting them in different scenarios. It also highlights the limitations and biases that must be considered when reviewing results.

Chapter or Section	Training set	Test set	Test set granularity	SVS
Section 3.5	DSing	DSing	Line	NA
Section 3.6	MulJam v2.0	MulJam v2.0, Multi-Lang Jamendo v1.0	Line	NA
Chapter 4	DALI v2.0	Multi-Lang Jamendo v1.1, Mauch MUSDB18 test	Word and Line (for alignment) Word (for alignment)	Spleeter [Hennequin et al., 2020] Open-Unmix [Stöter et al., 2019] HTDemucs [Rouard et al., 2023]
Chapter 5	DSing, TTS-generated DSing	NHSS (for conversion)	Line (for conversion)	NA
Chapter 6	DALI v2.0, MulJam v2.0	DSing	Line	Open-Unmix [Stöter et al., 2019]
Chapter 7	DALI v2.0 and MulJam v2.0 refined with Music.AI	Multi-Lang Jamendo v1.1	Song	Music.AI ^a

Table 3.7: Overview of datasets model training and evaluation granularity in this thesis. Unless specified, the default task is ALT. The SVS column indicates the Singing Voice Separation tool used for the task. NA indicates that the task does not involve SVS.

^a<https://music.ai/blog/research/source-separation-benchmarks/>

Chapter 4

The Role of Pitch: Incorporating Musical Attributes

4.1 Introduction

Lyrics alignment and automatic lyrics transcription (ALT) were historically approached using the same acoustic modeling frameworks. As discussed in Section 2.7.3 and Section 2.8.3, lyrics alignment was initially seen as a simpler task of ALT and both were often tackled with conventional ASR techniques. With the rise of end-to-end (E2E) methods later, the two tasks began to diverge, leading to separate research paths.

The work presented in this chapter was one of the earliest projects of this PhD and was conducted shortly after the first E2E models for lyrics alignment and ALT were proposed [Stoller et al., 2019a, Vaglio et al., 2020b]. We chose to focus on lyrics alignment due to its more constrained nature, which made it a suitable starting point for exploring new modeling techniques. Additionally, this work was motivated by the observation that phoneme and pitch onsets often occur around the same time. Since lyrics alignment explicitly evaluates temporal precision, it provides a more suitable setting for investigating the impact of our method than ALT. At the same time, we acknowledge that this decision was also influenced by the annotations available in the dataset, which included continuous pitch trajectories. While pitch was both conceptually and practically appropriate, other temporally precise annotations (such as note-level or vocal effort markings) could also have served a similar role had they been available.

One major challenge in training E2E models for lyrics alignment is the lack

of fine-grained annotations. A frame-level annotation of lyrics is hard to obtain. Some previous works trained a Gaussian mixture hidden Markov model (GMM-HMM) to predict the frame-level annotation and take it as the ground truth for training [Demirel et al., 2020, Gupta et al., 2019]. Others adopted the connectionist temporal classification (CTC) loss [Graves et al., 2006] and trained the model in an E2E way [Stoller et al., 2019a, Vaglio et al., 2020b]. However, the CTC loss is a weaker form of supervision as it only enforces that a symbol is observed but does not specify when [Liu et al., 2018].

In this chapter, we explore multi-task learning [Caruana, 1997] for lyrics alignment, where the auxiliary task is pitch detection. For more details on multi-task learning, please refer to Section 2.4. At first this seems to be an unusual combination as in traditional singing synthesis, pitch and lyrics are modeled as two independent attributes and thus it might not be possible to share many feature representations between the two tasks — an aspect that typically drives which tasks in multitask learning are combined. However, in our approach, the auxiliary pitch task is defined using the hierarchical annotations from the DALI dataset, where note-level pitches are aligned with word-level onsets. This provides frame-level pitch supervision that coincides with phoneme timings, implicitly constraining the temporal alignment of the lyrics predictions in multi-task learning. In other words, by adding a frame-level loss function through multi-task learning, it implicitly applies a stronger restriction about the timing as many of the phonemes and pitches share onsets. Additionally, previous studies have shown that joint learning of unrelated tasks might lead to sparser and more informative representations [Romera-Paredes et al., 2012], which could lead to improved accuracy as well. Hung et al. [2019] proposed a multi-task learning method for instrument detection which models pitch and instrument. Inspired by their work, we propose a model that outputs a representation for both pitch¹ and phoneme, where the task-wise loss can be computed by applying pooling along the other axis. We favor this method over a multi-head architecture commonly used in multi-task learning as the pitch information is kept close to the phoneme information all the way.

In real-world applications such as generating karaoke-style scrolling lyrics, cross-line misalignment is less tolerable than inline errors, and should therefore be avoided. Just as lyrics can be segmented line by line, the corresponding cut points in audio can be regarded as line-level boundaries, which can be estimated by a network. Previous works in audio-to-score alignment show that onset information can enhance the temporal precision by adding an onset term to the cost function [Ewert et al., 2009]. Similarly, we propose to incorporate the

¹In this study, pitch detection refers to the quantized pitch (as typically specified as the pitch of a note in MIDI) as opposed to the fundamental frequency.

boundary probability of audio frames into the alignment algorithm to further improve the alignment performance.

This is the first attempt to apply multi-task learning to lyrics alignment. Although multi-task learning has been extensively explored in ASR [Yang et al., 2018, Kim et al., 2017, Xiao et al., 2018], we choose a musically meaningful auxiliary task to take advantage of the correlation between lyrics and pitch [Nichols et al., 2009, Gong et al., 2015]. This is also the first time an audio-to-lyrics alignment system incorporates boundary information into the forced-alignment.

Initially, our method was tested on English datasets. This part of the work was published in the proceedings of International Conference on Acoustics, Speech, and Signal Processing (ICASSP) 2022 [Huang et al., 2022]. Following the release of the Multi-Lang Jamendo dataset [Durand et al., 2023], we further extend the method to multilingual lyrics alignment by expanding the phoneme vocabulary and training on a larger dataset. This allows us to evaluate the generalizability of the proposed approach under more challenging multilingual conditions. Interestingly, while our method proves effective in English, it is not effective in the multilingual setting. (We further explore multilingual adaptation in Chapter 6, which explores the design of multilingual ALT systems.) Finally, we investigate the sensitivity of our models to different source separation tools using the word-level annotation extension of MUSDB18 test set we introduce in Section 3.4.

The chapter is structured as follows: The proposed architecture and loss computations are discussed in Section 4.2. The proposed method to incorporate boundary information is explained in Section 4.3. The dataset, baseline and experimental settings are described in Section 4.4. The results for English models are presented and discussed in Section 4.5, followed by multilingual results in Section 4.6. We further evaluate our models on the MUSDB18 test dataset in Section 4.7 and compare across source separation tools. Finally, we conclude and reflect on the role of pitch in Section 4.8.

4.2 Joint Phoneme Recognition and Pitch Detection

A general overview of the various modules involved in our system is given in Figure 4.1 — our proposed extensions and their details will be discussed below. It consists of a singing voice separation module, an acoustic model, a grapheme to phoneme (g2p) module, and the alignment algorithm. The lyrics are first converted to a phoneme sequence via g2p (Section 4.4.1). The singing voice sep-

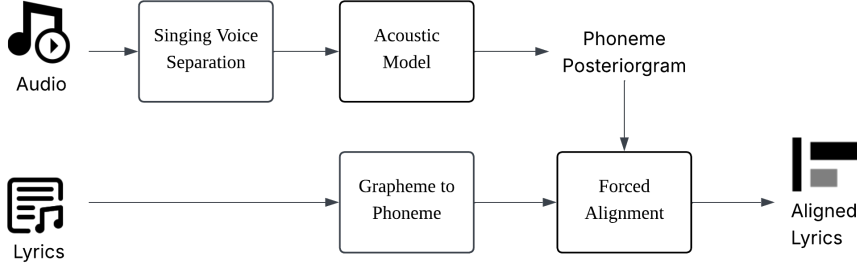


Figure 4.1: System overview.

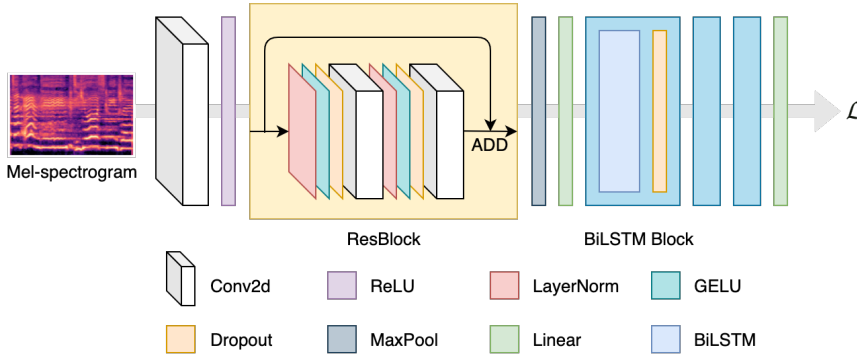


Figure 4.2: The network architecture for the baseline, boundary detection, and the MTL models. The only differences among them are the dimensions of the BiLSTMs and the last fully-connected layer.

aration model extracts the vocals to remove the effect of the background music (Section 4.4.1). The acoustic model takes the Mel-spectrogram of the separated vocals as input, and produces the phoneme posteriorgram (Section 4.2.1). Then the forced-alignment algorithm is applied on the posteriorgram and the corresponding lyrics (Section 4.3).

4.2.1 Acoustic Model

The acoustic model takes a 128-bin Mel-spectrogram with a fixed duration as input. The sampling rate is 22050 Hz, the Fast Fourier Transform (FFT) size is 512 and the hop size is 256. To fix notation, let $X = \{x_1, x_2, \dots, x_m\}$ be the frames of the input Mel-spectrogram, and $L = \{l_1, l_2, \dots, l_n\}$ be the phoneme sequence for the corresponding lyrics, where m and n are the lengths of the Mel-spectrogram and the phoneme sequence. The size of the phoneme set $\mathcal{S}_{\text{phone}}$ is N_{phone} , and the size of the pitch set $\mathcal{S}_{\text{pitch}}$ is N_{pitch} .

Figure 4.2 shows the network architecture. We employ a CRNN architecture for phoneme recognition (see Section 3.5), chosen for its recurrent structure and

efficient CTC training, which provides precise temporal alignment suitable for the lyrics alignment task. It consists of a convolutional layer, a residual convolutional block, a fully-connected layer, 3 bidirectional LSTM (Long Short-Term Memory) layers, a final fully-connected layer, and non-linearities in between. The kernel size of the convolutional layers is 3×3 , with stride and padding equal to 1. The number of filters for the three convolutional layers is 32. The dimensions of the bidirectional LSTMs (BiLSTM) are 256. Layer normalization is applied on the feature dimension, with a mini-batch size of 128. All dropout rates are set to 0.1. The last fully-connected layer is time-distributed (applied to each frame), with a target size of $N_{\text{phone}} \times N_{\text{pitch}}$. After that, the output is reshaped to an order-3 tensor $\mathcal{D}$ of size $N_{\text{time}} \times N_{\text{phone}} \times N_{\text{pitch}}$, where N_{time} is the output frame number. This representation $\mathcal{D}$ can be considered as the joint probability distribution before a **softmax** operation. This model is later referred to as **MTL** (Multi-Task Learning).

4.2.2 Loss Function

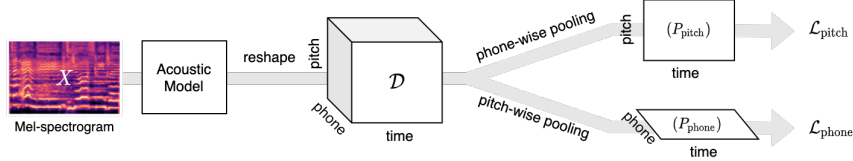


Figure 4.3: Multi-task loss computation.

Figure 4.3 describes the steps to compute the loss functions. Average pooling is applied on $\mathcal{D}$ along the pitch axis to get the (log-) posteriorgram of phoneme P_{phone} of size $N_{\text{time}} \times N_{\text{phone}}$, where the CTC loss [Graves et al., 2006] $\mathcal{L}_{\text{phone}}$ is computed. The same is applied along the phoneme axis to get the pitch posteriorgram P_{pitch} of size $N_{\text{time}} \times N_{\text{pitch}}$ and compute the frame-level cross entropy loss $\mathcal{L}_{\text{pitch}}$. The final loss is a weighted sum of these two:

$$\mathcal{L} = \mathcal{L}_{\text{phone}} + \lambda \cdot \mathcal{L}_{\text{pitch}} \quad (4.1)$$

where λ is an adjustable parameter. We tested different values in $\{0.5, 0.8, 1.0, 1.2, 1.5\}$ and found $\lambda = 0.5$ to yield the best results and thus we used this setting in our experiments below.

The baseline model shares the same architecture except for the dimension of the last fully-connected layer, but focuses on phoneme recognition only. The output is a $N_{\text{time}} \times N_{\text{phone}}$ posteriorgram, where the CTC loss is computed. This model is referred to as **Base**.

4.3 Alignment

4.3.1 Viterbi Forced Alignment

In the calculation of the CTC loss [Graves et al., 2006], the probabilities of all possible alignment paths are accumulated. These paths are generated by inserting blank (ϵ) and repeated labels to the original sequence. By applying the same rules to the lyrics to be aligned at inference time, the best path can be decoded via Viterbi forced alignment [Forney, 1973].

To be more specific, the phoneme sequence is expanded by inserting ϵ symbols in between the phonemes to $L^* = \{\epsilon, l_1, \epsilon, l_2, \dots, \epsilon, l_n, \epsilon\}$. Let $P_{\text{phone}}(t, p)$ be the log-probability of the input frame x_t being a phoneme $p \in \mathcal{S}_{\text{phone}}$. A path $\{(t_q, l_q^*)\}$ is a sequence of index pairs indicating x_{t_q} is aligned to l_q^* . The target is to find the path that maximizes the score:

$$\operatorname{argmax}_{\text{path}} \sum_{(t, l^*) \in \text{path}} P_{\text{phone}}(t, l^*) \quad (4.2)$$

The solution can be computed efficiently via dynamic programming.

4.3.2 Incorporating Boundary Information

A boundary detection model is trained independently to predict the line-level boundary probability. The input and the network architecture are the same as the baseline except that it has a smaller BiLSTM dimension 32 and output size is 1 per time step. The target label is a boundary activation curve (range: 0 $\sim$ 1). The start time of a line of lyrics is considered a boundary event. Each event is converted to a Gaussian window centered at the event time, at a size of 0.7 sec. Let $P_{\text{bdr}}(t)$ be the boundary log-probability of x_t predicted by the model.

We propose to add the boundary probability to the Viterbi score as a bonus at the line beginning of the lyrics, which is a ‘boundary’ in the text. This is to encourage aligning a phoneme at the beginning of a line to a boundary-like audio frame. Equation (4.2) is updated as follows:

$$\operatorname{argmax}_{\text{path}} \sum_{(t, l^*) \in \text{path}} P_{\text{phone}}(t, l^*) + \alpha \sum_{\substack{(t, l^*) \in \text{path} \\ l^* \text{ is a line start}}} P_{\text{bdr}}(t) \quad (4.3)$$

where α is an adjustable weighting parameter. We test different values in $\{0.5, 0.8, 1.0, 1.2, 1.5\}$ and found $\alpha = 0.8$ to yield the best results and thus we used this setting in our experiments below. This alignment method is referred to as **BDR** (**BounDaRy**) later in text.

4.4 Experiments

This section describes the dataset, training, and evaluation settings for the proposed models. We first test the proposed method on English data. We evaluate the model for lyrics alignment, boundary detection, and pitch detection. After a multilingual lyrics alignment evaluation set (Multi-Lang Jamendo [Durand et al., 2023]) is released, we train new models on multilingual data and evaluate the performance on the multilingual lyrics alignment task. To discriminate the language settings, multilingual models have a postfix **-Mul** in their names, while English models do not.

4.4.1 Dataset

All models are trained on the DALI v2 dataset [Meseguer-Brocal et al., 2020]. There are 7756 songs in total with word-level lyrics annotations and note-level pitch annotations. For the English experiments, we use the English subset according to the language label. The training set contains 4224 songs and the validation set contains 1056 songs for the baseline and the multi-task model ².

For the multilingual experiments, we use the song split created for multilingual lyrics transcription task described in Chapter 6 ³. The training set contains 6232 songs and the validation set contains 100 songs (20 each in 5 languages: English, French, Spanish, German, and Italian). The samples are generated by applying a 5.6 sec sliding window with a hop size of 2.8 sec. The target lyrics for an audio segment are the words fully covered within the window.

The boundary detection model is trained using the same English data, except that a subset of the English validation set is left out for evaluation. The subset contains 45 songs. It is not re-trained on the multilingual data because the boundary detection model detects line-level boundaries, which should be language-independent.

For lyrics alignment in English, we run evaluations on the Jamendo English set [Ramona et al., 2008, Stoller et al., 2019a] and the Mauch [Mauch et al., 2012] dataset. Each of them has 20 Western pop songs and word-level timestamps and boundary annotations. They are also used in the MIREX ⁴ lyrics alignment challenges.

For multilingual lyrics alignment, we evaluate the model on the Multi-Lang Jamendo v1.1 dataset [Durand et al., 2023], which is a multilingual extension of the above Jamendo English set. It contains 20 songs in English (EN), 19

²The code can be accessed through the link: <https://github.com/jhuang448/LyricsAlignment-MTL>

³<https://github.com/jhuang448/ALT-Preprocess/tree/main/input/preconstructed-split>

⁴<https://www.music-ir.org/mirex/>

songs in French (FR), 20 songs in Spanish (ES), and 20 songs in German (DE). Models trained on multilingual data are evaluated on this set.

In addition, we evaluate both English and multilingual models on the MUSDB18 test set (introduced in Section 3.4), which contains 45 songs across various genres. While MUSDB18 was originally designed for source separation, we use it here to test alignment performance under different vocal separation conditions. We evaluate using both the original vocal stems (clean vocals) and source-separated vocals via Spleeter [Hennequin et al., 2020], Open-Unmix [Stöter et al., 2019], and HTDemucs [Rouard et al., 2023].

For pitch detection, we evaluate the MTL model using the RWC Music Database - Popular Music [Goto, 2006]. It contains 94 popular songs (74 Japanese and 20 English) ⁵.

Vocals are extracted by Spleeter [Hennequin et al., 2020] for English experiments and Open-Unmix [Stöter et al., 2019] for multilingual experiments mentioned above before feeding to the network ⁶.

The open-source g2p tool ⁷ is used to convert the English lyrics to phoneme sequences. The phoneme set follows the convention of the CMU pronouncing dictionary ⁸ and has a size of 39. In practice, we added the space `_` and the epsilon ϵ (for the CTC loss) to the phoneme set so that the number of classes N_{phone} is 41. For multilingual experiments, we use Phonemizer [Bernard and Titeux, 2021] instead for multilingual support, following Vaglio et al. [2020b]. The phoneme set has a size of 69. We add the space `_`, the epsilon ϵ , and the unknown phoneme $\langle unk \rangle$ to the phoneme set, so that final number of classes N_{phone} is 72.

The target pitch range is D2-B5, therefore N_{pitch} is 47 (with one additional class for silence).

4.4.2 Training

All models are trained using a learning rate of $1e^{-4}$ and the ADAM optimizer [Kingma and Ba, 2015]. Early stopping is adopted when the validation loss does not decrease for 20 epochs after the 20th epoch. We employed this strategy as we found some early checkpoints to produce a relatively low loss value but which turned out to be underfitting. The boundary detection model is trained with binary cross entropy loss, and we do not wait for 20 epochs before early-stopping.

⁵The dataset contains 100 songs, but 6 of them (No. 3, 5, 8, 10, 23, and 66) are removed because they have multiple singers.

⁶Multilingual experiments are conducted later, when Open-Unmix becomes widely-used, causing this inconsistency in source-separation tools.

⁷<https://github.com/Kyubyong/g2p>

⁸<http://www.speech.cs.cmu.edu/cgi-bin/cmudict>

4.4.3 Evaluation

For lyrics alignment at the word level, we report the Average Absolute Error (AAE) [Mesaros and Virtanen, 2008] and Percentage of correct onsets with a tolerance window of 0.3 seconds (PCO) [Mauch et al., 2012]. These metrics are explained in Section 2.7.2. In addition to the word-level evaluation, we also report the line-level results to demonstrate the benefit from boundary information.

For boundary detection, we compute the precision, recall, and F-measure. A predicted boundary is considered a hit if there is a reference boundary within ± 0.5 seconds [Turnbull et al., 2007]. We also report the Area Under the Receiver Operating Characteristic Curve (AUC-ROC) between the boundary activation curve and the ground truth.

For pitch detection, we compute 3 F-scores (COn, COnP, and COnPOff) originally proposed in Molina et al. [2014]. COn only evaluates the onset times, COnP evaluates both onsets and the pitches, and COnPOff evaluates onsets, pitches, and offsets. The onset tolerance is 50ms, while the offset tolerance is $\max(50\text{ms}, 0.2 \cdot \text{note duration})$. Since the ground truth of RWC Pop contains many octave errors, instead of evaluating the absolute pitch, we compute the metrics on an octave-wrapped pitch with a tolerance of 50 cents.

4.5 Results and Discussion

Dataset	SS	FW	Jamendo		Mauch	
Metric			AAE	PCO	AAE	PCO
SDE2 [Stoller et al., 2019a]	Y	E2E	0.39	0.87	0.26	0.87
GC [Gupta et al., 2020]	N	STD	0.22	0.94	0.19	0.91
VHM [Vaglio et al., 2020b]	Y	E2E	0.37	0.92	0.22	0.91
DSE M6 [Durand et al., 2023]	N	ContrL	0.15	0.92	-	-
KGLW [Kick et al., 2025]	N	ContrL	0.20	0.94	-	-
Base	Y	E2E	0.31	0.94	0.20	0.89

Table 4.1: Comparison with the state-of-the-art systems for lyrics alignment. The **SS** column indicates if the input has been source-separated (Yes or No). The **FW** column indicates if the method takes an end-to-end (E2E), standard (STD) ASR, or contrastive learning (ContrL) framework. The dashed line indicates when our ICASSP paper [Huang et al., 2022] was published.

Metric	Precision	Recall	F-score	AUC
BDR	79.3%	55.9%	64.3%	89.9%

Table 4.2: Boundary detection results.

Metric	COnPOff	COnP	COn
pYIN [Mauch and Dixon, 2014]	5.9%	12.2%	43.5%
MTL	4.9%	16.3%	32.6%

Table 4.3: Pitch detection results.

4.5.1 English State of the Art Comparison

The comparison of our baseline with state-of-the-art lyrics alignment systems is listed in Table 4.1. Both GC [Gupta et al., 2020] and VHM [Vaglio et al., 2020b] are trained on DALI v1, while VHM takes an end-to-end approach on separated vocals and GC uses a standard ASR framework trained on polyphonic audio. SDE2 [Stoller et al., 2019a] is a Wave-U-Net model trained on an internal dataset consisting 44,232 songs. It is also trained on separated vocals. DSE M6 [Durand et al., 2023] and KGLW [Kick et al., 2025] are more recent methods trained with contrastive learning framework, published after the English experiments were conducted. More details on related work are discussed in Section 2.7.3.

The results show that our English baseline performs similarly to the state-of-the-art back then (2022) on both Jamendo and Mauch. Among the E2E models trained on separated vocals, our baseline outperforms SDE2 and VHM on all metrics except PCO on Mauch. Besides the network architecture, the key difference is the alignment unit. In our baseline, the CTC loss and the alignment are computed on phoneme sequences, while SDE2 and VHM are computed on characters. This further verifies that using phonemes over characters for alignment has better performance [Vaglio et al., 2020b].

4.5.2 Boundary Detection and Pitch Detection

The boundary detection results are listed in Table 4.2. An AUC of 89.9% indicates that the system performs well as a boundary detector. However, it is hard to provide comparison with existing systems because most of them detect structural boundaries (segmentation).

The pitch detection results of the MTL model are listed in Table 4.3. For comparison, we also list the pYIN [Mauch and Dixon, 2014] (note mode) results.

While pYIN achieves lower error rates for COnPOff and COn, the MTL model shows comparable performance for onset timing (COnP). This suggests

that although the MTL model is not optimized for general pitch detection, the onset-focused supervision from the hierarchical pitch annotations in DALI still provides useful temporal information. These onset cues are particularly valuable for lyrics alignment, supporting the use of pitch as an auxiliary task in the multi-task learning framework. The limited overall performance is likely due to the distribution mismatch between the training set (mostly Western music) and the testing set (80% Japanese songs).

The predicted pitch results and the ground truth on a short clip from Jamendo are visualized in Figure 4.4. It can be observed that our pitch tracker generally follows the singing voice, but produces some short notes that should be merged.

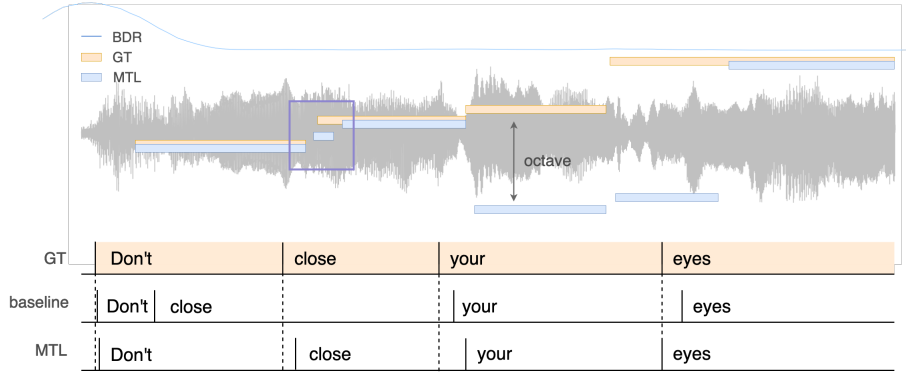


Figure 4.4: Lyrics alignment, pitch detection, and boundary detection results on a clip of `Average_-_Embers.mp3` from Jamendo. The GT (ground truth) notes are slightly shifted for visualization purposes. Two notes have the same pitch if there is an overlap. The MTL model successfully associates the onset of the word “close” with the corresponding note change.

Level	Word				Line			
	Jamendo		Mauch		Jamendo		Mauch	
Metric	AAE	PCO	AAE	PCO	AAE	PCO	AAE	PCO
Base	0.31	0.94	0.20	0.89	0.30	0.93	0.37	0.87
Base + BDR	0.29	0.94	0.20	0.90	0.25	0.95	0.33	0.88
MTL	0.23	0.94	0.21	0.90	0.27	0.93	0.43	0.86
MTL + BDR	0.23	0.94	0.20	0.91	0.25	0.94	0.42	0.89

Table 4.4: English lyrics alignment results. Substantial improvement in AAE on Jamendo for the MTL model can be observed for word-level metrics. All line-level metrics increase after introducing boundary information.

4.5.3 MTL Evaluations

Results of different combinations of Baseline / MTL with BDR are listed in Table 4.4. Comparing with the baseline, the improvement is significant in AAE on Jamendo for the MTL model, but limited in the other metrics. As can be observed in Table 4.1, results on Jamendo are worse than those on Mauch. This is probably due to more slurred pronunciation in Jamendo [Stoller et al., 2019a]. We argue that our MTL model alleviates this problem by estimating the pitch at the same time. Since pitch and phoneme often share onsets, a change in pitch can be a good indicator of a change in phoneme as well. The lyrics alignment results on the same clip are presented in Figure 4.4. Comparing to the baseline, MTL is able to place the word “close” at the right time with the help of changes detected in pitch (within the purple frame).

Though the benefit from adding boundary information is marginal in word-level metrics, it is clear in line-level. Both metrics on Jamendo and Mauch are better with BDR.

4.6 Multilingual Lyrics Alignment Results and Discussion

Level	Word				Line			
Dataset	Overall		English		Overall		English	
Metric	AAE	PCO	AAE	PCO	AAE	PCO	AAE	PCO
Base-Mul	0.15	0.96	0.21	0.95	0.26	0.94	0.22	0.94
Base-Mul + BDR	0.15	0.96	0.21	0.95	0.26	0.94	0.21	0.94
MTL-Mul	0.17	0.96	0.24	0.95	0.23	0.95	0.26	0.93
MTL-Mul + BDR	0.16	0.96	0.24	0.95	0.23	0.95	0.26	0.94

Table 4.5: Multilingual lyrics alignment results on the Multi-Lang Jamendo dataset (**Overall**) and the English subset (**English**). The English subset is equivalent to the Jamendo (English) set in Table 4.4.

4.6.1 Results on English Subset

Table 4.5 shows the results of models trained on multilingual data and evaluated on the Multi-Lang Jamendo dataset. Compared to models trained on English data (Table 4.4), the multilingual baseline model performs better in word-level metrics on the English subset. However, the multilingual MTL model does not show notable improvement over the English MTL model, and in fact, performs worse than the multilingual baseline.

4.6.2 MTL Evaluations

For the overall multilingual results, the MTL model also does not outperform the baseline in word-level metrics, though it achieves slightly better results in line-level metrics. Incorporating boundary information leads to consistent but marginal improvements, with limited impact on the final scores (limited to the third decimal place therefore not visible from the numbers).

Overall, unlike in the English experiments, the multilingual MTL model does not surpass the baseline. This may be attributed to the larger size of the multilingual training set. In this setting, the benefit of multi-task learning becomes less significant, potentially functioning more as a regularization mechanism. In some cases, this can lead to a slight performance drop. We further discuss the challenges of extending English models to multilingual lyrics transcription in Chapter 6, which is closely related to the findings here.

Model	Overall ⁹		EN	ES	DE	FR ⁹
	AAE	PCO	AAE	AAE	AAE	AAE
DSE M6 EN	0.35	0.89	0.15	0.37	0.41	0.47
DSE M6 Lang-Cond.	0.18	0.94	0.21	0.13	0.16	0.19
Base-Mul	0.15	0.96	0.21	0.14	0.08	0.15
Base-Mul+BDR	0.15	0.96	0.21	0.14	0.08	0.15
MTL-Mul	0.17	0.96	0.24	0.12	0.09	0.21
MTL-Mul+BDR	0.16	0.96	0.24	0.12	0.09	0.20

Table 4.6: Multilingual lyrics alignment results on the Multi-Lang Jamendo dataset, with comparison to the DSE models in Durand et al. [2023]. The **Overall** column is the overall result, while the other columns are the results on each language subset.

4.6.3 Multilingual State of the Art Comparison

Table 4.6 presents a comparison with the DSE M6 models proposed in Durand et al. [2023]. The DSE M6 model is trained using a contrastive learning framework, and has a language-conditioned variant (M6 Lang-Cond.). Although some columns are not directly comparable due to updates in the dataset⁹, our baseline models achieve significantly lower AAE scores on the Overall, DE, and FR subsets, even though the DSE models are trained on a much larger dataset (88k songs). Their performance on the English subset, however, remains the best reported in the literature.

⁹The Multi-Lang Jamendo dataset initially contained 20 songs per language. The DSE models were evaluated on this version. One French song was later removed due to legal issues, so our models are evaluated on 19 French songs. As a result, the **FR** and **ALL** columns are not directly comparable.

4.7 Results on MUSDB18 Test

Model	Stem		Spleeter		Open-Unmix		HTDemucs	
Metric	AAE	PCO	AAE	PCO	AAE	PCO	AAE	PCO
GC	0.28	0.93	3.71	0.63	1.68	0.82	1.02	0.86
Base	0.73	0.90	2.56	0.85	2.92	0.85	2.21	0.83
MTL	1.17	0.87	3.08	0.82	2.19	0.85	2.21	0.85
Base-Mul	0.38	0.96	3.47	0.86	1.69	0.88	0.74	0.92
MTL-Mul	0.26	0.97	3.66	0.86	1.50	0.88	3.55	0.90

Table 4.7: Lyrics alignment results on the MUSDB18 test set. We compare the model from Gupta et al. [2020], our Baseline models, and the MTL models. The header specifies the inference input source, indicating whether clean vocal stems or separated vocals obtained using the stated source separation method were used.

In this section, we evaluate our models on the MUSDB18 test set and compare them with the GC model [Gupta et al., 2020]. As introduced in Section 3.4, MUSDB18 was originally designed for source separation and provides clean vocal stems, allowing us to assess the sensitivity of lyrics alignment models to the quality of vocal separation. To ensure a fair comparison with the GC baseline and to avoid adding complexity to the system, boundary probability (BDR) is not applied during alignment, since BDR also operates on source-separated vocals.

We evaluate performance both on clean vocal stems and on vocals separated using three widely used source separation tools: Spleeter [Hennequin et al., 2020], Open-Unmix [Stötter et al., 2019], and HTDemucs [Rouard et al., 2023]. These tools represent successive improvements in separation quality, with signal-to-distortion ratios (SDR) for vocal stems of 6.86 dB, 7.21 dB, and 8.04 dB, respectively. SDR values are computed by taking the median of the per-song median SDRs over 1-second chunks. Higher SDR indicates better separation quality.

Table 4.7 presents the results. Performance on source-separated vocals is significantly worse than on clean stems. This substantial drop highlights the challenge of aligning lyrics when clean vocals are not available and emphasizes the impact of separation artifacts introduced by source separation models.

The metrics are further plotted in Figure 4.5 to better visualize the trends. Overall, the metrics improve as the quality of the vocals increases (i.e., fewer artifacts), although some exceptions exist. The AAE metric is more sensitive to outliers, whereas PCO exhibits smoother behavior, as reflected in the plots. Notably, the MTL-Mul model with HTDemucs separation yields a large standard

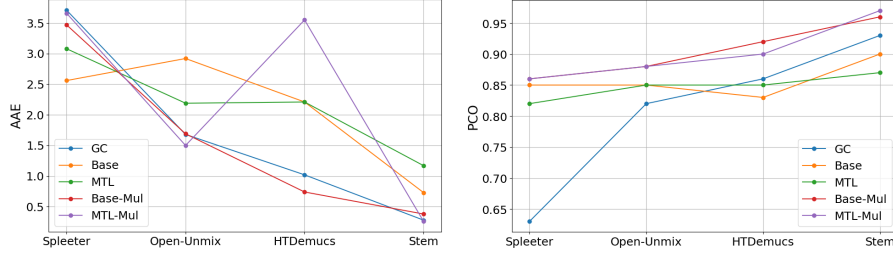


Figure 4.5: AAE and PCO across source separation methods and models.

deviation of 14.56, indicating considerable variability in alignment performance under this condition. No single model or vocal condition consistently outperforms the others. This observation also holds when comparing Baseline models to MTL models. Combined with the findings in Table 4.4 and Table 4.5, these results underscore the importance of evaluating lyrics alignment models across diverse datasets. Over-optimization on a small datasets (such as Jamendo English and Mauch) may not generalize well to real-world conditions.

Regarding the use of pitch information, our results suggest that it may provide benefits under certain conditions. However, its effectiveness appears sensitive to the quality of the input and the evaluation context.

4.8 Conclusion

In this chapter, we proposed a multi-task learning approach for lyrics-to-audio alignment by learning a joint representation of phonemes and pitch, and by incorporating boundary information to enhance alignment. Our method builds on a phoneme-based E2E alignment model, which already outperforms previous English state-of-the-art E2E systems for lyrics alignment. The MTL model further improves alignment performance over the baseline on both Jamendo English word-level and line-level metrics for English experiments.

In extending the approach to multilingual lyrics alignment, we expand the phoneme vocabulary to cover all training languages. The increased amount of training data leads to substantial improvements in the baseline model. However, the MTL setup does not yield further gains. While boundary information contributes to both models, the improvement is marginal. On the multilingual evaluation set, our baseline model outperforms the latest state-of-the-art system, which was trained on a significantly larger dataset using contrastive learning objectives.

Evaluation on the MUSDB18 test set reveals that our models struggle to generalize to external songs. Both our approach and the GC baseline are sen-

sitive to the quality of source separation, which remains a limiting factor for real-world applications. Nonetheless, we expect future improvements in accessible source separation tools to fill in the gap and to enhance performance in more practical scenarios. Similar sensitivity is observed for lyrics transcription, which is discussed in more detail in Chapter 7.

Pitch information proves beneficial in the English experiments on the Jamendo English dataset, supporting our motivation that pitch onsets may serve as indicators of phoneme onsets. However, this benefit does not extend to the multilingual setting. One explanation is that pitch and phoneme boundaries may not consistently align in real audio recordings. While they may coincide during composition or in symbolic scores, boundary agreement is often inconsistent in vocal recordings among human annotators [Proutskova et al., 2024]. This ambiguity makes learning a robust joint representation challenging. Despite these challenges, incorporating pitch remains a promising direction for future work. However, it demands careful attention to label quality and a more thoughtful approach to leveraging musical attributes.

Incorporating pitch into lyrics transcription may also provide complementary cues beyond alignment. pitch is directly tied to the construction of formants in spectrograms, since resonant peaks are shaped by the fundamental frequency and its harmonics. Providing models with pitch information could therefore guide attention toward spectral regions most relevant for phoneme discrimination. It might be more beneficial for music mixtures where background instrumentation can obscure vocal harmonics. On the other hand, one could argue that explicitly conditioning on pitch might complicate the task. Effective transcription often requires invariance to pitch variation, and the essential formant structure remains visible even if the spectrogram is blurred and individual harmonics are no longer resolved. This suggests that harmonics are not always crucial for identifying phonetic content.

Ultimately, whether pitch information helps or hinders may depend on the model’s existing ability to recognize phonemes. A system already trained to ignore pitch might be distracted by harmonics and forced to reallocate capacity. The performance will highly depend on the quality of pitch annotations. Conversely, if the model still struggles with phoneme discrimination, pitch could serve as a valuable guiding signal.

In conclusion, this chapter explored the use of pitch and boundary information as auxiliary signals for improving lyrics alignment through multi-task learning. While pitch proves useful in the English setting, it does not improve performance in the multilingual case. The best-performing model overall is the multilingual baseline, which achieves strong results with a simple training objective without auxiliary tasks.

We will return to the multilingual challenge in Chapter 6, where we explore advanced strategies for multilingual lyrics transcription. In the following chapter, however, the focus shifts to pronunciation modeling for singing voice. We introduce a new task, singing-to-speech conversion, to systematically investigate pronunciation variation and its implications for lyrics transcription.

Chapter 5

Pronunciation and Rhythm: Studying Singing Voice through Singing-to-Speech Conversion

5.1 Introduction

While ALT has seen significant progress, most studies have focused on challenges related to pitch, rhythm, and background music. However, an equally crucial but less-explored factor is pronunciation variation in singing, which can significantly impact transcription performance. Previous research has emphasized the frequency and dynamic characteristics of singing voice and the influence of the music accompaniment as the primary difficulties of lyrics alignment and transcription [McVicar et al., 2014, Gupta et al., 2020, Deng et al., 2022]. Mesaros [2013] suggests that prosody, position of formants (which determines vowel sounds), dynamic range, and the loudness of phonation, to be the main differences between singing and speech. Recently, Kruspe [2024] identifies the challenges as: larger pitch fluctuations, increased loudness variability, pronunciation variation, timing variations, different vocabulary, and background music interference. We also see a growing awareness of the importance of pronunciation in singing voice analysis [Demirel et al., 2021a, Gupta et al., 2018a, Wu and Lee, 2024].

Despite this growing recognition, the role of pronunciation in ALT remains difficult to study in isolation. Unlike pitch and rhythm, which can be straight-

forwardly manipulated using pitch shifting and time stretching, pronunciation lacks direct and interpretable modification techniques. Similarly, while background music effects can be studied using source separation, and vocabulary differences can be analyzed with ASR language models, there is no straightforward way to isolate and control pronunciation variations in singing.

To address this challenge, we propose a novel task: Singing-to-Speech Conversion (S2S). Rather than attempting to manipulate pronunciation directly, S2S transforms singing into speech while preserving phonetic information. This provides a structured approach to studying pronunciation effects in ALT. Instead of altering the pronunciation directly, we aim to convert singing to speech while preserving the phonetic information. Our study focuses on English, a non-tonal language, where phonetic information is primarily reflected in pronunciation. Unlike tonal languages, where linguistic tone may correlate with melody, English lacks lexical tone, allowing us to examine pronunciation variations in S2S without tonal interference. By converting singing to speech while preserving phonetic information, S2S provides a controlled environment to examine pronunciation effects on ALT. This conversion also removes confounding factors such as pitch and rhythm, allowing us to isolate the impact of pronunciation. Additionally, it promotes phonemic awareness (discussed in Section 2.8.4) in singing voice analysis, offering a direct comparison between singing and speech, which is often difficult to perceive in musical contexts.

In this chapter, we present the first deep learning-based S2S system, leveraging a generative flow to align phonetic content across domains while normalizing pitch and rhythm. To address the challenge of drastic timing differences between singing and speech, we adapt the monotonic alignment search (MAS) algorithm [Kim et al., 2020] for S2S conversion. Through objective and subjective evaluations, we demonstrate that our model outperforms signal processing-based approaches in naturalness and maintains phonetic similarity to the original recording better than a transcribe-then-synthesize baseline. Finally, we show that S2S conversion can serve as an effective data augmentation method for low-resource ALT, providing insights into the role of pronunciation variations in lyrics transcription. The codebase can be accessed online ¹. This chapter presents research published in the EURASIP Journal on Audio, Speech, and Music Processing [Huang and Benetos, 2025].

The primary motivation for introducing S2S in this thesis is to provide a way to study pronunciation variation in singing, an aspect that is difficult to isolate with existing methods. By converting singing into speech while preserving phonetic content, we can remove confounding factors such as pitch and rhythm, thereby allowing a more direct examination of how pronunciation dif-

¹<https://github.com/jhuang448/singing-to-speech>



Figure 5.1: Example of phonetic ambiguity in singing.

fers between singing and speech. Beyond this core objective, S2S also offers a secondary benefit as a data augmentation technique for ALT, since it can generate speech-like training material from singing recordings. In this way, the task serves both as a tool for analyzing pronunciation variation and as a potential means to improve ALT systems.

The rest of the chapter is organized as follows: We define the new task in Section 5.2. The proposed S2S model is introduced in Section 5.3. In Section 5.4, we demonstrate the application of the S2S model as a data augmentation method for ALT. Experimental details are provided in Section 5.5. Objective and subjective results for the proposed S2S model, and the data augmentation ALT results, are discussed in Section 5.6. Finally, conclusions are drawn in Section 5.7.

5.2 Singing to Speech Conversion

We propose Singing to Speech Conversion (S2S) as a novel task, which aims to reduce the variations in pitch, rhythm, and timbre in singing to create a signal that sounds like speech. Unlike speech synthesis, the goal of S2S is not to generate high-quality spoken output but to remove variations in pitch, rhythm, and timbre while retaining phonetic information. S2S could potentially improve the training process for lyrics transcription by providing more uniformly structured data.

In addition to its potential for analysis tasks, S2S could support singing manipulation as well. By converting singing to a normalized and interpretable format, an S2S-converted signal can serve as input for various manipulation tasks, such as singing identity conversion, pitch correction, and singing technique/style conversion. Furthermore, S2S could enhance the listening experience of individuals with hearing loss, particularly those with cochlear implants (CI). CI users perceive frequencies at a limited resolution, which may restrict their enjoyment of music [McDermott, 2004]. Since more stable pitch contours have been reported to be easier for CI users to perceive, the increased pitch stability from S2S conversion may also hold potential benefits for intelligibility, though this remains to be validated in future work.

As discussed in Chapter 2, singing and speech differ significantly in their acoustic properties. In addition to the pitch and rhythm introduced by melody, dynamics and pronunciations could be different as well [Demirel et al., 2021a, Gupta et al., 2018a]. A previous computational pronunciation analysis study found severe confusion between the plosives “D” and “T”, and in *approximants* with vowels, when predicting phonemes in singing with a phoneme recognition model [Demirel et al., 2021a]. It is also common for singers to modify vowels or omit consonants for fluency in singing. Figure 5.1 illustrates the phonetic ambiguity in ALT. These pronunciation differences affect the intelligibility of singing (i.e. words pronounced as they are in singing may be less understandable when read) but contribute to the expressiveness. We argue that the phonetic information is crucial to a singing performance, and essential to analysis and non-phonetic manipulation tasks. Therefore we focus on reducing the variations in singing attributes, while retaining the phonetic information, rather than generating high-quality, intelligible spoken lyrics.

Nowadays, automatic lyrics transcription (ALT) systems are able to achieve decent performance on monophonic singing, with a word error rate of around 10% [Ou et al., 2022] for English. Additionally, text-to-speech (TTS) systems can generate high-quality speech recordings [Tan et al., 2021]. While combining these two systems may appear to be a straightforward solution for singing-to-speech conversion, it comes with several limitations. First, this approach would result in a loss of expressiveness, as it compresses all phonetic information into text through the transcription bottleneck. Secondly, temporal information is also lost through the bottleneck, and alignment is not available between the singing and the converted signal. Thirdly, the effectiveness depends upon the quality of ALT and TTS systems, which might not be available for low-resource languages.

5.3 Proposed Method

To effectively tackle the challenges of singing-to-speech conversion (S2S), it is crucial to leverage methodologies that can manipulate duration in a manner suitable for speech synthesis. Duration manipulation is a key aspect, as speech and singing, even with identical linguistic content, exhibit significant differences in timing. Singing typically involves longer phoneme durations, and the ability to control this variation is essential for a seamless transformation between the two domains. Our model is based on the Glow-TTS architecture, which in its original form aligns input text and output spectrograms using Monotonic Alignment Search (MAS) based on latent features [Kim et al., 2020].

Flow-based models, such as those used in Glow-TTS, are particularly well-

suited for this task due to their ability to model data distributions directly and allow for efficient sampling. These models are capable of handling variations in phoneme duration by learning complex mappings between input and output sequences.

Furthermore, Glow-TTS explicitly predicts the duration for each input token, from which an alignment path can be generated. This exploits the inherent monotonic nature of TTS, where each input token corresponds to multiple output frames. A similar correspondence exists between speech and singing with the same linguistic content, where speech is often much shorter than singing. Given this context, we propose adopting the duration prediction from Glow-TTS, as it proves effective in handling cross-domain length mismatches.

To adapt it to the S2S task, we need to change various core components. First, we modify the MAS algorithm to address the issue of temporal alignment between the singing and speech audio. Next, unlike TTS, S2S takes singing Mel-spectrograms as input and produces speech-like Mel-spectrograms. Thus, the proposed model consists of a Mel Encoder f_{enc} , a flow-based decoder f_{dec} , a duration predictor f_{dur} , and a phoneme predictor f_{phone} . The diagram of the adapted singing-to-speech model is presented in Figure 5.2. An introduction to flow-based generative models can be found in Section 2.3.5.

We define x_{si} and x_{sp} to be a pair of singing and speech Mel-spectrograms which share the linguistic content. Let T_{si} and T_{sp} denote the lengths of the singing and speech Mel-spectrogram. Given the true data distribution $(x_{si}, x_{sp}) \sim p^*(x_{si}, x_{sp})$, the goal is to model $x_{sp} \sim p^*(x_{sp}|x_{si})$.

5.3.1 Training

The Mel Encoder f_{enc} maps the singing Mel-spectrogram x_{si} into the statistics of a latent space Z . z_{si} is a latent variable, sampled from multivariate Gaussian distribution $\mathcal{N}(\mu, \sigma)$ with a noise level ϵ as follows. The phoneme predictor f_{phone} is applied to z_{si} to produce a phoneme posteriorgram Q_{si} :

$$\mu = f_{enc}(x_{si}), \sigma = 1 \quad (5.1)$$

$$z_{si} = \mu + \epsilon\sigma \quad (5.2)$$

$$Q_{si} = f_{phone}(z_{si}) \quad (5.3)$$

During training, an inverse pass of f_{dec} maps the speech Mel-spectrogram x_{sp} to the same latent space Z , and the same phoneme predictor is applied to

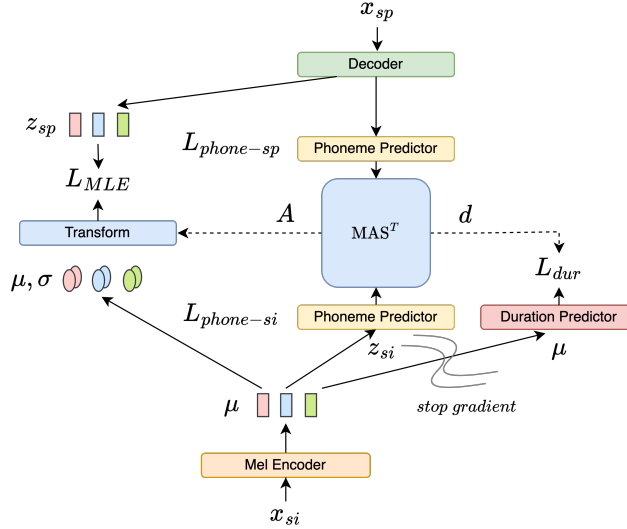


Figure 5.2: Diagram of the proposed S2S model.

produce a speech phoneme posteriorgram Q_{sp} :

$$z_{sp} = f_{dec}^{-1}(x_{sp}) \quad (5.4)$$

$$Q_{sp} = f_{phone}(z_{sp}) \quad (5.5)$$

Let S_{phone} be the phoneme sequence of the linguistic content. f_{phone} is trained with a CTC Loss [Graves et al., 2006] as follows:

$$L_{phone-si} = CTC(Q_{si}, S_{phone}) \quad (5.6)$$

$$L_{phone-sp} = CTC(Q_{sp}, S_{phone}) \quad (5.7)$$

Further, according to Eq. 2.5, we can calculate the log-likelihood:

$$\log p(x_{sp}|x_{si}) = \log \pi(z_{sp}|x_{si}) + \log \left| \det \frac{\partial f^{-1}(x_{sp})}{\partial x_{sp}} \right| \quad (5.8)$$

Given that z_{si} and z_{sp} differ in length, an alignment A is required in some form to relate them. In Glow-TTS, the phoneme alignment is trained in the latent space Z with MAS. In the original MAS algorithm for TTS, it searches for the most probable monotonic alignment between the latent features computed from speech and the statistical prior distribution derived from text. Essentially, it finds a monotonic alignment path between the two, using DTW with directional constraints to ensure monotonicity. In TTS, due to the property that one input phoneme would always produce 1 or more frames, MAS does not allow

multiple input tokens to be aligned with one output frame. In other words, the alignment only allows expanding the length for every token in the sequence. However, in the S2S scenario, this assumption no longer holds. Our preliminary study on the NUS-48E dataset [Duan et al., 2013] shows that around 80% of the phoneme counts are shortened in speech compared to singing. If we were to allow direct matching, cases would occur where one frame of singing is aligned to multiple speech frames. This leads to duration labels greater than one, which is inconsistent with the majority of training examples where the ground-truth duration is less than one. Such mismatches would introduce instability and confuse the duration predictor. To avoid this issue, we restrict the alignment to only allow shortening of the input sequence. We implement this variant of the MAS algorithm, denoted MAS^T , by swapping the direction restriction so that it enforces shortening rather than expansion.

In [Proszewska et al., 2022], the authors present a voice conversion model based on Glow-TTS and mention that MAS leads to challenging training dynamics. They avoid the problem by using pre-trained explicit phoneme durations and therefore no alignment is required. In internal tests, we also found alignments in Z -space failing to produce reasonable alignment paths. For that reason, we compute the alignment between the singing and speech phoneme posteriorgrams Q_{si} and Q_{sp} using MAS^T :

$$A = MAS^T(Q_{si}, Q_{sp}). \quad (5.9)$$

Here, $A(i) = j$ encodes that the i^{th} frame of Q_{si} is aligned to the j^{th} frame of Q_{sp} . From A we define d_j as the inverse of the number of singing frames being aligned to the j^{th} frame of Q_{sp} . This is used as the duration prediction target and the mean squared error (MSE) loss L_{dur} is defined as follows:

$$d_j = \frac{1}{\sum_{i=1}^{T_{si}} 1_{A(i)=j}}, j = 1, \dots, T_{sp} \quad (5.10)$$

$$L_{dur} = MSE(f_{dur}(\mu), d) \quad (5.11)$$

We can describe the prior distribution as follows:

$$\log P(z_{sp}|x_{si}; \theta, A) = \sum_{j=1}^{T_{sp}} \log N\left(z_{sp,j}; \frac{1}{d_j} \sum_{A(i)=j}^{T_{si}} \mu_i, \sigma\right) \quad (5.12)$$

where θ indicates the network parameters. The inner summation means that, when multiple singing frames are aligned with one speech frame, an average is

computed among the μ values. The objective is given by:

$$\min_{\theta} L_{MLE}(\theta) = \min_{\theta} -\log P(z_{sp}|x_{si}; \theta, A) \quad (5.13)$$

The final loss consists of L_{MLE} and L_{dur} , which share the same weight ω , while $L_{phone-si}$ and $L_{phone-sp}$ are included without additional weighting:

$$L = \omega(L_{MLE} + L_{dur}) + L_{phone-si} + L_{phone-sp} \quad (5.14)$$

where ω is a tunable parameter.

5.3.2 Inference

During inference, we follow the same process to calculate z_{si} (Eq. 5.1-5.2). Duration is predicted by the duration predictor:

$$d = f_{dur}(\mu + \epsilon\sigma) \quad (5.15)$$

The mapping A can be inferred by accumulating d until the sum is greater than a threshold h . The z_{sp} is sampled from the isotropic multivariate Gaussian distribution with the aligned mean values:

$$z_{sp,j} \sim \pi(z_{sp,j}) = \mathcal{N}\left(z_{sp,j}; \frac{1}{d_j} \sum_{A(i)=j}^{T_{si}} \mu_i, \sigma\right) \quad (5.16)$$

$$x_{sp} = f_{dec}(z_{sp}) \quad (5.17)$$

5.3.3 Architecture

The input x_{si} is an 80-bin Mel-spectrogram, computed at 16k Hz sampling rate, with an FFT size of 1024 and a hop size of 256, from 80 to 7600 Hz. This setting is adopted to be compatible with a pre-trained Parallel Wave GAN ², which is used for speedy Mel-to-audio conversion. The Mel-spectrograms are standardized using the global mean and standard deviation.

Mel Encoder. The first part of the Mel Encoder has 5 blocks of Conv1D layers with group normalization. The convolutional layers have 512 channels, with a kernel size of 5 and a stride of 1. Each group has 16 channels, resulting in 32 groups for normalization. The 5 Conv1D blocks are followed by another 3 Conv1D blocks, which alter the channel number from 512 to 128, 32 and 80, which is the embedding dimension and the hidden dimension for alignment. The second part uses the same architecture as Glow-TTS’s text encoder, except that

²<https://github.com/kan-bayashi/ParallelWaveGAN>

the text embedding layer is removed. The output μ (and z_{si}) is then passed to the MAS^T module, duration predictor, and phoneme predictor.

Phoneme Predictor. The phoneme predictor uses a CRNN with CTC loss (see Section 3.5) to capture precise temporal cues for duration modeling. It contains 1 linear layer, 3 Bidirectional LSTM layers, and 1 linear layer to map the dimension to the number of phonemes in the end. Log-softmax is applied before it is used for alignment and Connectionist Temporal Classification (CTC) loss [Graves et al., 2006] computation.

Duration Predictor. The duration predictor consists of 2 multi-head attention layers, accompanied by Conv1D layers and LayerNorms. Note that this is more complicated than the linear projection in Glow-TTS, because it needs to learn from the context to decide the shortening rate. Stop gradient is applied to avoid affecting other objectives.

Flow-based Decoder The flow-decoder is a stack of 12 blocks, each consisting of an activation normalization layer, invertible 1×1 convolution layer, and affine coupling layer [Kingma and Dhariwal, 2018].

5.4 S2S for data augmentation

As discussed in Section 2.8, automatic lyrics transcription (ALT) for singing voice is analogous to Automatic Speech Recognition (ASR) for spoken language [Demirel et al., 2020]. Due to the limited publicly available training data, ALT suffers from insufficient data, especially for non-English languages, as English dominates most of those datasets.

As one potential application, we propose S2S as a data augmentation method for ALT, which focus on low-resource settings. It has the following strengths:

1. Unlike traditional augmentation methods for audio such as speed perturbation and SpecAugment [Park et al., 2019], S2S is content-aware. Therefore it can reduce the variations in irrelevant aspects, and let ALT focus on the phonetic information.
2. S2S does not require access to external audio and metadata.
3. S2S can be applied to unseen languages, making it suitable for low-resource language scenarios. Since languages often share numerous phonemes, an S2S model can generalize to other languages by leveraging its knowledge of phoneme mapping.

In Section 5.5.5, we investigate the efficacy of S2S models as data augmentation methods across varying amounts of training data and different languages.

5.5 Experiments

5.5.1 Datasets

The experiments are conducted on the DAMP Sing! 300x30x2 dataset [Smule, 2015, Dabike and Barker, 2019] (DSing), which contains singing recordings with lyrics annotations collected from the Smule karaoke platform across 30 countries. In each country, 300 songs are included, each performed by one female and one male singer. We use the split configuration in [Dabike and Barker, 2019]. There are in total 81092 (from the *DSing30* split), 482, and 480 utterances from train, development, and test splits with lyrics annotations. Because the alignment depends on the phoneme posteriorgram, it works only after the phoneme predictor is trained to a certain level. To achieve faster and more stable training, we decide to use samples that are less than 5 seconds and longer than 3 seconds. After filtering, 20985 utterances are left for training. The lyrics are first converted to a phoneme sequence via a grapheme to phoneme tool ³.

For every singing utterance, a speech audio is generated by Tacotron2 ⁴ [Shen et al., 2018]. We used an off-the-shelf Tacotron2 model trained on the LJ Speech dataset [Ito and Johnson, 2017], which consists of read audiobook recordings by a single female speaker. All generated speech was resampled to 16k Hz to match our training setup. While Tacotron2 is known for producing relatively flat prosody and regular rhythm, this property aligns with our objective: the goal of S2S is to remove singing-specific variations such as expressive pitch and timing while preserving phonetic content. In this respect, the choice of TTS system is not critical, as long as it produces intelligible speech with minimal singing-related variation. Although Tacotron2 occasionally introduces pronunciation or prosodic errors, it provides a practical and scalable way to generate paired singing–speech data, which would otherwise be infeasible to collect. All of the audio files are resampled or generated at 16k Hz sampling rate.

For evaluation, we use both the test split from DSing and NHSS [Sharma et al., 2021]. NHSS is a speech and singing parallel database with 2791 pairs of utterances, each has a singing and a spoken version. Although there is no direct comparison between the proposed S2S and the spoken utterances in NHSS due to our S2S task definition, the spoken utterances can serve as a source of natural prosody (intonation and rhythm) for our baselines.

³<https://github.com/Kyubyong/g2p>

⁴<https://github.com/coqui-ai/TTS>

5.5.2 Baselines

As a first baseline we'll consider a straightforward approach to S2S: transcribing the lyrics, followed by synthesis using TTS. We choose the lyrics transcription model from [Ou et al., 2022]. The TTS model is Tacotron2 [Shen et al., 2018]. This baseline is referred to as **ALT-TTS**.

We build two more baselines with the WORLD synthesizer [Morise et al., 2016], one with duration adjustment and the other without. WORLD is a vocoder which parameterizes speech into pitch contour (F_0), harmonic spectral envelope (SP), and aperiodic spectral envelope (AP). It is commonly used for pitch manipulation on voices by editing the pitch contour and resynthesizing the signal [Morrison et al., 2021, Luo et al., 2018]. For baselines, we align the singing with its speech reference, and replace the F_0 of singing with that of the aligned reference speech. In this way, we approximate the intonation and the duration (rhythm) in the reference. We use different reference speech signals on DSing and NHSS. For DSing, the synthesized speech by **ALT-TTS** is used as a reference for intonation and duration adjustment. On NHSS, we use spoken utterances as the reference.

The steps to create the baseline with duration adjustment (**WORLD-dur**) are as follows: First, a standard DTW is applied between the MFCC features of singing and the synthesized speech, similar to the temporal alignment method adopted in [Vijayan et al., 2017]. Secondly, with the synchronization information, time-stretching is applied to the singing using rubberband⁵. Next, the F_0 contours are extracted by WORLD from the stretched singing ($F_{0_{si}}^{align}$) and the speech ($F_{0_{sp}}$). A linear interpolation on $F_{0_{sp}}$ is applied to compensate the voiced/unvoiced frame mismatch due to imperfect alignment, so that the voiced/unvoiced information is the same as $F_{0_{si}}^{align}$ for each frame. Finally, we re-synthesize the signal using the interpolated $F_{0_{sp}}$, the stretched singing SP , and the stretched singing AP . Similar to **WORLD-dur**, if the time-stretching is applied to the synthesized speech, the output would have no duration adjustment. This is denoted as **WORLD-nodur**.

SVC models are not considered as baselines since most of them are not capable of duration manipulation, making them invalid as S2S systems for comparison.

5.5.3 Training and Inference

We use the ADAM optimizer, at the learning rate of 0.0001. Parameter ω is set to 10. The model is first trained for 223 epochs at a noise level ϵ of 0, and

⁵<https://breakfastquay.com/rubberband/>

fine-tuned at a noise level of 0.3 for another 60 epochs. The models before and after fine-tuning are referred to as **M1** and **M2**.

During inference, the duration is predicted by the duration predictor. Additionally, it is possible to manually adjust the duration by entering duration rates. When all values are set to 1, the duration adjustment is disabled, and the rhythm information in the input is preserved. To demonstrate the impact of duration adjustment on the converted audio, we compare two models: one with duration adjustment, denoted as **M*-dur**, and one without duration adjustment, denoted as **M*-nodur**.

5.5.4 Evaluation

Objective Evaluation

To evaluate the audio quality and intelligibility of the S2S systems, we adopt two objective metrics: MOSNet [Lo et al., 2019] and SRMR (speech-to-reverberation modulation energy ratio) [Polyak et al., 2020, Santos and Falk, 2014]. MOSNet is a deep learning-based metric that predicts a mean opinion score (MOS) estimation for voice conversion tasks. SRMR, on the other hand, is a non-intrusive (i.e. without a reference signal) speech intelligibility metric that measures the ability to comprehend speech in a noisy environment. SRMR-CI is a variant of SRMR designed for CI (Cochlear Implant) users, and we use it to measure the intelligibility for people with hearing loss. It is worth noting that MOSNet and SRMR may not be the most suitable metrics for evaluating the intelligibility of S2S output, because the noise distribution in our setting differs from those used for a denoising or enhancement system. Nonetheless, we believe that the combination of MOSNet and SRMR provides some useful insights into the naturalness and intelligibility. Besides, we report the word error rate (WER) and character error rate (CER) from a pre-trained ASR model ⁶ and a pre-trained ALT model [Ou et al., 2022].

Subjective Evaluation

Our pilot subjective study shows that the model after fine-tuning (M2) generates perceptually better audio than the model without fine-tuning (M1). Due to the limited resources for a subjective test, M1 is excluded from the subjective study.

We conducted an online survey to evaluate the performance of S2S conversion methods, in which we compared five different methods: ALT-TTS, WORLD-nodur, WORLD-dur, M2-nodur, and M2-dur. We randomly selected one singing utterance between 3 to 5 seconds from each of the 8 singers (four female and four

⁶<https://huggingface.co/speechbrain/asr-transformer-transformerlm-librispeech>

male) out of 10 in the NHSS dataset, and applied the five methods to each of these utterances. Participants in the survey were asked to rate the naturalness, phonetic similarity to the singing reference, and intelligibility to recognize the given reference lyrics on a Likert scale ranging from 1 to 5. The survey was held on Qualtrics ⁷. The audio samples used in the survey can be accessed online ⁸.

In the context of this study, phonetic similarity refers to the extent of acoustic resemblance between two linguistic elements in terms of their pronunciations. However, it is important to note that a universally accepted definition does not exist, as pointed out in [Laver, 1994]. In our online survey, we define the phonetic similarity from a practical viewpoint, assessing the degree to which the given audio sample sounds like the reference in terms of pronunciation.

From our internal tests, we found that the rating of phonetic similarity can be easily influenced by audio quality and intelligibility. We observed that some participants disregarded the pronunciation and consistently chose the sample with the best quality or the most intelligible linguistic content. To reduce such biases in the analysis, we introduced two screening questions ⁹. Participants were asked to select the TTS-generated samples that demonstrated the highest phonetic similarity to a singing reference, where we modified the input phoneme prompt for TTS to mimic the singing pronunciation. Responses that did not pass the screening would be excluded from the analysis.

5.5.5 S2S for data augmentation

We propose S2S to be an effective augmentation method under low-resource settings. To approximate the low-resource condition and to study how the effectiveness varies over the training data size, different training subsets from the same datasets are used.

Data Splits

For English ALT, the three splits of DSing (DSing1, DSing3, and DSing30) are adopted. Additionally, we utilize OpenSinger [Huang et al., 2021a], a Chinese singing voice dataset, to conduct parallel experiments on a language unseen during training, reflecting the model’s generalization ability. We form the test set by leaving out four singers (2 males and 2 females) out of the total 76, and another 4 singers are reserved for the validation set. We randomly sample subsets, resulting in four training sets: OS2h, OS5h, OS12h, and OS23h (named according to their total durations).

⁷<https://www.qualtrics.com/>

⁸<https://jhuang448.github.io/singing-to-speech-samples/>

⁹Please check Appendix A for details.

S2S Methods for Comparison

Among the proposed S2S models, M2-nodur is selected for yielding the best ALT WER. S2S models (M2-nodur and WORLD-nodur) are compared against time warping and frequency masking from SpecAugment [Park et al., 2019], as well as no augmentation. ALT-TTS is not included due to its dependencies on ALT and TTS models, which assume a high-resource setting.

Alternative baselines such as singing-to-singing or speech-to-singing conversion were not considered, since these approaches would either retain singing-related variability or reintroduce it, whereas our goal is to reduce such variability by approximating speech-like characteristics for more effective ALT training.

Architecture

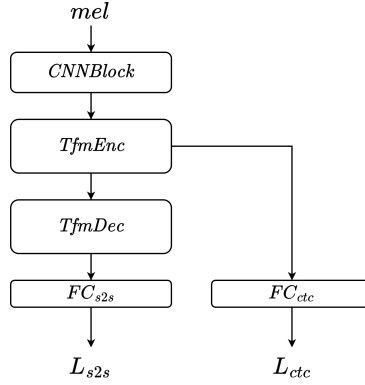


Figure 5.3: Architecture of the ALT model. It consists of a convolutional block $CNNBlock$, a Transformer encoder $TfmEnc$, a Transformer decoder $TfmDec$, and two fully connected layers FC_{ctc} , FC_{s2s} .

Experiments are conducted using the same Transformer ALT model and training pipeline implemented with the speechbrain [Ravanelli et al., 2021] toolkit. The ALT model is a hybrid CTC/attention [Watanabe et al., 2017] Transformer. We adopt the Transformer baseline (see Section 3.5) for its lightweight, transparent, and data-driven design, which allows us to focus on evaluating data augmentation and singing-to-speech adaptation without the influence of pre-trained speech models. Figure 5.3 illustrates the model architecture. The input is an 80-bin Mel-spectrogram, consistent with the proposed models. It is passed through a convolutional block and a Transformer encoder. The encoder’s output is directed to two branches: one for CTC loss [Graves et al., 2006], and the other for sequence-to-sequence (seq2seq) loss. The CTC branch comprises a single fully connected layer, whose output is used to compute a CTC loss L_{ctc} . The seq2seq branch consists of one Transformer decoder and another fully

connected layer. The seq2seq branch produces a cross-entropy loss L_{s2s} and performs inference autoregressively. The final loss is a weighted sum of the two:

$$L = \alpha L_{ctc} + (1 - \alpha) L_{s2s} \quad (5.18)$$

where α is the weighting parameter.

We follow the default setting in a speechbrain recipe ¹⁰ and make adaptations. The input x_{si} is an 80-bin Mel-spectrogram, computed at a 16kHz sampling rate, with an FFT size of 1024 and a hop size of 256, ranging from 80 to 7600 Hz. This configuration is consistent with that of the pre-trained Parallel Wave GAN model ¹¹ and the proposed S2S model output. The convolutional block comprises 3 CNN blocks, each with 64 channels. The initial two layers have a kernel size of 5 and a stride of 2, while the last layer has both the kernel and stride set to 1. The Transformer encoder consists of 12 layers, and the Transformer decoder consists of 6 layers. Each encoder layer consists of a multi-head attention and a position-wise feed-forward layer. Each decoder layer shares the same structure except that the attention layer is causal. The dimensions are configured as follows: attention dimension is set to 512, and the dimensional size of the position-wise feed-forward layer is 2048. The number of heads is set to 8, and the loss weighting parameter α is set to 0.3 according to [Gao et al., 2022a].

Training

All experiments on DSing datasets are trained for 100 epochs, and all experiments on OpenSinger datasets are trained for 300 epochs (except for OS2h is 600 epochs). The numbers are adjusted according to the convergence condition.

For both languages, we use pre-trained tokenizers from huggingface hubs ¹². Both tokenizers produce 5000 tokens (subwords) in total.

SpecAugment methods are applied at runtime, with the time-warping parameter set to 5, and the number of frequency masks set to 2, each with a width of 15. Augmented Mel-spectrograms using S2S methods are incorporated at training time, with each singing Mel-spectrogram having a 50% probability of being replaced by its S2S-converted counterpart.

We use the ADAM optimizer [Kingma and Ba, 2015] and employ the Noam learning rate scheduler [Vaswani et al., 2017], with the initial learning rate set

¹⁰<https://github.com/speechbrain/speechbrain/blob/v1.0.0/recipes/LibriSpeech/ASR/transformer/hparams/transformer.yaml>

¹¹<https://github.com/kan-bayashi/ParallelWaveGAN>

¹²`speechbrain/asr-transformer-transformerlm-librispeech` for English and `speechbrain/asr-transformer-aishell` for Chinese.

to 0.001. The number of warm-up steps is set to 25000. The final model is derived by averaging the best 5 validated end-of-epoch checkpoints.

For validation and testing, we apply beam search on the Transformer decoder to determine the best autoregressive prediction. The beam size is set to 10 during validation and 66 during testing. No language model is involved, and we report WER for English and CER for Chinese results.

5.6 Results and Discussion

5.6.1 Objective results

Table 5.1 shows the objective evaluation results on S2S methods and their reference audio on DSing¹³. It can be observed that the MOSNet and SRMR are degraded after conversion with WORLD models, while M1 and M2 models show improvements in SRMR. This suggests that the signal processing approach introduces artifacts, while the proposed models unintentionally achieve some level of denoising and dereverberation due to the clean training target.

All S2S methods show increased WER/CERs after conversion, especially for the proposed M* models. This can be attributed to a domain mismatch: ASR models are trained to handle variation in speech pronunciations, and ALT models are trained to handle variation in singing pronunciations. However, the S2S methods generate speech-like signals that still retain characteristics of singing pronunciations, a type of variation that neither ASR nor ALT models have been exposed to during training. As a result, the converted signals fall between the two domains, which impacts recognition performance. Appendix C provides further validation of this claim.

Comparing to M2, the M1 model yields superior MOSNet and SRMR results, but inferior WERs/CERs. This can be attributed to the $\epsilon = 0.3$ fine-tuning process, which encourages a sparser distribution of the centres of the multivariate Gaussian. This encourages the encoder to be more confident about the phoneme information, and leads to clearer pronunciation. Other ϵ values (0.1, 1.0) have been explored and we chose $\epsilon = 0.3$ empirically.

All methods show an increase in WER/CER after conversion, and these results are not satisfactory. This is not surprising, partly because the ASR/ALT models are not accustomed to speech-like signals being uttered in singing pronunciations. Moreover, proposed models have limited information about semantics, hence when they encounter ambiguous phonemes in singing such as ‘D’ or ‘T’, or noisy recordings, they cannot rely on a language model, but convert it

¹³The results on NHSS are presented in Appendix B and exhibit a similar trend.

Duration adjustment Method	Reference Sing	ALT-TTS	nodur			dur		
			WORLD	M1	M2	WORLD	M1	M2
MOSNet↑	3.01	2.86	1.99	2.89	2.84	2.42	2.96	2.94
SRMR↑	12.21	14.51	1.85	15.55	13.73	5.56	19.86	15.85
SRMR-CI↑	7.40	9.08	5.08	9.55	8.18	4.20	11.62	8.93
ASR WER/CER↓	0.54/0.31	0.31/0.18	0.66/0.40	1.00/0.66	0.97/0.63	0.85/0.62	0.88/0.64	0.84/0.58
ALT WER/CER↓	0.13/0.07	0.22/0.15	0.23/0.15	0.73/0.51	0.61/0.40	0.72/0.53	0.71/0.53	0.62/0.45

Table 5.1: Objective evaluation on the DSing test set. **Sing** indicates the original singing voice.

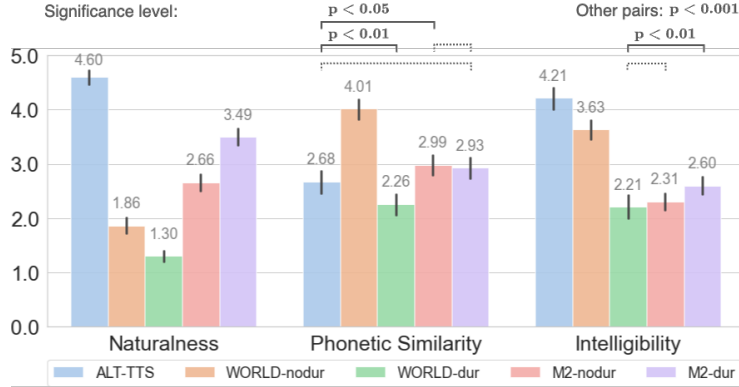


Figure 5.4: Comparison of different methods in subjective evaluation. All values are Mean Opinion Score (MOS). All pair-wise significance level is $p < 0.001$ except for those indicated above the plot. A dotted line means no significance.

as they are. As a result, some utterances may be challenging to detect by an ASR model but still make sense to a human listener.

The ASR and ALT error rates are heavily degraded for WORLD when applying the duration adjustment, while the opposite can be observed for M1. Relatively small degradation of ALT error rates can be observed for M2. This suggests that the proposed models are more effective at addressing substantial rhythm conversion.

5.6.2 Subjective results

We recruited 23 subjects who were Master’s or PhD students in Computer Science and Engineering or interdisciplinary fields involving AI ¹⁴. Among them, many had experience in music processing. Table 5.2 summarizes the participants’ self-reported gender, age, and singing experience. Figure 5.4 shows the MOS results of all methods and aspects, and the pair-wise Wilcoxon signed-rank test significance levels.

ALT-TTS achieves the highest score in naturalness, due to the advance of TTS technology. Our proposed M2 models outperform WORLD methods, probably because the neural network approach introduces fewer artefacts than the signal processing approach. The WORLD method performs worse with duration adjustment, while M2 performs better. This trend is consistent with the objective evaluation and is also observed in the intelligibility scores, suggesting that M2 is better at handling rhythm conversion. However, this improvement

¹⁴The ethics application for the subjective test was approved by the Queen Mary Ethics of Research Committee [reference number QMERC20.565.DSEEC23.009.]. All participants provided informed consent prior to their participation in the study.

Gender	
Man	11
Woman	6
Age Group	
18-25	3
25-35	13
35-45	1
Singing Experience	
Not trained / Do not sing	6
Little training but enjoy singing	9
Trained for years	2

Table 5.2: Demographic information and singing experience of participants, including gender, age, and amount of singing experience.

may not solely result from rhythm modeling. The neural approach in M2 also produces cleaner and more intelligible signals, which can further enhance perceived rhythm naturalness. On the other hand, the duration predictor in M2 produces a more natural speech rhythm which is easier for human perception and contributes to a higher level of intelligibility.

WORLD-nodur achieves the highest score in phonetic similarity, probably because this is the method with the least manipulation (only in pitch) therefore the pronunciation is well-preserved. ALT-TTS is rated slightly lower than M2 models. This suggests that, despite the high accuracy of the ALT model, the ALT-TTS method may sacrifice some pronunciation details in favour of preserving linguistic content. In contrast, the M2 models retain pronunciation details better, but at the cost of a degradation in intelligibility.

An unexpected challenge in our subjective evaluation was the varying phonemic awareness among participants. We observe that native English speakers appeared to focus more on decoding linguistic content rather than carefully distinguishing phonetic details. This tendency made it difficult for them to assess phonetic similarity accurately, as they unconsciously filled in missing or altered phonemes based on contextual understanding. This may explain why improvements in phonetic similarity appeared less pronounced than intelligibility gains. Since this effect was not explicitly tested, future studies could explore ways to better isolate phonetic perception, such as using foreign languages or targeted training sessions.

It is also important to acknowledge that some of the singing samples used in the subjective test were likely performed by non-native English speakers. As a result, part of the observed pronunciation variation may stem from accents rather than from the singing itself. This uncontrolled factor could have influenced the perceived phonetic similarity and intelligibility ratings. Future

evaluations could mitigate this by using accent-balanced datasets or focusing on languages where native and non-native singing samples can be clearly distinguished.

5.6.3 S2S for data augmentation results

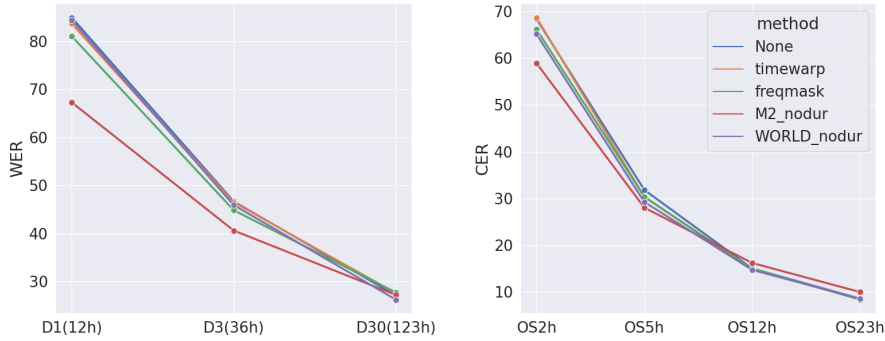


Figure 5.5: S2S for data augmentation results on different training sets. On the left: WERs on the DAMP test set, with numbers in parentheses indicating the duration of the subsets. The numbers in the parentheses indicate the duration of the subsets. On the right: CERs on the OpenSinger test set.

Figure 5.5 displays the ALT results when training on different sizes of subsets with different augmentation methods. M2-nodur shows a clear advantage over the others under low-resource settings for both languages. As the training data increases, it performs slightly worse than the others, likely due to the error propagated from M2. It is noteworthy that M2-nodur proves effective on OS2h and OS5h, even though Chinese samples were unseen during training. Additionally, the test set involves a tonal language (Chinese), while the training data (English) is non-tonal.

5.7 Conclusion

This chapter introduced the singing-to-speech conversion task and presents the first deep learning-based S2S system. By leveraging existing TTS models and technology to generate synthetic data for training and adapting the Glow-TTS model to S2S, we achieve a better balance of naturalness and phonetic similarity than the baseline models. Additionally, we demonstrate the new S2S task can benefit ALT through data augmentation, and the proposed model exhibits the ability to generalize to unseen languages. Moreover, S2S has potential as an

input for various manipulation tasks and could serve to anonymize singing voices before uploading them to a voice editing service. The proposed MAS^T may offer a solution to bridge timing differences between singing and speech, and enable drastic duration manipulation for controllable singing voice editing, which is traditionally difficult.

Currently, the model is trained with MAS^T under the assumption that all frames in singing are shortened in speech. It can sometimes lead to problematic conversion for consonants. Furthermore, glottal fricatives (“H”) are often skipped probably due to their resemblance to inhaling. To support more accurate phonetic representations, one promising direction would be to incorporate language models that leverage semantic context to inform pronunciation decisions. Another area with room for improvement is the audio quality of the converted samples. Narrowing the performance gap between M2 and ALT-TTS may involve refining the synthesis process to produce more natural and expressive speech.

Besides, the observed difficulty in phonetic evaluation suggests that pronunciation variation may have a larger impact on ALT than previously anticipated, making it a promising direction for future research. Further studies are needed to better understand these effects and explore methods to isolate phonetic perception in evaluations.

In the next chapter, we address the multilingual challenge in ALT. We conduct an investigative study on how to extend English ALT to multilingual settings. We also propose phonetic reinterpretation, a method inspired by the observation that pronunciation variation in singing leads to mishearings. Building on the insights from singing-to-speech conversion, this approach aims to tackle the challenges discussed in Section 2.8.4.

Chapter 6

Multilingual ALT: Addressing the Resource Imbalance

6.1 Introduction

The year 2022 marked a milestone in the literature of ALT, with the publication of the first self-supervised learning (SSL)-driven ALT model [Ou et al., 2022] and the release of OpenAI’s Whisper [Radford et al., 2023]. Both mark a significant step towards leveraging pre-trained ASR models for ALT. The former, fine-tuning Wav2Vec2 [Baevski et al., 2020] on singing data, achieves the new state-of-the-art performance on English. The latter, Whisper, is a large-scale robust ASR model trained on a diverse set of languages. Whisper’s performance is surprisingly high on multilingual ALT, despite its focus on speech. Since then, follow-up works on low-resource ALT have emerged with the help of pre-trained models [Zhuo et al., 2023, Wang et al., 2023], where the abundant resources available for ASR have helped boost the performance of ALT through transfer learning. A detailed description of these milestone works can be found in Section 2.8.3, while our own exploration of pre-trained models is presented in Chapter 7.

However, the computational costs and interpretability of these models remain a concern. The large size and complexity of pre-trained models make them less suitable for iterative development. They are also not interpretable, making it harder to understand the relationship between input and output. Moreover, these approaches, more or less, require an update of knowledge of the target singing domain for optimal performance, either through transfer learning [Ou et al., 2022] or post-processing [Zhuo et al., 2023]. Therefore, studying the multilingual nature in the singing domain is valuable.

In this chapter, we conduct a systematic study with a lightweight baseline on the multilingual nature of the task, revealing connections between language and ALT performance, as well as the low-resource and imbalanced data situation, in the context of singing voice. The proposed baselines offer advantages such as smaller size, faster inference, and lower VRAM usage. Their lower inference cost makes them more eco-friendly. We show a comparison of the computational costs as well.

We primarily focus on spectrogram-based models, which are trained from scratch on multilingual singing data. Additionally, we compare these models with models that take Wav2Vec2 [Baevski et al., 2020, Conneau et al., 2021] as input, where the features are pre-trained on large-scale speech data before being fine-tuned on the multilingual singing dataset. Our study comprises three key steps: Firstly, we explore the impact of training on multilingual data compared to monolingual models. For spectrogram models, leveraging multilingual training data is helpful. On the other hand, Wav2Vec2-based models benefit from pre-training on multilingual data. Secondly, we look into language conditioning methods, examining language embedding conditioning and Feature-wise Linear Modulation (FiLM) conditioning [Perez et al., 2018]. Introducing language conditioning during training significantly enhances performance. We propose optionally language-informed models based on them to extend the functionality. Thirdly, we combine both approaches through phonetic reinterpretation, inspired by the concept of transliteration. Transliteration involves mapping text between different writing systems (e.g., Cyrillic to Latin) while preserving pronunciation, whereas phonetic reinterpretation converts text from one language into another based on how it spells in the target language. By training on multilingual data with language conditioning and phonetic reinterpretation, the proposed model achieves the best overall WERs for blind inference while maintaining the capability for language configuration.

Our work makes several distinct contributions to the field: We provide practical guidelines covering training data, features, and language conditioning mechanisms. Our findings reveal the effectiveness of joint multilingual training and conditioning mechanisms. The proposed model can perform inference with and without language information. Building upon these guidelines, we propose a phonetic augmentation approach tailored to multilingual data. To the best of our knowledge, this is the first ALT work that incorporates phonetic reinterpretation. Through our research, we aim to advance the field of lyrics transcription by providing practical insights and innovative methodologies. The work in this chapter was extended and updated from our published paper at EUSIPCO 2024 [Huang and Benetos, 2024].

The remainder of this chapter is structured as follows: Related work is dis-

cussed in Section 6.2. Section 6.3 provides the motivation of the proposed methods. Section 6.4 introduces the datasets and experimental settings that apply to all subsequent sections. The proposed methods for multilingual joint training, language conditioning, and phonetic reinterpretation are detailed in Section 6.5, Section 6.6, and Section 6.7, respectively, along with the corresponding results and discussions. Finally, we discuss the limitations and present our conclusions in Section 6.8 and Section 6.9.

6.2 Related Work

Multilingual ALT introduces additional challenges beyond English ALT, such as implicit language identification, dataset imbalances, and pronunciation variations across overlapping character sets, as discussed in Section 2.8.4. This section reviews prior work on language conditioning mechanisms for multilingual ASR and phonetic reinterpretation techniques, both of which are the foundation of our proposed methods.

6.2.1 Language Conditioning for Multilingual ASR

Language configurability is favoured in multilingual ASR systems. In multilingual ASR, language conditioning is commonly enabled through architecture extension or prompt tuning [Lester et al., 2021]. The former would involve some conditioning layers/vectors applied at intermediate outputs [Zhou et al., 2018, Kannan et al., 2019] or building language-specific sub-networks [Tjandra et al., 2023, Yang et al., 2023, Thomas et al., 2020], while the latter embeds the language information as a language prompt [Li et al., 2024]. Feature-wise Linear Modulation [Perez et al., 2018] (FiLM) conditioning has also been adopted to improve multilingual ASR in air traffic control [Fan et al., 2024a].

6.2.2 Phonetic Reinterpretation

Transliteration is the process of converting text from one script into another script, while maintaining the phonetic correspondence between the original and the converted text [Prabhakar and Pal, 2018]. Our work extends this idea by introducing phonetic reinterpretation, which focuses on creating phonetic consistency across languages, even for those sharing the same writing system. Machine transliteration is usually text-based, either rule-based or data-driven. It has been used in multilingual ASR to perform data augmentation at the target label level and balance the languages [Khare et al., 2021, Thomas et al., 2020, Datta et al., 2020]. It has also been applied to address code-switched

ASR problems ¹ by avoiding error propagation from frame-level language identification [Yan et al., 2023]. Phonetic reinterpretation builds upon these ideas, taking a broader approach to phonetic alignment across languages for robust multilingual models.

6.3 Motivation

6.3.1 Joint Multilingual Training and Language Conditioning

Multilingual lyrics transcription presents unique challenges due to the linguistic diversity across languages, including variations in pronunciation, spelling, and grammar rules. However, many languages share common phonetic features, such as vowels, consonants, nasal sounds, and articulations, regardless of their writing systems. These phonetic commonalities, especially in languages that share linguistic roots or have been influenced by their neighbours, provide opportunities for cross-lingual knowledge transfer. Several studies in multilingual ASR demonstrate that high-resource languages like English can support low-resource languages [Huang et al., 2013, Kunze et al., 2017]. Languages using the Latin script (e.g., English, French, Spanish, German) exhibit spelling similarities that further enhance shared learning in multilingual models.

Building on successful approaches in multilingual ASR and lyrics alignment [Vaglio et al., 2020b], this chapter explores joint multilingual training to capture shared phonetic representations across languages. This approach enables the model to generalize effectively across languages with imbalanced training data, enhancing acoustic feature learning.

To address language-specific variations, language conditioning mechanisms are introduced. By incorporating language identity during training, the model can adapt its parameters to capture language-specific acoustic nuances. During inference, language conditioning ensures that the model switches to language-specific representations, preserving the distinct linguistic properties of each language.

6.3.2 Phonetic Augmentation for Multilingual Data

A significant challenge in multilingual ALT is the imbalance in training data across languages. For instance, English typically has significantly more annotated training data compared to other languages. To address this, we aim to leverage pseudo-labels generated from an initial training round as a way to

¹Code-switched ASR refers to the challenge in ASR where the speech switches between two or more languages within a single utterance.

enhance performance in low-resource languages. This method is conceptually related to self-training, but is extended in our approach to handle cross-lingual scenarios.

Self-training [Scudder, 1965] has demonstrated its effectiveness in reducing WER in low-resource ASR [Kahn et al., 2020, Singh et al., 2023] and ALT [Gao et al., 2023b]. The self-training process typically involves training an initial model with a limited set of labelled data, which is then used to generate pseudo-labels for unlabelled data. These pseudo-labels are subsequently used to retrain the model, improving its performance as more pseudo-labelled data becomes available.

Building on the principles of self-training, we propose an approach tailored to the multilingual ALT context, where languages often have imbalanced training data. In addition to generating pseudo-labels for unlabelled data, we extend the self-training framework by introducing phonetic reinterpretation. This technique converts a script into a phonetic equivalent in a different language, enabling the model to align similar sounds across languages more effectively.

By emphasizing phonetic similarity during conversion, the model can more effectively leverage high-resource language data to learn patterns that transfer to low-resource languages. We refer to this data augmentation method as phonetic augmentation, with the converted labels referred to as phonetic proxies. This technique enriches the training data by introducing additional variations, forcing the model to focus on shared acoustic and phonetic features.



Figure 6.1: Example of phonetic reinterpretation for ALT.

Another motivation for phonetic augmentation comes from a unique phenomenon in ALT: listeners often ‘mishear’ and reinterpret lyrics across languages in ways similar to phonetic reinterpretation [Scherling et al., 2022, Beck et al., 2014]. By incorporating phonetic augmentation, we enable the model to leverage human-like phonetic reinterpretations that are common in cross-lingual music contexts. Figure 6.1 illustrates this idea with an English-to-French example, where the phonetic reinterpretation preserves the original sound patterns while shifting the content. This approach introduces interpretative variations in the training data, encouraging the model to rely more on acoustic-phonetic cues rather than language-specific lexical patterns. As a result, the model becomes

less sensitive to spelling and pronunciation inconsistencies across languages, improving cross-lingual robustness and distinguishing this task from standard ASR.

This approach is detailed in Section 6.7, where we show that the model can facilitate cross-lingual phonetic alignment through phonetic augmentation to improve low-resource ALT.

By incorporating these strategies (multilingual joint training, language conditioning, and phonetic augmentation) we aim to address the imbalanced data challenge of multilingual lyrics transcription. Our proposed methods are grounded in transfer learning, cross-lingual phonetic learning, and data augmentation, which have shown to be effective in related ASR tasks.

6.4 Implementation and Experimental Setup

	Train		Valid	Test
	DALI	MulJam	DALI	Jamendo
English	259444	86549	1017	868
French	11959	33220	1059	809
Spanish	10317	15119	1111	881
German	26343	3208	1103	871
Italian	9164	7807	1193	0
Russian	0	1805	0	0

Table 6.1: The number of lines in the training, validation, and test sets for each language.

6.4.1 Datasets

Two datasets, MulJam v2.0 [Yuan et al., 2023] and DALI v2 [Meseguer-Brocal et al., 2020] (we will ignore **v2.0** and **v2** in the remaining text for simplicity), are involved in training. MulJam consists of six languages, which are also the languages with the most songs in DALI. For evaluation, we use the Multi-Lang Jamendo v1.0 dataset [Durand et al., 2023]. It comprises 80 songs in four languages: English, French, Spanish, and German. Line-level segments are created based on the timestamps provided in the annotation. All training, validation and testing are performed at the line level. Although the Jam-ALT benchmark [Cífka et al., 2024] provides a refined version of song-level annotation, we stick to line-level evaluation with the original annotation. We argue that line-level evaluation is the most suitable for our training setting, since song-level evaluation requires long-form decoding, making the results less interpretable

and harder to analyze. Characters not in a predefined character set ² have already been removed when the dataset is created. All characters are converted to uppercase. We compared several options for validation in the pilot study and chose only the DALI dataset for validation for its best performance. Table 6.1 lists the number of lines in the training, validation, and testing datasets.

For the training and validation sets, we apply additional filtering to eliminate lines that are likely to be problematic. We exclude lines with incorrectly annotated durations (such as timestamps that extend beyond the total audio length), excessively long durations (over 30 seconds), and abnormally high character rates (over 37.5 Hz). To isolate the vocal content, we apply source separation to all audio segments using Open-Unmix [Stöter et al., 2019].

The six languages are referred to as EN (English), FR (French), ES (Spanish), DE (German), IT (Italian), and RU (Russian) in the rest of the chapter.

6.4.2 Baseline and State-of-the-Art Models

In this section, we introduce the baseline models and state-of-the-art (SOTA) methods used for comparison. Although the models differ in their architectures, pre-training, and the type of data they were trained on, they are included to provide reference WERs and help demonstrate consistent trends in performance across languages.

The optimization goal is to maximize the probability of the predicted transcription Y given the input audio X , expressed as $P(Y|X)$. Under an end-to-end setting, this probability can be decomposed using the chain rule:

$$P(Y|X) = \prod_{t=1}^T P(y_t|X, y_1, \dots, y_{t-1})$$

where Y can be converted to a sequence of tokens $Y = (y_1, y_2, \dots, y_T)$ and T is the length of the predicted tokens.

The core experiments are conducted with a spectrogram-input baseline, which is a Transformer adapted from a previous English ALT work [Gao et al., 2023a]. The model does not have any pre-trained components. This baseline serves as a foundational comparison for our multilingual experiments.

We provide a comparison with two state-of-the-art methods. The first method is a Wav2Vec2-based model adapted from Ou et al. [2022]. It fine-tunes Wav2Vec2 features for English ALT. The pre-training for Wav2Vec2 was performed on a large speech dataset, and for our comparison, it is fine-tuned on multilingual extracted singing vocal data. The purpose of including this model

²<https://github.com/f90/jamendolyrics/blob/master/vocab/international.characters>

is to provide a reference for WER performance that can be achieved by adapting SOTA English ALT methods to multilingual singing data. The second method for comparison is Whisper [Radford et al., 2023]. This model is a pre-trained, large-scale speech-to-text system developed by OpenAI.

Table 6.2 shows a comparison in the number of parameters, required VRAM, and inference time among the three models. These comparisons provide additional insight into the trade-offs between model complexity and performance in the ALT task.

	SpecTfm	Wav2Vec2	Whisper
Num. of Params	249M	1454M	1550M
Required VRAM	0.7G	2.7G	9.4G
Inference Time	1×	~ 2.9×	~ 3.3×

Table 6.2: Comparison of model complexity, required GPU VRAM, and relative inference time. The Whisper model used here is **large-v3**. **SpecTfm** denotes the spectrogram-based Transformer. Inference time is the relative time to it.

Spectrogram-Based Transformer

The spectrogram-based model adopts a similar architecture to state-of-the-art Transformer models [Gao et al., 2023a], employing a hybrid CTC/Attention design [Watanabe et al., 2017]. It takes as input an 80-dimensional Mel-spectrogram computed at a sampling rate of 16kHz, with an FFT size of 400 and a hop size of 10 ms. The model includes a convolutional block, a Transformer encoder, a Transformer decoder, and two fully connected layers. The output dimension of these fully connected layers matches the size of the target character set N . Let y denote the list of target lyrics tokens, y^{bos} represent y with a **<bos>** (begin-of-sentence) token added at the beginning, and y^{eos} represent y with a **<eos>** (end-of-sentence) token appended at the end. During training, the Mel-spectrogram is passed through the convolutional block before being forwarded to the Transformer encoder and decoder. To speed up convergence, teacher forcing is employed, where the model receives the ground truth tokens y^{bos} and predicts the subsequent ones y^{eos} :

$$feat = CNNBlock(mel) \quad (6.1)$$

$$h = TfmEnc(feat) \quad (6.2)$$

$$o = TfmDec(h, y^{bos}) \quad (6.3)$$

where h and o represent the outputs from the encoder and decoder respectively. Subsequently, h and o are fed into the two fully connected layers denoted as

FC_{ctc} and FC_{s2s} . These layers generate the posteriorgram for the CTC branch and the probability matrix for the sequence-to-sequence (seq2seq) branch.

$$p_{ctc} = FC_{ctc}(h) \quad (6.4)$$

$$p_{s2s} = FC_{s2s}(o) \quad (6.5)$$

The loss function is a weighted combination of two parts: the CTC loss for alignment-free training (computed from the CTC branch), and the Kullback-Leibler (KL) divergence loss to encourage smooth predictions (computed from the seq2seq branch):

$$Loss = \alpha L_{ctc}(p_{ctc}, y) + (1 - \alpha) L_{s2s}(p_{s2s}, y^{eos}) \quad (6.6)$$

where α is the loss weighting parameter.

The convolutional block of the Transformer model consists of three CNN blocks, each with 64 channels. The first two layers have a kernel size of 5 and a stride of 2, whereas the final layer has both the kernel size and stride set to 1. Positional encoding [Vaswani et al., 2017] is incorporated into the Transformer input before the encoder processes it. The Transformer encoder contains 12 layers, while the Transformer decoder contains 6 layers. Each encoder layer includes a multi-head attention layer and a position-wise feed-forward layer. Similarly, each decoder layer contains these components, except that the attention layer is causal. The attention dimension is 512, there are 4 attention heads, and the position-wise feed-forward layer has a dimension of 2048. The loss weighting parameter α is set to 0.3.

Wav2Vec2-Based Transformer

To build a comparable Wav2Vec2 ALT model, we replace the convolutional block in the spectrogram-based model with a Wav2Vec2 feature extractor frontend. The number of encoder layers in the Transformer is reduced to 6, to reduce the backend capacity when using a powerful feature representation. The target character set is modified for the multilingual scenario, resulting in the multilingual vocabulary $\mathcal{C}_L$. All other configurations remain consistent with those described in Vaswani et al. [2017]. The Wav2Vec2 is fine-tuned together with the backend with a smaller learning rate, following the settings in Ou et al. [2022].

Whisper

As mentioned in Section 2.8.3, Whisper has been proven effective on ALT [Zhuo et al., 2023]. Whisper is an encoder-decoder Transformer-style model trained

with a multitask training format. A set of special tokens specifies the target task. Besides ASR and ALT, it can also perform speech translation, language identification, and voice activity detection. For comparison with baseline, we use the latest **large-v3** model. During inference, we do not specify the target language and let the model itself infer it automatically.

Justification for the Baseline Model

We intentionally choose not to use ALT models or feature extractors pre-trained with speech data, such as Wav2Vec2 and Whisper, although we are aware that incorporating these could potentially enhance performance. This decision is made to avoid the influence of the pre-trained models’ data distribution, complicating the assessment of whether any observed differences are due to training on a broader range of languages’ data or the knowledge gained during pre-training.

Instead, we rely on the benchmarking results presented in Section 3.5 to guide the selection of a Transformer with hybrid CTC/attention loss as the baseline architecture, as it provides a transparent data-driven setup and remains lightweight, making it easier to track how the model learns from the multilingual training data. The model’s simple architecture allows us to better understand the relationship between the input and output. Since the model lacks the complexity introduced by pre-training on large and diverse datasets, it is easier to trace how specific features influence the performance, making error analysis more straightforward. It is also flexible for deployment and takes fewer resources for training and inference. Although this simplicity limits the performance in comparison to state-of-the-art models, our aim is to provide insights into the task and benefit multilingual ALT model development in the long run. Table 6.2 shows that the baseline model has the advantages of much smaller model size, lower VRAM requirements, and shorter inference time, making it preferable for iterative development.

6.4.3 Training and Inference

Let $\mathcal{C}_l$ be the vocabulary of a language l . For English ALT, $\mathcal{C}_{EN}$ contains the 26 letters in the alphabet, the space token, the apostrophe, the **<bos>** and **<eos>** tokens, the epsilon token ϵ (for CTC loss computation), and the unknown **<unk>** token.

We take a straightforward approach to generalize an English ALT model to multilingual ALT by expanding the vocabulary. Let L denote the set of languages supported by a given multilingual ALT model. The vocabulary of the multilingual ALT model is the union of the vocabularies of all languages in

L .

$$\mathcal{C}_L = \bigcup_{l \in L} \mathcal{C}_l \quad (6.7)$$

The target dimensions of the last layers of the two branches increase to match the size of $\mathcal{C}_L$ correspondingly.

All experiments are conducted with the speechbrain toolkit [Ravanelli et al., 2021], using one NVIDIA A100 40GB GPU. The codebase can be accessed online ³.

Spectrogram-Based Transformer

We adapt the speechbrain recipe ⁴ to build the spectrogram-based Transformer model. The Adam optimizer is adopted for model training, with an initial learning rate of 0.001. The Noam learning rate scheduler is used with a warmup step of 25000. Validation is conducted every epoch, and the end-of-epoch checkpoint with the best WER is selected for evaluation.

All multilingual models are trained for 50 epochs to ensure convergence. For monolingual experiments, FR, ES and DE models are trained for 70 epochs, while the EN model is trained for 50 epochs. During validation and testing, beam search is utilized on the Transformer decoder to autoregressively choose the optimal prediction. The beam size is set to 10 for validation and 66 for testing, following the recipe.

Wav2vec2-Based Transformer

The Adam optimizer is used as well. Similar to Ou et al. [2022], the initial learning rate for the Wav2Vec2 frontend is 0.00001, while the initial learning rate for the backend is 0.0003. Learning rates are scheduled using the new-bob technique ⁵. The annealing factors for the frontend and the backend are 0.9 and 0.8. The improvement threshold is 0.0025 for both. All models are trained for 20 epochs. Beam search setting is the same as that for the spectrogram-based models.

6.4.4 Language Model

Language models (LMs) are commonly used in ASR and ALT inference. They help to refine the output of the ASR/ALT model by providing context and

³<https://github.com/jhuang448/MultilingualALT-trsl>

⁴<https://github.com/speechbrain/speechbrain/blob/develop/recipes/LibriSpeech/ASR/transformer/hparams/transformer.yaml>

⁵<https://speechbrain.readthedocs.io/en/latest/API/speechbrain.nnet.schedulers.html#speechbrain.nnet.schedulers.NewBobScheduler>

knowledge about grammar and syntax. Integrating LMs enhances the quality and naturalness of the transcription.

Although LMs are not the primary focus of our study, we report WER when utilizing language-specific LMs during inference for language-configurable models in certain experiments. Training models with phonetic proxies (Section 6.7) brings the risk of generating nonsensical text, as the phonetic proxies may not follow any grammar and syntax rules. Integrating LMs into the inference process is expected to enhance the comprehensibility of the transcriptions.

The language models are trained with the default settings of the speechbrain Transformer LM recipe ⁶. We train four LMs, for English, French, Spanish, and German. The training corpus is a combination of MulJam, DALI and a subset taken from the Multilingual LibriSpeech (MLS) [Pratap et al., 2020]. In [Gupta et al., 2020], the authors compared n-gram LMs trained on English lyrics corpus and speech corpus and found the in-domain (lyrics) corpus to be superior. Moving to multilingual, our pilot study shows that the LMs are better when trained with speech corpora, compared to using only the limited lyrics in the singing datasets. To be more specific, we compute the number of lines in English in MulJam and DALI, and sample lines (less than 500 characters long) in the other three languages from MLS so that the four languages have an equal size of LM corpora. Because the lyrics annotations in DALI do not include accented characters (they are converted to the Latin alphabet), we exclusively use the MulJam validation set for validation purposes.

The Transformer LM consists of 12 Transformer layers. The attention dimension is set to 768. The number of heads is 12 and the positional-wise feed-forward layer dimension is 3072. The LMs are trained using cross-entropy loss. The same optimizer and Noam learning rate scheduler are adopted as the spectrogram-based Transformer, with the initial learning rate set to 10.

When utilizing an LM at inference, a log-probability score from the LM is added to the beam search (log-probability) score produced from the S2S branch output, at a weight of 0.1.

6.4.5 Evaluation

We report Word Error Rate (WER) on the Multi-Lang Jamendo v1.0 evaluation dataset, which consists of four languages (EN, FR, ES, DE). Therefore the core experiments are conducted on the same language set. For language-configurable models, conditioning is specified in each table. When computing the WER, we do not apply further text normalization on the ground truth (such as removing

⁶<https://github.com/speechbrain/speechbrain/tree/develop/recipes/LibriSpeech/LM>

punctuation).

To better understand the results, we conduct statistical significance tests among some of the core experiments with a Python toolkit [Ferrer and Riera, 2023] that uses the bootstrapping technique [Efron and Tibshirani, 1993]. The confidence intervals are computed based on individual WERs of all lines (all languages). We use a 5% confidence interval to determine significance and the number of bootstrap sets is 1000. All reported results include statistical significance analysis indicated in each table caption.

6.5 Multilingual Joint Training

Low-resource language ALT suffers from having limited labelled data for training. Many languages share common phonemes, and those with the same writing system also share characters. In this section, we explore a straightforward way to utilize the commonalities among multilingual data through multilingual joint training.

6.5.1 Method

Training on Multiple Languages

We select six languages for the training set and create subsets for monolingual, 4-language (EN, FR, ES, DE), 5-language (EN, FR, ES, DE, IT), and 6-language (EN, FR, ES, DE, IT, RU) datasets. The inclusion of four languages is based on the languages available in the Jamendo evaluation dataset. Italian is chosen as the fifth language because it also belongs to the Latin writing system, while Russian represents a different writing system (Cyrillic). The vocabulary sizes for the 4-language, 5-language, and 6-language experiments are 56, 58, and 91, respectively.

pre-training on Multiple Languages

For Wav2Vec2 ALT models, we compare two checkpoints for Wav2Vec2 feature extractor initialization: `large-960h-1v60-self` [Baevski et al., 2020] and `large-xlsr-53` [Conneau et al., 2021]. The checkpoint `large-960h-1v60-self` is the feature used in the state-of-the-art English ALT [Ou et al., 2022] and `large-xlsr-53` is pre-trained on cross-lingual data. This allows us to demonstrate how pre-training language settings can affect performance.

	Mono	4 Lang	5 Lang	6 Lang
English	53.15	51.06	50.27	50.92
French	78.27	71.13	70.08	70.69
Spanish	80.51	69.71	68.55	65.55
German	82.78	71.67	70.93	68.24
All	73.32	65.68	64.74	63.71

Table 6.3: WER (%) of monolingual, four-, five-, and six-language models. Significant differences are found between monolingual and 4Lang models, and between 5Lang and 6Lang models, but not between 4Lang and 5Lang models.

6.5.2 Results and Discussion

Table 6.3 shows the WERs of spectrogram-based models trained on subsets comprising one, four, five, and six languages. It can be observed that, by training on more languages’ data, overall WER is improved, as well as the individual WERs. Note that Russian uses the Cyrillic writing system, which has no overlap with the Latin alphabet.

There is a clear gap between the WERs for English and those for other languages. For monolingual models, Spanish has better WERs than German, despite having a similar amount of training data. This corresponds to the characteristics of the two languages: Spanish has more phonetic and consistent spelling and pronunciation than German [Delattre and Olsen, 1969]. Furthermore, German more frequently employs compounding, resulting in greater variations in pronunciation and stress patterns. This suggests that the complexity of pronunciation and spelling rules differs among languages, thereby affecting the difficulty and data requirements for training ALT models in various languages. By joint training ALT models on multilingual data, non-English languages experience more significant improvement compared to English, and the gap between English and other languages is largely reduced. This indicates that leveraging high-resource data (English) can enhance the performance of low-resource language ALT and mitigate performance imbalances by taking advantage of shared spelling/pronunciation patterns.

Table 6.4 presents the WERs of our multilingual baseline, two Wav2Vec2 models, and Whisper, as introduced in Section 6.4.2. Whisper results are computed on mixture, as opposed to source-separated vocals with Open-Unmix, since it performs better. The impact of background music on transcription is later discussed in detail in Chapter 7. As expected, the Wav2Vec2 models and Whisper outperform our multilingual Transformer. Wav2Vec2 cross-lingual achieves better WERs than Wav2Vec2 English for all metrics except English, indicating that the pre-training configuration significantly influences performance.

	Baseline 4lang	Wav2Vec2 Cross-lingual	Wav2Vec2 English	Whisper large-v3
English	51.06	38.63	34.96	36.80
French	71.13	55.17	69.91	49.33
Spanish	69.71	45.17	63.24	41.15
German	71.67	46.62	64.82	44.52
All	65.68	46.42	57.98	42.95

Table 6.4: WER (%) of two Wav2Vec2-based multilingual models and Whisper. The initial Wav2Vec2 feature extractors are **large-xlsr-53** for Wav2Vec2 Cross-lingual and **large-960h-1v60-self** for Wav2Vec2 English.

Even with the use of pre-trained models, the WERs for non-English singing remain higher than those reported for the English monolingual model in Ou et al. [2022], suggesting greater challenges of multilingual ALT.

6.6 Language Conditioning

6.6.1 Method

Language-Informed Models

We condition our multilingual model with language information to examine its impact. By informing the model of the target language, it can learn language-specific features via the encoder and accurately predict characters from the target language’s alphabet through the decoder. We explore two methods to condition models on language information.

Language embedding A learnable embedding layer is used to convert language IDs into embeddings, with a dimension of 5. This embedding is then broadcast along the time axis to match the dimensions of the model’s input features before concatenation. During training, we investigate three methods: appending the language embedding to the encoder input (**emb1**), to the decoder input (**emb2**), and to both inputs (Figure 6.2). These three conditioned models are referred to as Emb1, Emb2, and Emb12, respectively.

FiLM conditioning layers The second conditioning method we explore is Feature-wise Linear Modulation [Perez et al., 2018] (FiLM). A FiLM layer applies linear modulation to intermediate output whose parameters are generated according to the conditions. We explored four conditioning positions and their combinations, as indicated in Figure 6.3: FiLM1 applies to the input of the Transformer encoder, FiLM2 applies to the input of the Transformer decoder, FiLM3 applies to the output of the CTC branch, FiLM4 applies to the output of the seq2seq branch. Note that, FiLM1 and FiLM2 apply to the same position

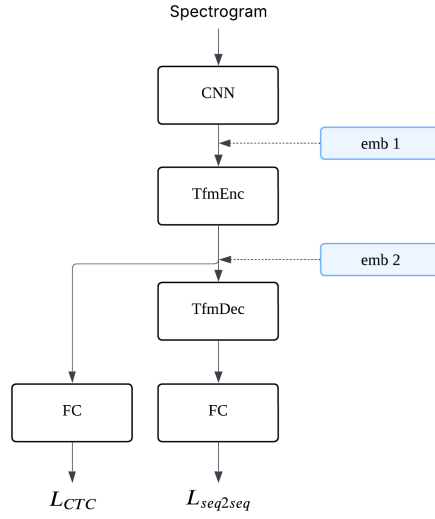


Figure 6.2: Language conditioning through language embeddings.

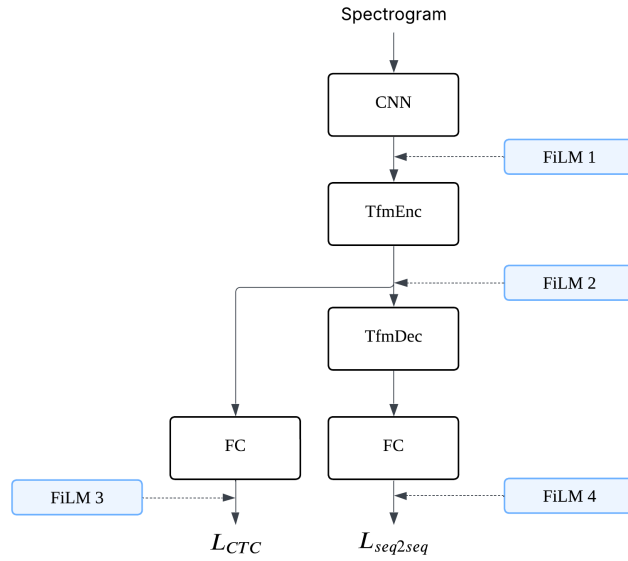


Figure 6.3: Language conditioning through FiLM layers.

as emb1 and emb2 in the network, while FiLM3 and FiLM4 are specific to the FiLM conditioning mechanism.

Optionally Language-Informed Models

Despite the improved performance, the above language-informed models are limited to scenarios where language information is accessible, narrowing their use cases. Taking one step further, we extend the language-informed models to enable language-agnostic (blind) inference while retaining language configurability. This can be achieved through straightforward methods, including adding an unknown language condition and bypassing the conditioning layers. These two methods are respectively combined with the language embedding and FiLM conditioning methods.

Unknown language embeddings When the language information is unavailable, an embedding representing ‘unknown language’ would replace the standard language embedding.

FiLM layers with a switch A switch is incorporated into the FiLM layer to allow bypassing the feature modulation. During training, the switch is randomly activated and deactivated for each batch. As a result, the model gains the ability to infer with and without language information.

6.6.2 Results and Discussion

	Baseline	Emb			FiLM						
	no cond.	1	2	12	1	2	3	4	12	123	1234
English	51.06	52.49	49.92	50.48	53.05	51.05	51.64	50.94	51.27	49.66	51.10
French	71.13	66.89	68.71	68.48	68.71	68.94	70.92	71.24	67.92	67.68	68.24
Spanish	69.71	63.71	66.52	63.54	63.58	65.61	68.46	68.89	62.80	63.10	64.36
German	71.67	62.40	63.82	62.13	62.87	63.53	67.91	68.24	61.47	61.90	61.95
All	65.68	65.68	62.12	61.09	62.00	62.19	64.58	64.67	60.82	60.51	61.34

Table 6.5: WER (%) of language-informed models. The numbers indicate the activated conditioning positions depicted in Figure 6.2 and Figure 6.3. Significant differences are found between baseline~Emb12 and baseline~FiLM123.

Experiments are conducted with single or multiple conditioning layers. Table 6.5 lists the results of language-informed models. The following observations could be made:

- The best overall WER is produced by FiLM123. Applying conditioning at multiple positions outperforms single-layer conditioning.
- When comparing encoder-level conditioning (position 1) with others, later conditioning tends to favour English results more, sacrificing other languages’ results.
- Compared to the Transformer baseline, improvements are often observed in some languages, while others may see no change or a slight decrease

in performance. Besides, lower-resource languages (Spanish and German) benefit more from language information than higher-resource languages (English and French). Specifically, FiLM1234 achieves a 9.7 percentage point reduction in WER for German.

	Emb12		FiLM123	
	unk		switch	
Conf.	on	off	on	off
English	51.26	52.20	50.13	50.64
French	67.70	70.19	66.63	69.85
Spanish	62.86	67.60	63.77	69.03
German	61.94	67.68	63.45	69.81
All	60.88	64.28	60.88	64.63

Table 6.6: WER (%) of language-configurable models. On/off in the configuration row means whether language conditioning is activated for inference. No significant difference is found between Emb12 and FiLM123 under the same conditioning settings.

Emb12 and FiLM123 are extended to optionally language-informed models because they perform the best in the two conditioning categories. The results are shown in Table 6.6.

- After introducing the mixed training, the overall language-informed WERs drop slightly but the language-agnostic WERs are better than the baseline.
- The overall WERs for Emb12-unk and FiLM123-switch is very close, and the absolute WER improvement brought by language information for each language is similar, too.

This suggests that both embedding-based and FiLM-based conditioning capture language information effectively, and that the choice of conditioning mechanism may be less critical than the presence or absence of explicit language cues.

6.7 Phonetic Reinterpretation

Phonetic reinterpretation builds upon the concept of transliteration, which traditionally involves representing the sounds of words in a different script. While transliteration focuses on systematic conversions between writing systems (e.g., Latin to Cyrillic), phonetic reinterpretation extends the concept to include cross-lingual phonetic variation within the same script (e.g., between English and French). This approach, while not a strict transliteration in the conventional sense, allows for phonetic alignment across languages.

In the context of lyrics transcription, phonetic augmentation through reinterpretation serves as a data augmentation method, as it allows us to improve low-resource language ALT with high-resource language data. Together with language conditioning, the model is equipped with the ability to generate phonetic proxies according to the requirement for the target language.

6.7.1 Method

Generating Phonetic Proxies

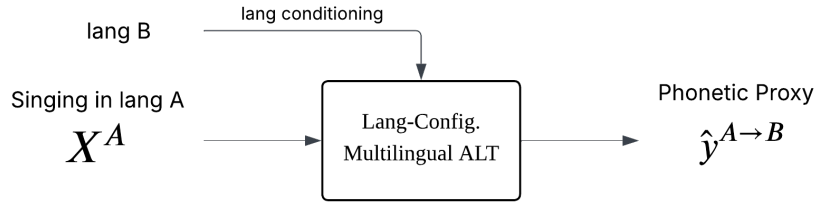


Figure 6.4: Process for generating phonetic proxies for singing samples in language A to language B using a language-configurable multilingual ALT model.

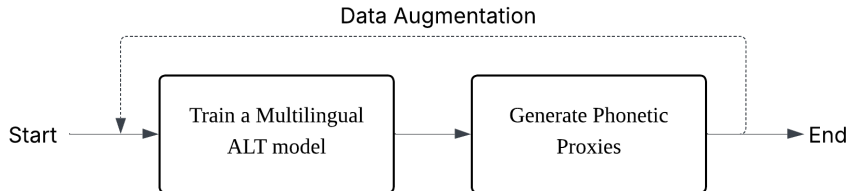


Figure 6.5: Process for training with phonetic augmentation.

Common practices of transliteration apply to text without reference to audio. Such an approach assumes that the pronunciation in the audio follows the standard (e.g. the dictionary). However, singing pronunciations vary greatly depending on the pitch, technique, and singer’s habit. Therefore, to obtain phonetic proxies that best reflect the pronunciation, we use ALT models to transcribe the singing voice in the target language. Figure 6.4 illustrates the process.

For each line in the training data, phonetic proxies in the other three languages are generated by a language-conditioned model as introduced in Section 6.6.

Training with Phonetic Augmentation

Phonetic proxies are used for phonetic augmentation on the text target. Figure 6.5 illustrates this process: phonetic proxies are generated and added to the training data, after which the model is retrained. During augmented training, at a probability of `aug_rate`, the text target would be randomly chosen from the four phonetic proxies (including the original target). Otherwise, the original target is used. The language conditioning would be adjusted accordingly. For example, if a French proxy is selected, the language conditioning would be set to French. At validation, no phonetic proxies are used.

6.7.2 Experiments

Since embedding and FiLM conditioning methods perform similarly, in this section, we apply phonetic augmentation with FiLM123 and FiLM123-switch to demonstrate the effectiveness. `aug_rate` is set to 0.3, which is determined from our pilot study.

At inference, we report the results with the conditioning switch on and with it off. Training with phonetic proxies raises the risk of predicting nonsensical text, as mentioned in Section 6.4.4, we also show the results that incorporate LMs. We refer to language conditioning and LM as language support.

In addition to training on the 4-language subset with phonetic augmentation, we trained two models using 5- and 6-language sets, where additional languages are paired only with their phonetic proxies in the original 4 languages. For example, in the 5-language experiment, the training data is the 5-language subset. For each Italian audio sample, a phonetic proxy is generated in each of the 4 other languages using the corresponding FiLM123 or FiLM123-switch model, with one proxy randomly selected as the target during training. This setup is designed to assess whether phonetic augmentation with previously unseen languages can improve performance.

6.7.3 Results and Discussion

Results for phonetic augmentation are presented in Table 6.7 and Table 6.8. After training with phonetic augmentation, the overall WERs improved for both FiLM123 and FiLM123-switch across all inference settings, with more significant improvements observed when inference involved less language support. For FiLM123-switch, phonetic augmentation is most effective for blind inference.

Comparing FiLM123-switch blind inference before and after phonetic augmentation, all WERs including individual ones show a clear improvement. While with more language support, individual WERs exhibited diverse patterns. It

Tran. Aug.	Lang	All	English	French	Spanish	German
N	Cond.	60.51	49.66	67.68	63.10	61.90
	Cond. & LM	58.65	49.48	65.92	59.25	60.09
Y 4lang	Cond.	59.79	51.20	65.43	60.81	61.97
	Cond. & LM	58.47	50.92	64.05	60.68	58.37
Y 5lang	Cond.	59.24	50.20	64.29	61.49	61.33
	Cond. & LM	57.88	49.50	63.91	59.29	59.03
Y 6lang	Cond.	59.25	50.13	65.26	61.66	60.21
	Cond. & LM	58.14	49.96	64.19	60.49	58.10

Table 6.7: WER (%) of the FiLM123 models trained with phonetic augmentation. Under the **Tran. Aug.** column, **N** means no phonetic proxy used, which is the same model as the column **FiLM123** in Table 6.5. **Y** means that the model is trained with phonetic augmentation generated by **FiLM123**, followed by the number of languages involved. The **Lang** column indicates if language conditioning and LMs are used during inference. After introducing phonetic augmentation, significant differences are observed in conditioned overall (*Cond. All*) WERs only between **N** and **Y 5lang**, and between **N** and **Y 6lang**.

Tran. Aug	Lang	All	English	French	Spanish	German
N	None	64.63	50.64	69.85	69.03	69.81
	Cond.	60.88	50.13	66.63	63.77	63.45
	Cond. & LM	58.66	49.45	63.68	60.68	61.24
Y 4lang	None	62.65	47.94	68.29	67.17	68.04
	Cond.	60.37	48.71	67.36	63.79	62.01
	Cond. & LM	58.37	48.29	64.99	60.67	59.82
Y 5lang	None	62.71	49.25	70.45	65.40	66.28
	Cond.	60.53	49.10	67.87	62.71	62.81
	Cond. & LM	59.27	50.59	65.85	59.84	61.00
Y 6lang	None	63.42	49.50	70.17	66.07	68.63
	Cond.	60.39	48.94	66.20	62.97	63.97
	Cond. & LM	58.74	48.46	64.56	60.20	62.17

Table 6.8: WER (%) of the FiLM123-switch models through phonetic reinterpretation iterations. Under the **Tran. Aug.** column, **N** means no phonetic proxy used, which is the same model as the column **FiLM123-switch** in Table 6.6. **Y** means that the model is trained with phonetic augmentation generated by **FiLM123-switch**, followed by the number of languages involved. The **Lang** column indicates if language conditioning and LMs are used during inference. **None** means blind inference. After introducing phonetic augmentation, significant differences are observed for blind inference (**None**) in overall (*All*) WERs between **N** and the augmented versions **Y *lang**.

indicates that using phonetic reinterpretation probably brings some confusion to the model on the understanding of each language.

The significant improvement observed in the blind inference setting likely results from several factors. First, phonetic augmentation introduces phonetic diversity that enhances generalization across languages, which appears especially beneficial in the blind inference setting. Second, the relatively low augmentation rate provides a balanced approach, adding helpful variation without confusing the model. When language-specific support is added, the model is more guided by spelling and grammar rules specific to each language, which may reduce the influence of phonetic diversity introduced through phonetic reinterpretation.

Going from no language support to language conditioning, to language conditioning with LM, most WERs improved, with the only exception being FiLM123-switch trained with phonetic reinterpretation on the English subset. The same model, FiLM123-switch, gives the best blind inference WER so far.

6.7.4 With Unseen Languages

The rows named **Y 5lang** and **Y 6lang** in Table 6.7 and Table 6.8 present the results when applying phonetic augmentation to Italian and Russian, languages that were not seen during training for the non-augmented models.

When moving from **Y 4lang** to **Y 5lang**, we observe improvements in most WERs for FiLM123, but extending further to **Y 6lang** yields limited benefits. Only German WERs show a modest improvement of around 1 point, while other languages’ WERs remain in the same range or worsen. This outcome contrasts with the findings in Section 6.5, where multilingual joint training is generally beneficial. It is likely that the models generating phonetic proxies encountered challenges with phonetic alignment when faced with an unfamiliar language.

Italian, the fifth language, aligns more closely with the original four linguistically, while Russian differs in origin and phonetics. Although phonetic proxies mitigate the effect of its Cyrillic script, Russian’s unique phonetic patterns and linguistic structure may reduce reinterpretation quality, impacting performance.

For FiLM123-switch, adding additional languages either does not improve or slightly worsens performance, with Spanish being the only exception where improvement is observed. However, in Section 6.5, both Spanish and German saw gains when adding the fifth and sixth languages. This inconsistency suggests that Russian and Italian may not align as closely with Spanish and German linguistically. Instead, the observed improvement in Spanish and German may relate more to their limited data resources rather than linguistic similarity.

6.8 Limitations and Key Insights

The proposed methods for multilingual ALT show promise, but the overall WERs are still quite high. This is mainly due to our choice of baseline models, which allow for a wide range of exploratory experiments. Although our methods improve performance, the accuracy is not suitable for production use. Additionally, the imbalance in training data across languages makes fair comparisons difficult. Languages with more training data tend to perform better, which can hide the true effectiveness of our methods for low-resource languages. This issue affects both acoustic and language models.

It is important to note that this work is primarily investigative. Our goal was to explore the potential of existing techniques in the context of multilingual ALT. This focus enabled us to thoroughly evaluate the impact of language information and phonetic augmentation.

Some findings align with ASR research, particularly regarding the benefits of multilingual joint training and language conditioning in enhancing performance for low-resource languages. For instance, as in ASR, incorporating high-resource languages improves model generalization across languages.

Our study offers insights specific to ALT, particularly regarding the challenges posed by pronunciation variation, singing-specific distortions, and cross-lingual phonetic patterns. We are inspired by the phenomenon that listeners often ‘mishear’ and reinterpret lyrics across languages, and our phonetic augmentation facilitates the alignment of phonetic patterns in ways not commonly addressed in ASR, especially within complex musical contexts. Additionally, our work provides key insights for future research. Multilingual joint training, while effective for languages with shared linguistic origins, could be further explored for languages with different writing systems, such as Russian, and potentially extended to more diverse language families. The success of language conditioning at multiple levels also offers a reusable insight for improving model performance in similar multilingual tasks. Finally, the phonetic reinterpretation we propose has the potential to be applied to other low-resource languages, including those with different origins, making it a promising area for future investigation.

6.9 Conclusion

In summary, this chapter provides practical guidelines for developing a multilingual ALT model with publicly available data: Joint training on data from across a wide range of languages is beneficial. Language information could enhance performance through conditioning. Conditioning at multiple positions is usually beneficial, while the best combination depends on the conditioning

mechanism.

We propose optionally language-informed models based on language-informed models to allow inference when language information is not available. This model performs slightly better in blind inference than the multilingual baseline, and slightly worse than the language-informed prototype. Alongside this, we introduce phonetic reinterpretation as a data augmentation strategy applied on the text target side, which, together with the language conditioning methods, proves particularly effective for low-resource languages. These findings contribute to the broader goal of extending ALT beyond English, enhancing its applicability to diverse languages.

There remains a noticeable gap between our models and those utilizing pre-trained models. Future advancements in architectures that learn from data more efficiently, such as OWSM v3.1 [Peng et al., 2024], may help bridge this gap further. The shorter development cycle provided by a lightweight model is a great advantage for iterative development.

In addition, we have demonstrated how phonetic augmentation could boost the performance of a spectrogram-based model using publicly available labelled datasets. Future work includes exploring the use of unlabeled datasets with phonetic augmentation and extending our study to other languages, including those from different writing systems. Further investigation is needed to refine phonetic reinterpretation techniques for balancing multilingual data, as well as to explore alternative tokenization strategies, such as phoneme representations or subword units, which may further enhance transcription performance.

The next chapter focuses on utilizing foundation models for multilingual ALT, which achieve better performance at the cost of higher computational demands and reduced flexibility in fine-tuning for specific target setups.

Chapter 7

Enhancing ALT for Music Mixtures: Adapting Foundation Models

7.1 Introduction

Previous research on the ALT task often distinguishes between studies that involve background music and those that do not. Music accompaniment, and similarly backing vocals introduce additional information within the acoustic scene, which can make it difficult for the model to focus on the lyrics and the singing voice. Many ALT studies have explored ways to mitigate this challenge, such as training on both music mixtures and their vocal-separated versions [Demirel et al., 2021c]. Alternative approaches, including joint training with source separation techniques, have been proposed to enhance model performance in these complex acoustic environments [Gao et al., 2023a].

Many recent ALT systems still rely on singing voice separation (SVS) as pre-processing for improved performance. Our multilingual ALT system presented in Chapter 6 also has an SVS component. In Gao et al. [2023a], ALT performance is directly influenced by the SVS model’s vocal separation quality, posing a challenge in optimizing the overall ALT pipelines as SVS models are trained separately. Moreover, SVS adds additional computational complexity during inference, making it less ideal for real-world applications. To reduce the reliance on SVS for improved performance, we handle music mixtures by treating background music as noise in this chapter. This requires the model to learn vocal representations that are invariant to music accompaniment. However, achieving this alignment is challenging due to the mismatch between these two domains,

which can hinder performance generalization.

As discussed in Section 2.6 and Ma et al. [2024], foundation models have demonstrated exceptional performance in ASR and MIR. Their success comes from large-scale pre-training on diverse speech datasets, allowing them to effectively generalize across various domains and tasks. When dealing with music mixtures, foundation models can leverage their robustness to diverse acoustic conditions, especially when they are exposed to various types of background sounds during training. This makes them suitable for the ALT task, where the model needs to focus on the singing voice while ignoring irrelevant accompaniment.

To better understand the challenges of adapting ASR foundation models, we begin with probing experiments using Wav2Vec2, a widely used SSL model (described in Section 2.4.3), under different acoustic domains. This analysis helps frame the generalization challenges when applying speech-trained models to ALT, especially when the background music is present. By freezing the Wav2Vec2 front-end and only training a lightweight back-end, we gain insight into domain sensitivity and the inherent difficulty of the datasets.

We then move on to explore parameter-efficient fine-tuning of Whisper, a foundation model originally developed for ASR known for its strong noise robustness. Previous studies that leverage Whisper can be found in Section 2.8.3. Given the computational challenges of full model fine-tuning and the limited size of ALT datasets, we employ Low-Rank Adaptation [Hu et al., 2022], a parameter-efficient fine-tuning (PEFT) method described in Section 2.4.5. We investigate single and dual-domain fine-tuning strategies, where models are trained on either with or without source separated vocals. Our results highlight the challenge of optimizing dual-domain training, as naïve approaches underperform compared to vocal-only fine-tuning. To address this, we propose using a consistency loss to better align vocal and mixture representations. While the improvement is modest, it demonstrates the potential of structured dual-domain training. Additionally, we release new annotated data for benchmarking ALT in two new languages, Italian and Portuguese ¹. The work on Whisper was conducted during my internship at Moises, which is accepted at INTERSPEECH 2025 [Huang et al., 2025].

The remainder of this chapter is organized as follows: Section 7.2 revisits the impact of background music on ALT by reinterpreting the results of previous studies alongside our preliminary experiments. Section 7.3 presents our experiments probing Wav2Vec2 with a lightweight back-end, including the method, experimental details, results and discussion. Section 7.4 describes the

¹The released data are available at: <https://github.com/weAreMusicAI/alt-datasets-interspeech2025>

fine-tuning of Whisper using consistency loss, aiming to improve performance on music mixtures. Both Section 7.3 and Section 7.4 include a method description, experimental details, and results with key findings. Finally, Section 7.5 summarizes the findings and suggests directions for future work.

7.2 Revisiting the impact of background music

We want to revisit some points discussed in Section 1.1 and Section 2.8.4, and examine studies that do not use singing voice separation (SVS) as a preprocessing step during inference, i.e., performing inference directly on music mixtures. Gupta et al. [2020] conducted the first study to directly examine the impact of background music on ALT. Although their best model achieved a word error rate (WER) of 59.6% on Jamendo (which is significantly higher than the state-of-the-art though), they found that models trained on mixtures outperformed those trained on vocals alone. This aligns with the intuition that training with in-domain data leads to better generalization. Similarly, both Gao et al. [2022c] and Demirel et al. [2021c] conducted comprehensive studies on input feature domains when performing inference on music mixtures. Their findings consistently show that models trained on both vocal and mixture data perform best, followed by those trained on mixtures alone, with models trained exclusively on vocals yielding the poorest results.

However, when running inference with Whisper, results does not follow the trend. Previous studies have applied Whisper to ALT [Zhuo et al., 2023, Cifka et al., 2024]. In Jam-ALT [Cifka et al., 2024], the authors investigated the impact of using HTDemucs [Rouard et al., 2023] for SVS before Whisper’s **large-v2** and **large-v3** models. Their results indicate that both versions of Whisper performed worse after applying vocal separation.

We argue that Whisper’s performance on ALT can vary depending on the SVS method used for preprocessing. Our preliminary experiments suggest that, under certain conditions, vocal separation can lead to improved performance. Table 7.1 presents the WER results on the Multi-Lang Jamendo v1.1 dataset [Durand et al., 2023] for three input types: raw mixtures, HTDemucs-separated vocals, and vocals separated using the internal Music.AI SVS model ². The dataset and evaluation methodology are further discussed in Section 7.4.2.

These findings suggest that Whisper is already highly robust to background noise, so improvements from SVS are only observed when the separation quality is sufficiently high. Given that previous studies [Gao et al., 2022c, Demirel et al., 2021c] found that training on both mixture and vocal domains further improves

²A report of the SVS performance is available at <https://music.ai/blog/research/source-separation-benchmarks/>

WER	English	Overall
No SVS	36.80	35.42
HTDemucs SVS	37.50	39.09
Music.AI SVS	32.36	31.86

Table 7.1: A comparison of Whisper **large-v2** performance on mixtures, HTDemucs-separated vocals, and vocals separated by Music.AI SVS model.

the performance, we extend this direction in Section 7.4 by proposing a fine-tuning approach that incorporates both domains to better leverage foundation models.

Notably, while there has been extensive work applying Whisper to mixtures, adapting Wav2Vec2 to work directly with mixtures has not been studied in the literature, despite its strong performance on clean vocals and SVS vocals [Ou et al., 2022]. In the next section, we aim to address this gap through our probing experiments. These experiments provide insights into the impact of background music on Wav2Vec2 features and reveal its generalizability to music mixtures.

7.3 Probing Wav2Vec2 with a Lightweight Back-End

Before diving into our Whisper fine-tuning experiments, we take a step back to examine how another popular SSL model, Wav2Vec2, performs in the context of ALT under different audio domains. Both Whisper and Wav2Vec2 have demonstrated strong performance on ALT, but their generalizability across clean vocals, singing-voice-separated (SVS) vocals, and mixtures is not equally understood. In this section, we conduct probing experiments by freezing Wav2Vec2 and fine-tuning a lightweight back-end, to evaluate its cross-domain generalizability. This analysis provides a complementary perspective to the Whisper-based experiments and highlights the challenges of domain shift in adapting ASR foundation models to ALT.

As introduced in Section 2.4.3 and Section 2.6, probing is a common technique for assessing the usefulness of SSL features in downstream tasks. It typically involves freezing the SSL model as a feature extractor and attaching a relatively simple back-end, such as fully-connected or attention-based layers. This setup allows us to isolate and assess the quality of the learned representations without end-to-end fine-tuning. Given the success of Wav2Vec2 in ALT [Ou et al., 2022], it is worthwhile to explore its performance when exposed to mixture conditions.

In addition, probing across multiple datasets allows us to better understand

dataset-specific factors that influence performance. The large variance in WERs reported in the literature is shaped not only by domain (e.g., mixture vs. vocals), but also by dataset characteristics such as genre diversity, recording and annotation quality, and context window length. Our experiments make this distinction visible by comparing results on two commonly used datasets: DSing (clean vocals) and MulJam v1.0 (mixtures and SVS vocals). Please refer to Section 3.2 for dataset descriptions.

7.3.1 Method

Our experiments build upon the work of Ou et al. [2022], and we adapt their repository ³ to probe Wav2Vec2. In our setup, we freeze the Wav2Vec2 and train only the back-end to focus on evaluating its generalizability. Figure 7.1 illustrates the model architecture. Wav2Vec2 serves as the feature extraction front-end, while the back-end adopts a hybrid CTC/attention design. The total loss is a weighted sum of the CTC loss and the seq2seq loss:

$$L = \lambda L_{CTC} + (1 - \lambda) L_{seq2seq} \quad (7.1)$$

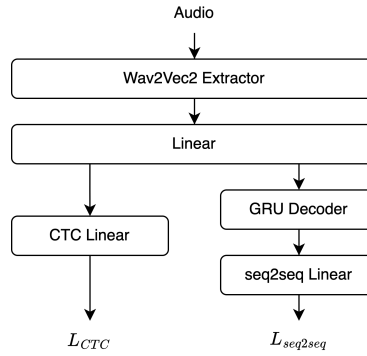


Figure 7.1: Diagram of Wav2Vec2 probing for ALT.

We provide a systematic study on the choices involved in training and architecture design in Section 3.5, which also demonstrate the impact of freezing the Wav2Vec2 front-end.

To examine cross-domain generalizability, we train and evaluate models on three domains:

- **Clean vocals**, sourced from datasets with vocal-only recordings.
- **SVS vocals**, obtained by applying singing voice separation (SVS) to mixture recordings.

³https://github.com/guxm2021/ALT_SpeechBrain/

- **Mixtures**, containing background music.

While SVS vocals and clean vocals are often treated equally in the literature, this is only an approximation. As discussed in Section 7.2, SVS tool quality has a significant influence on transcription performance. Therefore, we treat them as distinct, but closely related domains in our evaluation.

7.3.2 Experiments

Datasets

The training data for the three domains are sourced from two datasets: DSing and the English subset of MulJam v1.0. Clean vocals are taken directly from DSing, while the mixture is drawn from MulJam v1.0, and its corresponding SVS vocals are obtained by applying Open-Unmix [Stöter et al., 2019] to the mixtures. For evaluation, we use the DSing test split for models trained on clean vocals, and the MulJam v1.0 validation set for those trained on mixtures and SVS vocals. We choose the MulJam validation set over its test set due to the annotation granularity: the test set is labeled at the song level, whereas the validation set provides line-level annotations. To ensure consistency in evaluation across domains, we report line-level metrics on the MulJam validation set.

Training and Evaluation

All models are trained for 10 epochs, and the final checkpoint is used for evaluation. Following the setup in Ou et al. [2022], the back-end is trained using the Adam optimizer [Kingma and Ba, 2015] with an initial learning rate of 10^{-5} , scheduled using the Newbob technique. The batch size is set to 4, and the weighting factor for the loss function is set to $\lambda = 0.2$. Throughout the experiments, the Wav2Vec2 front-end is kept frozen, using the pre-trained `facebook/wav2vec2-large-960h-1v60-self` checkpoint.

We evaluate model performance at the line level, as defined by the annotation of the datasets. Both ground truth and predicted lyrics are converted to lowercase, with digits replaced by their corresponding words. Out-of-vocabulary characters are discarded, and redundant spaces are removed. We report both CERs and WERs.

7.3.3 Results and Discussion

Table 7.2 presents the results of probing Wav2Vec2 with a lightweight back-end across three audio domains. We can make the following observations:

Model	Test on	CER (%)	WER (%)
Clean Vocals	Clean Vocals	11	19
	SVS Vocals	88	129
SVS Vocals	Clean Vocals	22	36
	SVS Vocals	71	96
Mixture	Mixture	81	103

Table 7.2: The performance of Wav2Vec2 on different acoustic domains during probing.

First, the model trained on clean vocals achieves better performance on clean vocal data compared to the model trained on SVS vocals. Conversely, the SVS-trained model performs better on SVS vocals than the clean-trained counterpart. These results are expected, given that we only fine-tuned the back-end.

When evaluating on the MulJam v1.0 English subset, with or without SVS, the model trained on SVS vocals outperforms the one trained on mixture data. This supports the idea that SVS data can be beneficial when clean vocals are unavailable. Since SVS vocals are a closer domain to clean vocals than mixtures, Wav2Vec2 performs better with SVS vocals, leveraging the similarity to clean vocals despite the lack of direct access to clean data.

Across all models, word error rates (WERs) on the MulJam validation set remain above 90%, which indicates that MulJam is a significantly more challenging dataset than DSing. Performance gaps were observed in Ou et al. [2022] between DALI and Jamendo when fine-tuning Wav2Vec2.

In summary, these experiments reveal two key insights. First, the generalizability of Wav2Vec2 features across different domains is limited when the front-end is frozen, particularly for mixture data. Second, existing ALT datasets vary widely in difficulty, and such variance must be considered when interpreting performance metrics. These findings provide complementary context to our Whisper-based study in the second part of this chapter.

7.4 Fine-Tuning Whisper with a Consistency loss

In the recent literature, dual-path adaptive training is proposed as an alternative training framework for noise-robust ASR. According to these approaches, an auxiliary loss is included as penalty which is computed as the distance between the intermediary representations of encoder layers. We refer to this auxiliary loss as the *consistency loss*. Shi and Kawahara [2024] proposed this in the feature extraction module to handle data mismatch between pre-training and evaluation, where a mean squared error (MSE) is used as the consistency loss between the noisy and the clean features. In Zhu et al. [2022], an MSE loss between clean

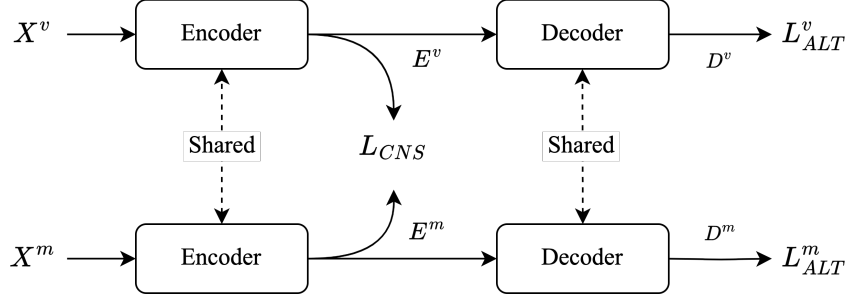


Figure 7.2: Diagram of dual-domain fine-tuning with consistency loss.

and noisy features are combined with the contrastive loss for Wav2Vec2.0 pre-training. Beyond feature pre-training, similar techniques have also been applied directly to ASR. Hu et al. [2023] introduced speech enhancement as a front-end to noisy ASR, and proposed a dual-path style learning approach to enhance robustness. Consistency losses are computed both at intermediate encoder layers, and the decoder output. Recently, Yao et al. [2025] proposed using a consistency loss between two CTC distributions from different augmented versions of the input speech, using the Kullback-Leibler divergence.

Due to their large-scale pre-training feature, the encoder representations of the open-source checkpoints for the Whisper models are shown to capture rich background sound information, making them useful for tasks such as audio tagging [Gong et al., 2023]. When a singing voice is accompanied by background music, this additional musical information is also encoded, even though it is irrelevant to the ALT task, which focuses solely on transcribing the singing voice. This suggests potential improvements by refining Whisper’s ability to focus more on vocal features. To achieve this, we propose using a consistency loss that encourages the model to produce similar encoded representations for both separated vocals and mixture inputs. The aim is for the model to emphasize the singing voice while minimizing the influence of the accompaniment.

7.4.1 Method

Figure 7.2 illustrates the dual-path fine-tuning process with consistency loss. Let $X^d, \forall d \in \{v, m\}$ represent the Mel-spectrograms with v and m , represent the vocal and the mixture paths of the overall architecture. y represents the tokenized ground truth lyrics. Both inputs are processed by the Whisper encoder, enc and decoder, dec . The encoded features produced by the encoder are denoted as $E = enc(X^d)$, while the corresponding decoder outputs are represented as $D = dec(E^d)$.

Following Whisper’s standard training procedure, we use the cross-entropy

Dataset	Split	Songs	30s Segments
DALI v2.0	train	6672	41451
DALI v2.0	dev	100	623
Muljam v2.0	train	5599	32393
Muljam v2.0	dev	114	682
Multi-Lang Jamendo	test	79 (+ 39)	-

Table 7.3: Dataset Statistics of DALI v2.0, MulJam v2.0, Multi-lang Jamendo, and the new Italian (IT) and Portuguese (PT) test sets.

loss, L_{CE} to compute the ALT losses based on the path-specific decoder outputs:

$$L_{ALT}^d = L_{CE}(D^d, y), \quad (7.2)$$

To encourage the model to learn similar representations for vocal and mixture inputs, we introduce a consistency loss computed between the encoded features:

$$L_{CNS} = l(E^v, E^m) \quad (7.3)$$

where the l is the loss term which can be either an L1 or L2 loss. The final loss is computed as:

$$L = \frac{L_{ALT}^v + L_{ALT}^m}{2} + wL_{CNS} \quad (7.4)$$

where w is a tunable parameter that balances the ALT loss and consistency loss. We conduct a series of experiments to determine the best loss function and the weight w .

7.4.2 Experiments

Datasets

For training, we use two datasets commonly used in recent ALT research, namely DALI v2.0 [Meseguer-Brocal et al., 2020] and MulJam v2.0 [Yuan et al., 2023]. DALI v2.0 is a multimodal, multilingual dataset for polyphonic music retrieved from YouTube, with time-aligned lyrics. To ensure language diversity, we selected the five languages with at least 200 songs each (English, French, Spanish, German, and Italian). MulJam v2.0 is a recently released multilingual lyrics transcription dataset in six languages (the above-mentioned five and Russian), sourced from MTG-Jamendo [Bogdanov et al., 2019]. The dataset properties can be seen from Table 7.3.

For evaluation, we used the Multi-Lang Jamendo dataset (described in Section 3.2) which has 79 songs in 4 languages (English, French, German, Spanish). Additionally, we release new evaluation sets in Portuguese (PT) and Italian (IT)

languages, containing 20 and 19 songs respectively. These sets are curated following the principles outlined in Cifka et al. [2024], ensuring consistency with the data properties of the older version. The song selections were sourced from the Jamendo website, associated with proper open-source licensing that would allow sharing for open-science. The lyrics annotations are prepared by three trained annotators who manually corrected identified errors which occasionally involve missing/repeated song sections or typos. Upon completion of the initial annotations, a peer review was conducted, wherein each annotator assessed the accuracy and consistency of the annotations performed by the others. In the rest of the chapter, “overall” refers to the Multi-Lang Jamendo dataset together with the new IT and PT sets.

Pre-processing

Initial inspection on the training sets revealed that the noise in lyrics annotations were non-negligible. To improve data quality, we first remove the over-repeated vowels which represent musical cues embedded in the lyrics. A more significant source of noise was related to inaccurate timestamp annotations and language metadata. To address this, we exploited an internal Music.AI lyrics alignment tool to generate more precise alignments at the lyrics-line level. Prior to generating these alignments, we transcribed the lyrics with an internal Music.AI ALT ⁴ for MulJam v2.0 and relabeled the languages accordingly.

Furthermore, we identified a formatting issue within the annotations provided by DALI v2.0 and Muljam v2.0, specifically on lyrical expressions. It was observed that several songs contained annotations that exaggerated the ground truth lyrics. This was manifested primarily through the repetition of characters, which aimed to simulate an extended pronunciation of certain words, as typically performed by the singer during a performance. To address this undesirable behaviour in our models, we implemented a normalization process targeting these kinds of samples. This approach involves the removal of redundant characters, retaining only those that contribute to the formation of valid words.

This results in a total of 38 languages. Notably, English is the dominant language, accounting for 70.27%, while Italian and Portuguese make up 4% and 0.27%, respectively. Furthermore, tracks without any singing voice content were discarded. For preparing the training data, we merge consecutive lyrics line-level samples to form segments up to 30 seconds. The internal Music.AI SVS is applied to generate separated vocal tracks.

⁴A report of the ALT performance is available at <https://music.ai/blog/research/lyric-transcription-benchmark/>

Training and Inference

All models are fine-tuned using the ADAM optimizer. The learning rate is set to $5e^{-7}$ for most models, except for **voc only**, which uses $1e^{-6}$. These values are selected through a parameter search within $[1e^{-7}, 5e^{-7}, 1e^{-6}]$. The learning rate follows a linear warmup and decay schedule, with the first 10% of training for warmup and the remainder for decay. The batch size is set to 64. For parameter-efficient fine-tuning, we employed Low-Rank Adaptation (LoRA) [Hu et al., 2022], with a rank of 8, a dropout rate of 0.5, and $\alpha = 8$. As discussed in Section 2.4.5, LoRA updates a small subset of parameters while keeping the majority of the model frozen, reducing computational cost without significantly compromising performance.

We employ a long-form decoding method similar to [Radford et al., 2023] that processes audio with sliding windows of 30 seconds. Prompts were not used as priors during inference, as they did not yield consistent performance improvements as reported in [Zhuo et al., 2023]. The model operates without any indication of whether the input segment is a vocal track or a mixture.

Evaluation

We assess model performance on both mixtures and separated vocals. Both the ground truth and predicted lyrics are normalized by converting to lowercase, removing punctuation, and eliminating consecutive whitespace. We report results for English subsets of the Multi-Lang Jamendo, and the newly introduced Portuguese and Italian. We also include the overall WER scores (in Table 7.4) summarized over all languages (i.e. all 118 songs in total).

Baselines

To evaluate the effectiveness of the proposed methods, we compare them against multiple baselines:

Whisper (no fine-tuning): The first baseline is Whisper **large-v2** without any fine-tuning, using the inference pipeline described in Section 7.4.2.

Single-domain fine-tuning: Vocals only fine-tuning (**voc only**) trains exclusively on separated vocals, without any mixture input. It serves for comparison with models trained with mixture data. Mixture only fine-tuning (**mix only**) trains only on mixture data, directly adapting Whisper to the target domain without seeing separated vocals during training.

Dual-domain fine-tuning: Random input fine-tuning (**random**) trains on both vocals and mixture data, with each input randomly selected as either vocals or mixture. This approach allows the model to learn from both data types without structured pairing. Dual input fine-tuning (**both**) processes paired

vocals and mixture samples, computing the loss as the average of L_{ALT}^v and L_{ALT}^m . Unlike **random**, where inputs are chosen independently, this baseline processes paired vocals and mixture samples during fine-tuning. This setup provides a direct comparison to our proposed approach without incorporating the consistency loss.

WER	L_{CNS}	w	EN Mix	Overall Mix	EN Voc	Overall Voc	IT Mix	PT Mix	IT Voc	PT Voc
raw whisper	-	-	36.80	35.42	35.80	33.49	35.98	35.53	34.72	32.68
voc only	-	-	<u>37.31</u>	32.85	<u>30.78</u>	29.67	31.98	33.09	32.28	31.21
mix only	-	-	37.23	33.67	35.62	33.67	33.76	36.13	36.73	34.90
random	-	-	34.90	33.33	33.20	33.35	34.15	35.10	37.36	34.01
both	-	-	36.67	33.79	33.40	32.72	34.48	36.71	35.96	34.37
w/CNS	L1	0.1	35.75	33.26	34.31	32.73	35.11	35.00	35.63	33.24
	L1	1.0	35.53	32.80	34.70	32.76	35.41	34.35	35.41	32.54
	L1	10.0	35.85	32.48	34.55	32.21	34.09	34.20	36.32	32.37
	L2	0.1	34.07	32.94	33.70	32.67	34.48	35.89	35.82	33.86
	L2	1.0	<u>34.53</u>	32.36	34.47	32.47	35.77	34.44	35.65	32.56
	L2	10.0	35.80	<u>32.68</u>	34.41	32.31	34.86	34.11	36.15	32.32

Table 7.4: Word Error Rates (WER) evaluated on both mixture (**Mix**) and separated vocals (**Voc**) conditions. The bold text highlights the best performance within each section, whereas underlined bold text indicates the overall best performance. The L_{CNS} column specifies the consistency loss function used (L1 or L2), and the w column denotes the corresponding weighting parameter.

7.4.3 Results and discussion

Single-Domain Fine-Tuning

The top section in Table 7.4 presents the WERs for the baseline models. We can make several observations. Both **voc only** and **mix only** outperform raw Whisper on overall mixture and overall vocals, indicating that fine-tuning on singing voice data is beneficial. However, for the English subset, performance sometimes worsens, suggesting potential language-specific variations.

The **voc only** model demonstrates strong performance on overall mixture, indicating that fine-tuning only on vocals enhances the model’s ability to generalize to mixture tasks. On the other hand, **mix only** improves mixture performance but does not perform as well on vocals, showing that training exclusively on mixtures does not sufficiently capture vocal features.

Dual-Domain Fine-Tuning

While the **random** model benefits from exposure to both data types, it does not outperform **voc only** on mixture tasks. This suggests that focused fine-tuning on vocals is more effective for mixture transcription than random exposure to both domains. The **both** model, trained with paired vocals and mixture, performs worse than **random** and **voc only**, but better than **mix only** on overall mixture. This suggests that simply averaging the loss without enforcing consistency constraints does not sufficiently enhance performance on mixture data. Among all fine-tuned models, only **random** and **both** show consistent improvement across English and Overall metrics.

Dual-Domain Fine-Tuning with Consistency Loss

The bottom section of Table 7.4 compares the WERs for models fine-tuned with and without consistency loss. Introducing consistency loss, whether using L1 or L2, consistently improves WERs for both English mix and overall mix compared to the **both** model without consistency loss. Notably, all consistency loss models outperform all baselines except **voc only** on overall mix, confirming the effectiveness of the proposed consistency loss approach.

Most models trained with consistency loss also show improvements on overall vocal compared to **both**. This demonstrates that consistency loss benefits both mixture and vocal performances, although the impact is more pronounced for mixture data. Among the models utilizing consistency loss, L2 loss with a weight of 1.0 achieves the best WER on overall mix, outperforming all other models, including **voc only**. This highlights the effectiveness of L2 loss in aligning mixture and vocal representations, though other configurations remain

competitive.

Comparison Between Single and Dual-Domain

The best overall mixture WER is achieved by a consistency loss model, but the improvement over the **voc only** is marginal. Meanwhile, **voc only** remains the most effective approach for other metrics, highlighting the importance of strong vocal representations for ALT performance.

Naïve dual-domain strategies (**random** and **both**), do not show clear advantages over single-domain fine-tuning, suggesting that simply exposing the model to both data types does not effectively bridge the gap between vocals and mixtures. This underscores the challenge of optimizing dual-domain fine-tuning. While incorporating mixture data is intuitively beneficial, it does not easily lead to better generalization. One possible reason is that exposure to mixture data may reduce the model’s acoustic generalizability, making it more sensitive to music styles. In contrast, **voc only** focuses solely on vocals, avoiding this issue. Since correlations between language and genre have been observed [Patel, 2010], dual-domain fine-tuning may introduce genre-dependent biases, leading to worse performance for underrepresented languages. Further investigation is needed to assess the impact of music accompaniment and to explore strategies for effectively leveraging both domains. Consistency loss provides a slight improvement, marking a step in that direction.

Results on Italian and Portuguese

For Italian and Portuguese, the results indicate that consistency loss does not improve WERs. In fact, the best performance is consistently achieved by the **voc only** model. Although consistency loss proves beneficial for English and overall mixture, it does not offer the same advantage for Italian and Portuguese. This discrepancy could be attributed to language-specific characteristics, differences in data distribution, or genre-dependent biases. Notably, Italian and Portuguese account for less than 5% of the training data. As noted in the previous section, correlations between language and music genre may introduce biases, which could contribute to the lower performance observed for Italian and Portuguese. These findings suggest that for underrepresented languages, the **voc only** baseline demonstrates the greatest robustness, consistently achieving the best performance across all conditions.

7.5 Conclusion

This chapter explores the impact of background music for ALT by efficiently adapting Wav2Vec2 and Whisper models across different acoustic domains. We first examine Wav2Vec2’s generalizability through probing experiments, revealing limited performance on mixtures when its front-end is frozen. This indicates the challenge of adapting foundation ASR models to mixtures for ALT.

Next, we explore the use of LoRA fine-tuning on Whisper for improving ALT and proposed a consistency loss for further improvements on music mixtures. While naïve dual-domain fine-tuning is ineffective, consistency loss provides modest but consistent improvements by better aligning vocal and mixture representations. Among different configurations, L2 loss with a weight of 1.0 yields the best performance on mixtures while maintaining vocal transcription quality. Moreover, we introduced a new ALT evaluation set in Italian and Portuguese.

While the publicly available singing data is sufficient to show improvements, it remains limited in scale. This is likely due to the small amount of data available for fine-tuning, which restricts the potential gains. In addition, as shown in Section 7.2, the quality of the SVS tool significantly affects whether using SVS is beneficial for ALT (A similar conclusion regarding lyrics alignment is drawn in Section 4.7). Unfortunately, current open-source SVS tools do not produce separation quality that is reliable enough, and this work depends on access to higher-quality tools. This limits reproducibility and broader applicability. However, we are optimistic that open-source SVS tools will continue to improve and eventually close this gap.

Future directions include leveraging larger datasets, including unlabeled singing data, to enhance model robustness and generalization. Additionally, extending the idea of using a consistency loss to other pre-trained models and exploring its effectiveness in self-supervised learning frameworks remains a promising direction.

This concludes our efforts to address the challenge of background music. The next chapter summarizes the whole thesis, presents the key challenges investigated, insights gained, and contributions made to ALT research.

Chapter 8

Conclusions and Further Work

This chapter summarizes the key findings and contributions of this thesis on automatic lyrics transcription (ALT). After summarizing each chapter in Section 8.1, we revisit the challenges identified in the background chapter (Section 2.8.4) and discuss how they have been addressed. We then highlight the insights gained and the contributions made across various aspects of the research. Section 8.2 provides a reflective discussion on the research question, *What makes singing fundamentally more challenging for ALT?*, based on the observations from our experiments. Finally, we outline the limitations and potential future directions in Section 8.3 and Section 8.4. This chapter provides a comprehensive overview and reflection on the work and place it within the broader context of computational singing voice analysis.

8.1 Summary

This thesis identifies and addresses key challenges in ALT, including melodic pitch, expressive timing, pronunciation variability, multilingual content, and background music. It introduces new datasets and benchmarks, and proposes novel techniques across lyrics alignment, phonetic modeling, multilingual transcription, and music-robust adaptation. The work offers both practical guidelines and conceptual insights into the nature of singing voice, with contributions spanning methodology and dataset development to support ALT research.

Chapter 2 introduces the background of the thesis. It examines the characteristics of the singing voice, reviews the machine learning models, architecture and training paradigms, and the data resources relevant to automatic lyrics

transcription (ALT). It also discusses related research topics, highlighting recent trends and their relevance to the work in this thesis. Specifically, it lists the trends in recent year’s research on ALT, and identifies the key challenges in this field. These challenges are revisited in Section 8.1.1.

Chapter 3 focuses on the dataset and benchmarking for ALT. The development of the MulJam dataset is presented, from its initial release to subsequent updates, explaining the motivations behind them. It also introduces a new English lyrics alignment dataset (word-level lyrics alignment annotations for MUSDB18 test set). Benchmarking experiments are conducted to assess the impact of architecture and training strategies, and provide empirical insights for the design of ALT.

Chapter 4 proposes a multi-task learning approach for lyrics alignment. This work emphasizes the relationship between the onsets of notes and phonemes in singing. By jointly modeling pitch and phoneme predictions, and incorporating structural information into the alignment algorithm, the method achieves higher alignment precision in English experiments. The proposed approach outperforms state-of-the-art end-to-end systems for lyrics alignment, demonstrating the benefit of leveraging musical attributes to improve performance. It is then extended to the multilingual setting. However, the results reveal different trends, leading to further reflection on the role of pitch in ALT. Pitch helps in English but not consistently multilingual, likely due to onset misalignments. It can still provide useful cues for phoneme discrimination, especially in mixtures, but its effectiveness depends on model capacity and annotation quality.

Chapter 5 aims to explore the role of pronunciation in ALT by introducing the singing-to-speech (S2S) conversion task. A generative flow-based S2S system is proposed to transform singing into speech while preserving phonetic information. The approach is evaluated with both objective and subjective metrics and shows effectiveness in modeling phonetic information. It is further applied to low-resource ALT as a data augmentation strategy, showing promising results even on unseen languages. This chapter highlights the importance of phonetic modeling as both an analysis tool and a practical aid for ALT.

Chapter 6 addresses the challenge of extending ALT to multilingual settings by adapting monolingual models to handle multiple languages. The study begins with a straightforward expansion of training data and vocabulary, followed by experiments with language conditioning techniques. To further address cross-lingual variation, phonetic reinterpretation is proposed, inspired by text transliteration. This approach demonstrates potential in improving ALT performance across languages, highlighting again the importance of phonetic modeling, and language-aware design in multilingual ALT.

Chapter 7 investigates how existing ASR models can be adapted to ALT,

with a focus on Wav2Vec2 and Whisper. It examines lightweight adaptation techniques such as probing and LoRA fine-tuning, and the use of the consistency loss to improve feature robustness against background music. These methods are evaluated on vocals, source-separated vocals, and music mixtures. It also introduces new evaluation sets in Italian and Portuguese, expanding the multilingual evaluation scope of ALT. This chapter highlights the benefits and limitations of leveraging ASR foundation models for ALT and provides insights into the challenges in dealing with background music.

Next, we reflect on the challenges, insights, and advances gained from our work, and conclude by outlining potential future directions for research.

8.1.1 Challenges

This thesis addresses several key challenges in ALT identified in Section 2.8.4. Below we reflect on the progress made for each challenge, as well as the remaining open issues.

Dealing with Phonetic Variations

Pronunciation variability in singing presents a major difficulty for ALT, as singers often alter phonemes in ways that differ significantly from speech. Chapter 5 investigates this by introducing the singing-to-speech (S2S) conversion task, which aims to convert singing into speech while preserving phonetic information. By converting singing to speech, we can better reveal phonetic variations that often go unnoticed during singing. When the lyrics are read with singing pronunciation, the phonetic variations become more apparent. This highlights the importance of phonetic modeling in singing voice analysis, and the need for specialized approaches to handle these variations. Experiment results demonstrate that pronunciation variations have a substantial impact on ALT, confirming that phoneme modeling in singing remains an open research questions.

Additionally, Chapter 6 proposes phonetic reinterpretation, which is another attempt to address phonetic variations. This method accounts for pronunciation differences across languages by generating alternative transcriptions that reflect varying spelling and phonetic rules. This remains an early exploration, and further work is needed to robustly handle the diversity of singing pronunciations across languages and styles.

These two approaches are complementary: S2S conversion provides a controlled way to study and isolate pronunciation variations in singing, revealing the impact of phonetic differences on ALT, while phonetic reinterpretation leverages this insight to augment training data, helping models generalize across

languages and spelling variations. Together, they form a framework for both analyzing and mitigating the challenges posed by phonetic variability in singing.

Addressing Data Scarcity

This thesis tackles the issue of data scarcity in ALT through multiple strategies. First, we contribute to the development of large-scale multilingual ALT resources by curating and expanding the MulJam dataset. This dataset provides a diverse and comprehensive resource for training and evaluating multilingual ALT models.

Second, we investigate data augmentation techniques that operate on both the audio and text sides. The singing-to-speech conversion introduced in Chapter 5 facilitates the model to learn singing pronunciation patterns in a more familiar context. Phonetic reinterpretation in Chapter 6 augments text targets by generating alternative transcriptions following rules of different languages. Both methods show promising results in improving ALT performance, especially in underrepresented languages.

Finally, we explore leveraging large pre-trained ASR models as a powerful way to address data limitations. These models, trained on massive amounts of speech data, offer strong feature representations. In this thesis, we explore the use of Wav2Vec2 and Whisper for ALT. While this approach continues to be promising, our experiments in Chapter 7 and 3 indicate the domain gap between speech and singing remains a challenge, especially when dealing with music mixtures. Successful adaptation requires more than simple fine-tuning or probing.

Expanding Beyond English

Extending ALT systems from English to a multilingual setting introduces significant challenges, including data availability and language imbalance. To address these issues, this thesis investigates several strategies for effectively using multilingual data. We explore multilingual training through data mixing, shared model architectures, and language conditioning techniques in Chapter 6. We also propose phonetic reinterpretation as a data-driven augmentation strategy to achieve a balanced language set with synthetic text target.

Dealing with Background Music

Whether background music helps or hinders ALT remains an open question. Current state-of-the-art performance is typically achieved when inference is performed on source-separated vocals using high-quality source separation tools. However, these tools are not always available. From a practical perspective,

performing ALT directly on music mixtures is more efficient and more suitable for large-scale or on-device applications.

Our findings suggest that a model’s exposure to music during pre-training plays a critical role in its robustness to background accompaniment. For example, Wav2Vec2, pre-trained on clean speech, struggles with music mixtures but performs reasonably well on isolated vocals. In contrast, Whisper’s training data includes podcast recordings that often contain background music. It shows better resilience to mixtures. These observations highlight the need for pre-training strategies that incorporate musical context for robust ALT systems.

8.1.2 Insights

End-to-End Model Design

In Chapter 3, we conduct experiments that reveal practical considerations for designing end-to-end ALT systems. Hybrid CTC/seq2seq architecture is consistently effective, which aligns with the findings in ASR literature [Watanabe et al., 2017], confirming its advantage in handling uncertainty in both alignment and sequence modeling. This makes the hybrid design a strong default for ALT, especially in noisy settings.

Additionally, when leveraging pre-trained models such as Wav2Vec2, aligning the downstream task (e.g., character- or phone-level CTC) with the objective used during pre-training significantly improves transferability. This suggests that not only the model architecture, but also the unit of representation and training objective, need to be selected with pre-training compatibility in mind.

Multilingual Adaptation

As explored in Chapter 6, we verify that the straightforward multilingual adaptation approach, which is simply expanding the training data and vocabulary, is effective. Beyond that, conditioning the model with language identifiers at multiple levels of the network can enhance performance in multilingual scenarios.

Our experiments highlight how linguistic properties interact with model behavior: for instance, German compounding or silent letters in French pose unique transcription challenges. Such phenomena suggest that multilingual ALT systems must be sensitive not just to phonetics, but also to language-specific spelling rules and word formation patterns. Understanding the nuances is crucial for building robust multilingual systems.

8.1.3 Advances

We summarize the technical contributions of this thesis as follows.

Datasets and Benchmarking

In Chapter 3, we contribute to the development and ongoing updates of the MulJam dataset, as well as the release of a 45-song English lyrics alignment dataset. In Chapter 7, we release new evaluation sets for lyrics transcription in Italian and Portuguese, further expanding the multilingual evaluation scope of ALT.

In Chapter 3, we contribute the ALT task to the MARBLE benchmark. We also explore the impact of commonly used architectures and training strategies on ALT performance. In Chapter 7, we revisit the ALT task to analyze the effect of singing voice synthesis (SVS) on ALT performance.

Methods

In Chapter 4, we propose a multi-task learning approach for lyrics alignment, which learns a joint representation of pitch and phoneme, and incorporates boundary information to enhance alignment accuracy. The proposed method outperforms a phoneme-only baseline in English experiments, and the boundary-aware alignment algorithm helps avoid cross-line misalignments. The multilingual baseline outperforms the state-of-the-art methods on the multilingual evaluation set.

In Chapter 5, we propose a generative flow-based singing-to-speech (S2S) conversion system. This method leverages a duration prediction mechanism inspired by Glow-TTS to handle the large timing differences between singing and speech. The proposed approach achieves higher phonetic similarity than a transcribe-then-synthesize baseline, and shows promise for data augmentation in low-resource ALT. The cross-domain duration prediction strategy can also be applied to other singing voice manipulation tasks where precise duration control is required.

In Chapter 6, we introduce phonetic reinterpretation, inspired by text transliteration. This approach offers an alternative way to define ALT targets by accounting for phonetic variation across languages. It improves performance, especially in low-resource languages. This novel method addresses the challenge of phonetic variation in multilingual ALT and enables more effective use of available multilingual data.

In Chapter 7, we adopt LoRA fine-tuning and consistency loss to improve ALT performance on music mixtures. LoRA enables efficient adaptation of foundation models by updating only a small subset of parameters, while consistency

loss encourages alignment between vocal and mixture representations. These methods achieve modest but consistent improvements on music mixtures.

8.2 What Makes Singing Harder than Speech

As outlined in the introduction 1.1, this thesis aims to answer why singing voice presents greater challenges than speech for ALT using contemporary machine learning methods. While speech recognition techniques provide a natural starting point, singing introduces unique factors, including wide pitch ranges, expressive rhythm, pronunciation variability, multilingual complexity, and background music. So far, we have discussed the observations and insights from our experiments in the individual chapters, with each chapter designed to isolate one or two specific sources of difficulty. Building on these findings, we now take a step back to interpret what they collectively suggest about the nature of singing voice and its impact on ALT.

Across the experiments, pronunciation variability emerges as the most significant factor complicating ALT compared to ASR. It is important to note that these challenges are interrelated and cannot be directly compared quantitatively. For example, pitch and timing variations can influence pronunciation, and background music can obscure phonetic details. Nevertheless, the evidence across multiple chapters (especially 5 and 6) suggests that pronunciation variability deserves particular attention.

The phonetic variations introduced by singing such as elongated vowels, altered consonants, and expressive articulation, pose a fundamental challenge. These variations frequently create mismatches between the acoustic signal and expected phoneme sequences, leading to higher error rates. This is illustrated in the singing-to-speech conversion experiments (Chapter 5), where the converted speech exhibits high WERs. Listening to these conversions, we often observe ambiguous pronunciations, indicating that recognizing singing pronunciation is inherently difficult without relying on linguistic context. It also suggests that singing-to-speech conversion could benefit from having access to the lyrics during conversion.

Considering whether singing-to-speech models could assist ALT in more direct way than data augmentation, one might imagine converting singing to speech first and then applying a speech recognition model (S2S + ASR). However, the high WERs on converted speech indicate that this approach is unlikely to be effective. The conversion itself is a difficult task, as pronunciation ambiguities in singing are challenging to resolve without linguistic context. In other words, for the purpose of ALT, training a dedicated model directly on singing proves more effective than relying on an S2S conversion plus ASR pipeline.

Addressing pronunciation variability can follow two main directions. The first is increasing the amount of training data to cover a wider range of variations, though this is often limited by dataset availability. The second is improving phoneme modeling to better capture the variations, potentially by leveraging the relationship between phonemes and musical attributes such as pitch and timing.

These insights contribute to ALT research by identifying core challenges specific to singing voice transcription, demonstrating the limitations of speech-based approaches, and providing concrete directions for future work. Researchers can build on these findings to design more robust ALT systems, especially for low-resource languages and complex musical contexts, by informing dataset construction, model design, and evaluation strategies.

8.3 Limitations

Despite the progress made in this thesis, several limitations must be acknowledged, relating to data availability, methodological scope, and computational resources.

Data limitations. A major constraint throughout this work is the limited availability of large-scale, high-quality, and diverse datasets for automatic lyrics transcription. The experiments primarily rely on existing datasets, many of which are relatively small, language-imbalanced, and restricted in genre and recording conditions. These limitations constrain the generalizability of the findings and the robustness of the models developed. The scarcity of extensive and accurately annotated singing data remains a central bottleneck for advancing ALT research.

Methodological limitations. Many modeling approaches explored in this thesis depend on the amount and diversity of available data. The lack of sufficient training data prevents the investigation of more complex architectures or advanced training strategies that may yield further improvements. This limitation also restricts the ability to fully explore multilingual or cross-domain modeling approaches, which could potentially generalize better across varied singing styles and recording conditions.

Computational limitations. All experiments were conducted in an academic research setting with limited computational infrastructure. Most models were trained on a single GPU (NVIDIA V100 or A100), which constrains the feasible model size, batch size, and overall training scale. Access to larger-scale

computational resources would allow for more extensive hyperparameter tuning, longer training schedules, and larger model configurations. Some exploratory experiments with larger models were only possible during an internship at an industrial lab with better hardware support.

Scope of the thesis. Due to the combined impact of data and resource constraints, the scope of this thesis is necessarily limited. The primary goal was not to build a state-of-the-art ALT system, but to understand the underlying challenges and investigate potential solutions within feasible research boundaries. Nevertheless, the approaches proposed here could be further validated and extended when larger datasets and computational resources become available. Researchers with access to such resources are encouraged to apply and adapt the methods discussed in this thesis to larger-scale settings, which may lead to stronger results—or alternatively, reveal new limitations that further refine our understanding of the ALT task.

8.4 Future Work

8.4.1 Learning Better Data Representations

Self-supervised learning (SSL) features have shown great success in various audio tasks, including speech recognition and music information retrieval. However, according to the Wav2Vec2 experiments in Chapters 3 and 7, existing SSL features trained on speech-only data perform much better after fine-tuning, but they do not work well on music mixtures. According to the MARBLE results, SSL features trained on music data, such as MERT, do not perform well on ALT. MERT performs very well on various music tasks, indicating its effectiveness in encoding musical information, but it fails to capture the phonetic information required for ALT.

Our results reveal that current SSL features are not yet suitable for ALT, especially in the presence of background music. On isolated vocals, speech-trained SSL features are competitive, and fine-tuning brings significant benefits. We suggest that future work on data representation should focus on developing features that can jointly capture both phonetic and musical information, and enhance the robustness of vocal features against background music.

8.4.2 Leveraging Data More Effectively

This thesis explored various ways to leverage the available data effectively, including data augmentation techniques such as S2S conversion and phonetic reinterpretation, as well as using related information from other tasks, such as on-

set information from pitch detection, and utilizing pre-trained models such as Wav2Vec2 and Whisper. These methods have shown promising results in improving ALT and lyrics alignment. The best ALT results are achieved by using pre-trained models, which take advantage of large-scale pre-training data in speech. This is likely to remain the trend in future work, as copyright issues limit the amount of available music data.

In this thesis, we did not explore the use of speech data extensively, in order to better understand the challenges specific to the music domain. However, to improve performance, future work should investigate how to leverage speech data effectively, focusing on transferring knowledge from speech to singing, and from noisy speech to singing with background music.

Since the release of Whisper in 2022, much of the ALT work using pre-trained models has relied on Whisper. While it currently offers the best performance, it is important to explore alternative options. Otherwise, advances and insights might become specific to Whisper alone. The OWSM series of pre-trained models [Peng et al., 2024] is worth exploring. These models are trained entirely on open-source speech data, which could provide greater control over training data. The version 3.1 is based on E-Branchformer [Kim et al., 2022], which is more efficient and faster than Whisper.

8.4.3 Understanding and Modeling Singing Voice

In Chapter 5, we propose methods for modeling duration and phonemes in singing voice, which can be applied more broadly to singing voice research and ultimately benefit ALT. Connecting ALT with other singing voice tasks may help us better understand how singing works and how to create or manipulate it. For example, generated music with lyrics can be used to create additional training data for ALT, while better ALT models could, in turn, improve the annotation quality of labeled data for music generation.

Once we better understand pronunciation variation in singing, researchers should take this into account when designing models that involve explicit phoneme modeling, such as pronunciation modules in GMM/DNN-HMM systems or phoneme recognition for lyrics alignment. Instead of relying solely on discrete phoneme labels, it may be useful to explore continuous phoneme representations. The effectiveness of recent contrastive learning approaches [Durand et al., 2023, Kick et al., 2025] may partly stem from their ability to capture such continuous variation.

Future work could further benefit from introducing auxiliary tasks in a multi-task framework. For instance, predicting musical or acoustic attributes such as energy, note onsets alongside phoneme targets could help the model learn

how pronunciation interacts with musical expression. Other auxiliary tasks, such as vowel quality estimation or articulation strength prediction, may also enhance the model’s capacity to generalize across different singing styles and pronunciations. Achieving this, however, requires high-precision annotations of phonetic and musical features to ensure that models learn from accurate and temporally aligned targets.

Alternatively, with sufficiently large and diverse training data, such modeling could be learned implicitly through a powerful language model, reducing the need for explicit phonetic supervision.

8.4.4 Reconsidering the Role of Background Music

In Chapter 7, we mention that the proposed method may cause the model to associate specific languages with specific genres, which becomes a potential issue when training with music mixtures. Genre information has been shown to be beneficial in Gupta et al. [2020] and Gao et al. [2022a]. In Gao et al. [2023a], the authors report the best performance when a small amount of background music is present.

This may also be due to the correlation between language and genre. I interpret the function of background music as providing context for the singing voice, similar to how background noise serves as context in speech recognition. Genre can indicate likely languages, while chords and beats provide cues about pitch and rhythm. Future work should revisit the role of background music in ALT, not as a source of noise to be removed, but as a contextual signal that could help the model better understand and transcribe the singing voice.

Appendix A

Screening questions for phonetic similarity ratings

In the context of the study in Chapter 5, phonetic similarity refers to the extent of acoustic resemblance between two linguistic elements in terms of their pronunciations. However, it is important to note that a universally accepted definition does not exist, as pointed out in [Laver, 1994]. In our online survey, we define the phonetic similarity from a practical viewpoint, assessing the degree to which the given audio sample sounds like the reference in terms of pronunciation.

From our internal tests, we found that the rating of phonetic similarity can be easily influenced by audio quality and intelligibility. We observed that some participants disregarded the pronunciation and consistently chose the sample with the best quality or the most intelligible linguistic content. To reduce such biases in the analysis, we introduced two screening questions.

In each screening question, participants were asked to listen to a short singing reference and choose, from two or more TTS-generated samples, the one that exhibited the highest phonetic similarity. To guide participants’ attention toward phoneme-level pronunciation, the TTS inputs for the correct answers were carefully constructed to mimic the phonetic characteristics of the singing reference. Participants were explicitly instructed to disregard intelligibility, linguistic correctness, and audio naturalness, and to focus solely on how closely the pronunciation matched the reference. The explanation and interface for the screening questions are illustrated in Figure A.1 and Figure A.2. Responses from participants who failed either of the two screening questions were excluded from the main analysis.

For example, the following singing shares the same linguistic content "**WHISPER WORDS OF WISDOM LET IT BE**", but each of them is pronounced very differently.

The pronunciation of the lyrics in standard British English is:
wɪs - pə wɒdz əv wɪz - dɛm lɛt ɪt bi

Below each sample, a transcribed phoneme sequence is provided. The transcription is subjective and you might perceive them differently. This is just to demonstrate what the differences might be:

▶ 0:00 / 0:06 🔊 ⋮

wɪs - pə wɒdz əv wɪs - dɛm lɛt ɪt bi

▶ 0:00 / 0:06 🔊 ⋮

wɪs - pə wɒst əv wɪs - tɛm lɛt ɪt bi

▶ 0:00 / 0:06 🔊 ⋮

wɪs - pə wɒdz ə wɪs - dɛ lɛ dɪ bi

Figure A.1: Example shown to participants to explain the concept of phonetic similarity.

Warmup exercise 1/2: Please select the sample with the highest phonetic similarity to the reference:

▶ 0:00 / 0:01 🔊 ⋮

▶ 0:00 / 0:01 🔊 ⋮

▶ 0:00 / 0:01 🔊 ⋮

Explanation of Phonetic Similarity

Phonetic Similarity refers to how much it sounds like the reference in terms of pronunciation. Please try to avoid recognising the lyrics/words and avoid taking into account if the linguistic content makes sense or not. Please try to also avoid taking into account the naturalness.

Figure A.2: Interface of a screening question.

Appendix B

Objective evaluation on NHSS

In this work, we present the objective evaluation results for the DSing test set. For completeness, results for NHSS are listed in Table B.1. Since the spoken lyrics utterances are available, we use the real spoken utterances by the singer as the reference speech signal instead of TTS for WORLD baselines. Generally, Table B.1 exhibits a similar trend to those on DSing test, with the exception that MOSNet and SRMRs are much higher for WORLD-nodur. This difference may be due to the use of real spoken utterances. The pitch contour extracted from the same singer’s speech provides a more natural intonation than TTS, and aligns with the normal pitch range of that speaker. For the same reason, intelligibility is better retained after duration manipulation, as reflected in WER/CERs. However, in a real-world scenario, access to the same singer’s spoken utterances is not always available.

Duration adjustment Method	Reference Sing	ALT-TTS	nodur			dur		
			WORLD	M1	M2	WORLD	M1	M2
MOSNet↑	2.96	2.88	2.56	2.82	2.79	2.26	2.96	2.98
SRMR↑	15.20	14.76	6.86	13.68	12.75	4.28	18.68	15.59
SRMR-CI↑	9.54	9.17	6.28	8.74	7.86	3.78	10.97	8.89
ASR WER/CER↓	0.56/0.32	0.33/0.18	0.65/0.39	1.35/1.00	1.07/0.70	0.70/0.49	0.84/0.60	0.84/0.60
ALT WER/CER↓	0.23/0.12	0.29/0.17	0.31/0.18	0.70/0.49	0.68/0.46	0.61/0.42	0.68/0.49	0.61/0.43

Table B.1: Objective evaluation on the NHSS set. **Sing** indicates the original singing voice.

Appendix C

Ablation study on Intelligibility

In Section 5.6.1, we report the WERs and CERs with existing ASR and ALT models on the original singing, and S2S-converted singing via ALT-TTS, WORLD and proposed models $\mathbf{M}^*$. We observe that all S2S methods exhibit higher WER/CER after conversion, particularly in the case of the proposed $\mathbf{M}^*$ models. We argue that the degradation is partly due to ASR/ALT models not being well-acquainted with converted speech-like signals uttered in singing pronunciations. In this section, we verify this claim by quantitatively measuring the intelligibility loss caused by conversion.

Each proposed method is applied to the whole training, validation and test sets of DSing. Then an ALT model, same as described in Section 5.5.5, is trained on the converted samples. By training ALT models in their respective domains, the intelligibility metrics can accurately reflect the linguistic information extracted from the converted signals. We refer to these models as in-domain trained ALT models. The results are displayed in Table C.1. The ASR and ALT results from Table C.1 are also included for comparison.

In comparison to pre-trained ALT and ASR models, which sometimes yield WERs close to 1, the in-domain-trained ALT models exhibit lower error rates. A WER degradation of approximately 10% is observed after conversion without duration manipulation ($\mathbf{M}^*$ -**nodur**), and approximately 15% for conversion with duration manipulation ($\mathbf{M}^*$ -**dur**). Interestingly, the nodur conversions outperform dur conversions for in-domain trained ALT models, contrary to the pattern observed in existing ASR and ALT models. The in-domain training results reveal that most linguistic information survived after the nodur conversion and is further degraded through duration manipulation. The poor existing AS-

R/ALT results can be attributed to the domain mismatch between their training and the S2S target.

Method	Sing	M1-nodur	M2-nodur	M1-dur	M2-dur
In-domain	0.27/0.19	0.37/0.26	0.36/0.25	0.44/0.32	0.42/0.30
ALT and ASR rows from Table C.1					
ASR	0.54/0.31	1.00/0.66	0.97/0.63	0.88/0.64	0.84/0.58
ALT	0.13/0.07	0.73/0.51	0.61/0.40	0.71/0.53	0.62/0.45

Table C.1: WER/CERs↓ of ALT models trained in their own domains. **Sing** indicates that the model is trained and tested on singing samples. **M1-nodur** means that the model is trained and tested on **M1-nodur**-converted samples. This table displays the change in linguistic information before and after conversion.

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